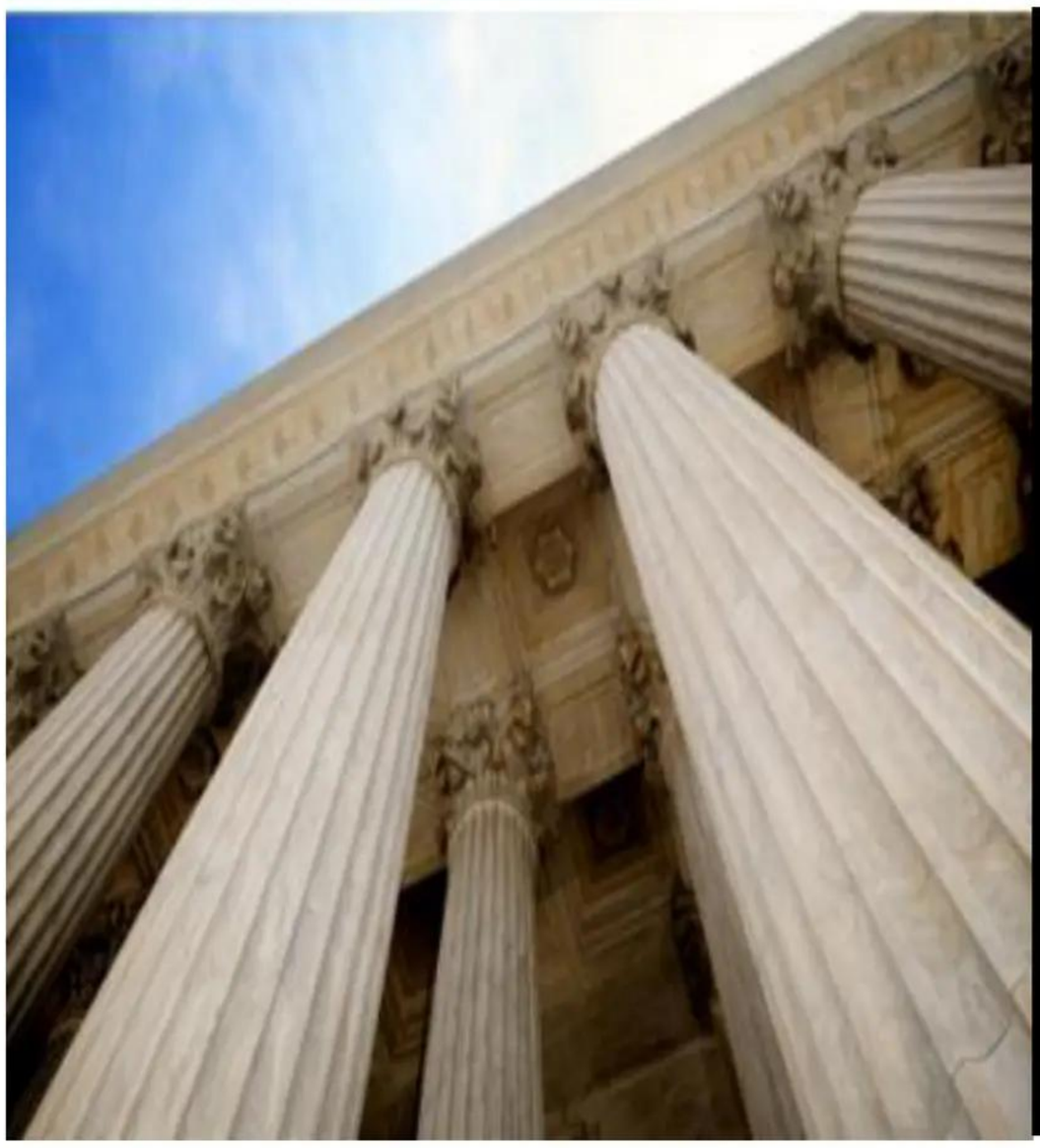

SAP MASTER GOVERNANCE (MDG)
Overview of the integration into the business processes for
financial (MDG-F)
customer (MDG-C)
supplier (MDG-S)
business partner (BP)
- ARIBA

Dipl.-Wirtsch.-Ing. Hans-G. Emrich

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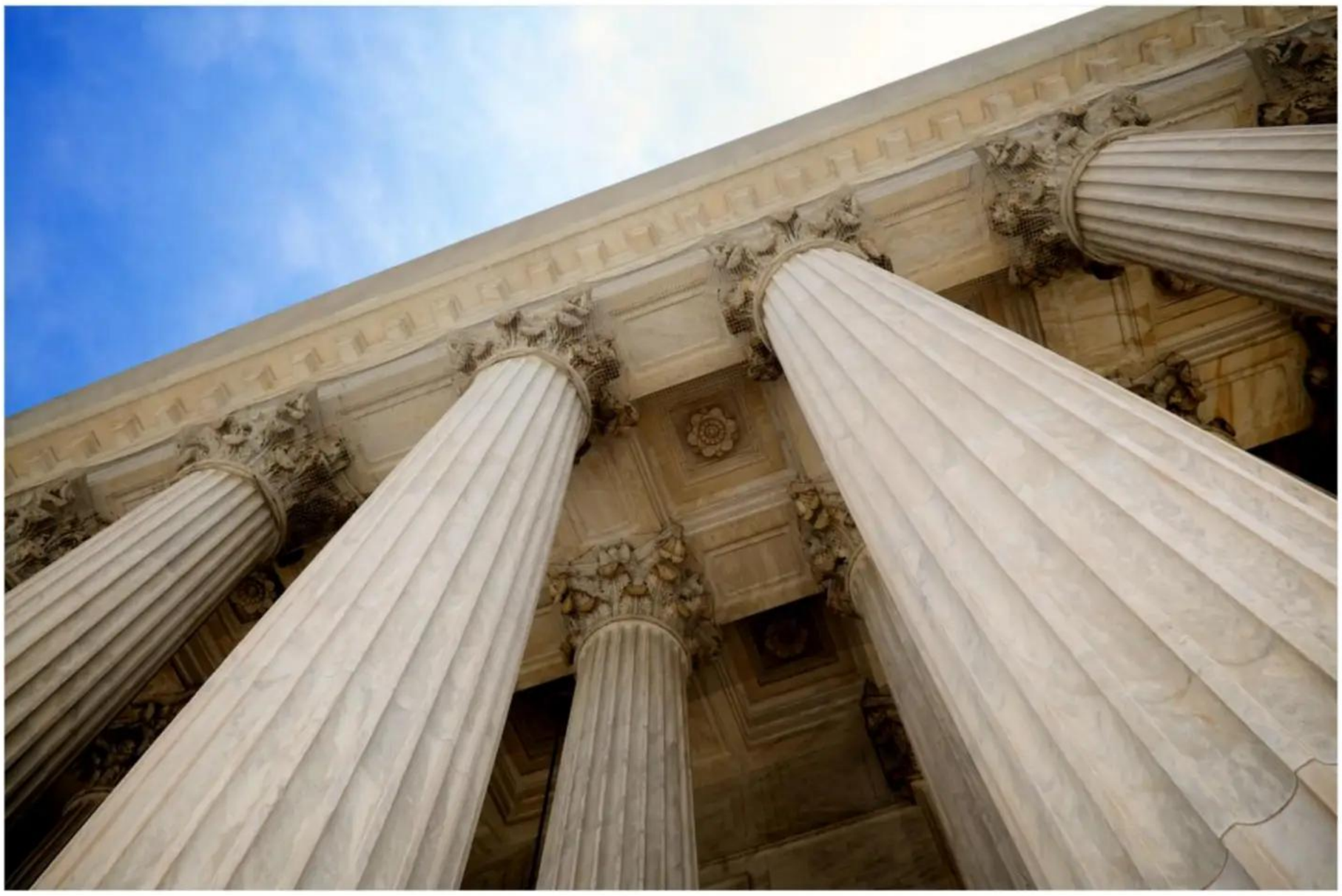
SAP MDG



SAP MASTER GOVERNANCE (MDG)
Overview of the integration into the business processes for
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May 2018

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Table of content

1 Preface 12

2 Introduction SAP Master Data Governance (MDG) 13
2.1 Central governance or consolidation for consistent master data 15
2.2 Integration and re-use 15
2.3 Data quality 16
2.4 Central governance & distribution / de-central ownership & consolidation 17
2.5 One single application for Enterprise master data management 18
2.6 Key trends and impacts in business and technology 18
2.7 Master data governance capatilities 19
2.7.1 Establish Best Quality Master Data to Optimize Business Processes 20
2.7.2 Deliver Out-of-the-Box 20
2.7.3 Integrated Object Model 21
2.7.4 Master data governance in hybrid landscapes 22
2.7.5 End-to-end integration in heterogeneous landscapes 23
2.7.6 Governance, Collaboration 24
2.7.7 Data quality 24
2.7.8 Robust Data Replication 26
2.7.9 Reusability & Extensibility 26
2.7.10 MDG and S/4HANA 27
2.8 Master data excellence for the digital business 28
2.9 MDG road map overview – key innovations 30
2.10 New MDG functions 30
2.10.1 The impact of digitization on master data 31
2.10.2 MDG > 2020+ Product vision - High quality master data for every use case 32
2.10.3 SAP Master Data Governance and cloud consumption models 32
2.10.4 Master data services on SAP Cloud Platform 33
2.10.5 MDG – Business Partner Service Consumption 33
2.11 Strategic direction, using connectivity in business networks 34

3 SAP MDG for Financials (MDG-F) 35
3.1 Specifics of SAP MDG-F: 36
3.2 MDG for Financials Addresses Main Action Areas 37
3.3 The Solution Building Block Overview 37
3.4 MDG-F Process Framework 38
3.5 The Change Request 38
3.6 Editions 38
3.7 Work with Changeable Keys 39
3.8 Process Features 40
3.9 Hierarchy Assignment in Single-Object Processing 40
3.10 Integrated Change Request 41
3.11 Hierarchy Interlocking 41
3.12 Usability Features in Single-Object Maintenance UI 42
3.13 Usability Features in Hierarchy Processing 42
3.14 SAP MDG Supporting Central Finance (S/4 HANA Finance) 43

3.15 Standard Objects in Data Model 0G 44
3.16 Deployment 44
3.17 Replication 45
3.18 Integration 45
3.19 User Interface 46
3.20 Request Cost Center or Profit Center With FioriApp 46
3.21 Approve Cost Center, Profit Center or G/L Account With Fiori App 47
3.22 Copying of accounts 47
3.23 Enhanced BCV content to display hierarchy changes 48
3.24 New functions (planned Enhancements in MDG Application Framework) 49
3.24.1 Integrated Object creation in Hierarchy Processing UI 49
3.24.2 Merge Cost Element into G/L Account 50
3.24.3 Internal Order Central Governance 51
3.25 MDG-F road map overview – key innovations 51
3.26 Benefits of SAP Master Data Governance for Financials 52
3.27 Financial Master Data 54
3.27.1 MDG for Financials 54
3.28 Data Model 0G 56
3.29 Accounting 59
3.30 Controlling 60
3.31 Consolidation 61
3.32 User Interface 63
3.33 Generic Interaction Layer (genIL) 63
3.34 Caution 66
3.35 Context Based Adaptations 67
3.36 Work Centers, Roles and Landing Pages 67
3.37 MDG-F User Interfaces 71
3.37.1 Accounting 71
3.37.2 Controlling 73
3.37.3 Consolidation 75
3.37.4 User Interfaces for Collective Processing 77
3.37.5 Search and Value Helps for Single Object Maintenance User Interfaces 77
3.38 Data Transfer 80
3.39 Object-based Replication 80
3.40 Change-Request based Replication 80
3.41 Edition-based Replication 81
3.42 Replication of Deletions 81
3.43 Replication of Accounts 82
3.44 Replication Monitoring 84
3.45 Validations and Data Derivation 85
3.46 Data Modeling 86
3.46.1 Q: Why is there an active data model 0F in the MDG hub? 86
3.46.2 Q: Is it recommended to copy data model 0G? 86
3.46.3 Q: How do I add custom Data Dictionary search helps to attributes of data model 0G? 86
3.47 User Interface Design 87
3.47.1 Q: Is it recommended to copy existing user interfaces? 87
3.47.2 Q: How can I start the user Web Dynpro application configurator for user interfaces? 87

3.47.3 Q: How can I start the user Web Dynpro component configurator for user interfaces? 89
3.47.4 Q: How can I start the user Web Dynpro customizing configurator for user interfaces? 90
3.47.5 Q: Why is there a short dump when I try to customize a user interface? 92
3.47.6 Q: What does the error message “Feeder class does not provide a valid field catalog” mean? 92
3.47.7 Q: Why is there still a dump / error message although I have defined the Standard Data Model as 0G? 92
3.47.8 Q: Why does the context based adaptation (CBA) not change the layout of the overview page (OVP)? 94
3.47.9 Q: Why are recent changes applied to the user interface not visible for other users? 95
3.47.10 Q: Create a new User Interface for custom Entity with Storage and Use Type 1? 95
3.48 Data Maintenance 96
3.48.1 Q: What does the error message “Relation <ABC> is not supported by Object <XYZ> mean?” 96
3.48.2 Q: Why does the Undo button not work for key fields on the Overview Page? 96
3.48.3 Q: How can I search for an Edition instead of a Valid On date? 96
3.48.4 Q: Why does the Search not find my Object that I have saved as Draft? 97
3.48.5 Q: Why does the Print Preview not work in my copied Application Configuration? 98
3.49 Data Replication 99
3.49.1 Q: How does the MDG Hub handle key mapping for MDG-F entities? 99
3.49.2 Q: Why are attachments not replicated to ERP? 99
3.49.3 Q: Why don’t ERP transactions show a change history for master data changed in MDG? 99
3.49.4 Q: Replicate MDG-F entities into the same client of my system? 99
3.49.5 Q: Why are some of my Account IDocs not processed immediately? 99
 IDoc: <IDoc Number> Status: IDoc ready to be transferred to application 100
3.50 SAP Notes 100
3.51 ALE Replication using the same Client in MDG-F 100
3.52 Steps for the ALE Scenario Configuration 101
3.52.1 Define Logical Systems 101
3.52.2 Define a RFC Connection 101
3.52.3 Define an ALE tRFC Port 101
3.52.4 Define an ALE Distribution Model 101
3.52.5 Define Partner Profiles 101
3.52.6 Steps for the MDG Data Replication Framework (DRF) Configuration 103
3.52.7 Define Business Systems 103
3.52.8 Define a Replication Model 104
3.52.9 Confirmation IDocs (ALEAUD) for setting replication status in MDG 104
3.52.10 SAP Notes 104
3.52.11 Example 104
3.53 New function - Master Data Governancefor FI Contract Account 124
3.53.1 Prozess modelling – new function 127
3.54 Planned functionality 128

4 SAP MDG for Customers (MDG-C) and MDG for Suppliers (MDG-S) 129
4.1 Overview 130
4.1.1 Master data quality impacts all Lines of Business 130
4.1.2 Centrally govern and consolidate master data for your enterprise 131
4.2 SAP Master Data Governance , Central Governance 131
4.2.1 Process Flow 131
4.2.2 Master Data – A multitude of processesand facets> Supplier Master Data 132
4.2.3 Master Data – A multitude of processesand facets > Customer Master Data 132
4.3 When to Use Which Process? 133
4.3.1 Navigation to Create / Change Process 133
4.3.2 Workflow Template: Create Supplier (Distributed) 134
4.3.3 Workflow Template: Change Supplier (Distributed) 134
4.3.4 Workflow Template: Block / Unblock Supplier (Distributed) 135
4.3.5 Workflow Template: Set Flag for Deletion Supplier (Distributed) 135
4.3.6 Workflow Template: Create / Change / Block / Set Flag for Deletion Supplier (Simple) 136
4.3.7 Workflow Template: Create Customer (Distributed) 136
4.3.8 Workflow Template: Create / Change / Block / Set Flag for Deletion Customer(Simple) 137
4.3.9 Create / Change / Block / Set Flag for Deletion Business Partner 137
4.3.10 Workflow Templates out-of-the-box 138
4.3.11 Multiple Change Requests per Customer / Supplier at the Same Time138
4.3.12 Workflow-Step Dependent Configuration Enables Flexible UI / Process Design 139
4.4 User Interface 140
4.4.1 Landing Page for MDG on S/4HANA 140
4.4.2 Workflow Inbox 141
4.4.3 UI Configurations: Overview out-of-the-box configurations 141
4.4.4 UI Configurations: BP Maintenance UI 142
4.4.5 UI Configurations: ERP Customer-like UI 142
4.4.6 UI Configurations: ERP Vendor-like UI 143
4.4.7 UI Configurations: Specialized UI, Maintain / approve Company Code data 143
4.4.8 UI Configurations: Specialized UI, Maintain / approve Sales Areadata144
4.4.9 UI Configurations: Specialized UI, Maintain / approve Purchasing Organization data 144
4.4.10 UI Configurations: Fiori UI, Lean Request for new Customer 145
4.4.11 UI Configurations: Fiori UI, Lean Request for new Supplier 145
4.4.12 UI Configurations: Fiori UI, Lean Request for new Business Partner146
4.4.13 UI Configurations: Fiori UI, Lean Request for change of Customer 147
4.4.14 UI Configurations: Fiori UI, Lean Request for change of Customer (Change Data) 147
4.4.15 UI Configurations: Fiori UI, Track My Request 148
4.4.16 UI Configurations: Fiori UI, Approve Customer 149
4.4.17 UI Configurations: Fiori UI, Approve Supplier 149
4.4.18 Field changes 150
4.4.19 Table changes, Create / Update 150
4.4.20 Highlight Changes: Table Changes, Delete 151
4.4.21 Changes on Detail Pages 151

4.4.22 Undo Changes 152
4.4.23 Create with reference 152
4.4.24 Copy segments 153
4.4.25 Flexible UI via Configuration / Personalization: Call personalization 153
4.4.26 Flexible UI via Configuration / Personalization: Change layout 154
4.4.27 Flexible UI via Context Based Adaptations (CBA) 154
4.4.28 Governance Scope - Deactivate Entity Types / Fields 155
4.4.29 Integrate Additional Information via BCV Side Panel: Open Side Panel 155
4.4.30 Integrate Additional Information via BCV Side Panel: Content for MDG Customer 156
4.4.31 Integrate Additional Information via BCV Side Panel: Content for MDG Supplier 156
4.5 Data Quality 157
4.5.1 Search Overview 157
4.5.2 Search Capabilities 157
4.5.3 Search Options 158
4.5.4 Enterprise Search 159
4.5.5 HANA Fuzzy Search capabilities 159
4.5.6 Duplicate Check, Validations, Address Enrichment 160
4.5.7 Duplicate detection via HANA Fuzzy Search 160
4.5.8 Validations via BRFplus 161
4.5.9 Framework for Data Enrichment Spots 161
4.6 Client Maintenance 162
4.6.1 Process Steps 162
4.6.2 Client Maintenance ERP Customer: Search on Hub & Copy From Hub 162
4.6.3 Maintenance on Hub & on Client can be Used in Unison 163
4.7 Cleansing 163
4.7.1 Overview 163
4.7.2 Identifying Duplicates 164
4.7.3 Cleansing Case 164
4.7.4 Merge 165
4.8 Process of Multiple Objects 166
4.9 Benefits for MDG-S and MDG-C Solution 168

5 SAP MDG for Materials (MDG-M) 169
5.1 Master Data Governance for material data 170
5.2 Capabilities 172
5.2.1 Establish Best Quality Master Data to Optimize Business Processes 172
5.2.2 Deliver Out-of-the-Box 172
5.2.3 Integration Model 173
5.2.4 Governance, Collaboration 173
5.2.5 Data Quality 174
5.2.6 Replication 175
5.2.7 Reusability & Extensibility 175
5.3 Governance Principle 176
5.3.1 Central Maintenance rules 176
5.3.2 Central Maintenance of Materials 176
5.3.3 Material Governance Homepage 177
5.3.4 Fiori Launchpad in S/4 177
5.3.5 Workflow Inbox 178
5.3.6 Workflow-Step Dependent Configuration Enables Flexible UI / Process Design 178

5.3.7 Out-of-the-box configurations 179
5.3.8 Change Requests With Built-In Approval Process 179
5.3.9 Change Request Header 180
5.4 Workflow rules 180
5.4.1 Standardized but Flexible Processes Incl. Rules Management 180
5.4.2 Rule-Based Workflow Template 181
5.4.3 Rule-Based Workflow Components 181
5.5 Architecture Rules 182
5.5.1 Staging Area to Maintain Inactive Versions of Material Data 182
5.5.2 Handling of Active and Inactive Data 182
5.6 Function in Detail 183
5.6.1 Process Flow 183
5.6.2 Material Processes 184
5.6.3 SAP MDG Fiori Apps: Request new material and Approve new or changed material 195
5.6.4 Multi-Record Processing 196
5.6.5 Mass Processing of materials 198
5.6.6 Consolidation - Process flow 198
5.6.7 MDG Framework Capabilities 200
5.6.8 Master data remediation (only S/4HANA) 207
5.6.9 Integration Scenarios 208
5.7 Benefit from the Solution MDG-M 210
6 Business Partner (BP) 213
6.1 Goals for Central Governance of Business Partners 213
6.2 Structure 214
6.3 Integrated Object Model: Business Partners 215
6.4 Business Partner Tables 216
6.5 BP Identification 216
6.6 Business Partner Capabilities 217
6.6.1 Integrated Object Model: SAP Business Partner Fundamentals 217
6.6.2 Integrated Object Model: Solutions for Customers, Supplier, and Business Partners 217
6.6.3 Integrated Object Model: Data Model MDG-BP 218
6.6.4 Integrated Object Model : MDG-C with BP 219
6.6.5 Integrated Object Model: Data Model MDG-S with BP 220
6.6.6 Integrated Object Model: Data Model Customer / Vendor Integration220
6.6.7 Governance, Collaboration 221
6.6.8 Data Quality 222
6.6.9 Data Replication 222
6.6.10 Reusability & Extensibility 223
6.6.11 Deployment Options: Consciously Decide for a Central hub or Co-deployment 223
6.7 Function in Detail Business Partner Data 224
6.7.1 Data Model Business Partner 224
6.7.2 Central Data 224
6.7.3 Addresses 225
6.7.4 Address Usages 225
6.7.5 Address Versions 226
6.7.6 Business Partner (BP) Roles 226
6.7.7 BP - Bank Accounts 227
6.7.8 BP - Payment Cards 227

6.7.9 BP - Identification Numbers 228
6.7.10 BP - Tax Numbers 228
6.7.11 Tax Numbers, Support of numbers with more than 20 characters 229
6.7.12 Industries 229
6.7.13 BP Relationships 230
6.7.14 BP Relationships, Add Relationship in UI 231
6.7.15 BP Relationships, Time Dependency 231
6.7.16 BP - Data Model ERP Customer / ERP Vendor 233
6.7.17 Assign ERP Customers / ERP Vendors, Use Cases & Concept 233
6.7.18 Assign ERP Customers / ERP Vendors, Implementation 234
6.7.19 Assign ERP Customers / ERP Vendors, Usage of Multiple BP addresses (UseCase) 234
6.7.20 Assign ERP Customers / ERP Vendors, One BP perAddress 235
6.7.21 Assign ERP Customers / ERP Vendors, One BP per LegalEntity 235
6.7.22 Assign ERP Customers / ERP Vendors, One BP per Legal Entity, User Interface 236
6.7.23 General Data (Customer and Vendor) 236
6.7.24 General Data (Customer and Vendor), Select account group during creation 237
6.7.25 Documents (Customer and Vendor) 237
6.7.26 Lean Classification (Customer and Vendor), 237
6.7.27 Unloading Points (Customer) 240
6.7.28 Company Code data (Customer and Vendor) 240
6.7.29 Dunning Areas (Customer and Vendor) 241
6.7.30 Withholding Tax Types (Customer and Vendor) 241
6.7.31 Sales Area Data (Customer) 242
6.7.32 Partner Functions (Customer) 242
6.7.33 Tax Indicators (Customer) 243
6.7.34 Purchasing Organization Data (Vendor) 243
6.7.35 Different Purchasing Organization Data (Vendor) 244
6.7.36 Partner Functions (Vendor) 244
6.7.37 Business Partner Screening 245
6.8 Business Partner – Customer-Vendor Integration S/4 HANA 254
6.8.1 Setting up Business Partner S/4 HANA 254
6.8.2 Customizing 259
6.9 Excel Upload for Business Partner 264
6.9.1 OVERVIEW 264
6.9.2 INSTALLATION AND CONFIGURATION OF THE EXCEL UPLOAD264
7 Integration MDG and ARIBA 270
7.1 Add-On deployment in SAP ERP system 271
7.2 Hub deployment of Ariba Network Integration 271
7.3 Ariba Integration for SAP SRM 272
7.4 Add-On Deployment in SAP SRM System Connected to SAP ERP 272
7.5 SAP SNC System Connected to SAP ERP 273
7.6 SAP Ariba Solution Components 276
7.7 SAP Ariba Integration 277
7.7.1 ARIBA NETWORK 277
7.7.2 ARIBA Management solution portfolio overview 279
7.7.3 Enhanced SAP ERP integration with access to SAP ERP master data including material master 280
7.7.4 Master Data Integration 280

7.7.5 Master Data replication (single ERP case) 281
7.7.6 Option 1: SLP – SAP ECC Integration (CI-7) Process Flow 282
7.7.7 Option 2: SLP – SAP ECC / MDG-S Integration (CI-7) Process Flow 282
7.8 Different methods to integrate Ariba and SAP 283
7.8.1 1. File-Based integration: 283
7.8.2 2. Web services based integration: 284
7.8.3 3. Direct connectivity integration: 284
7.8.4 4. Mediated connectivity integration: 284
7.9 SAP ARIBA integration into SAP® ERP and SAP S/4HANA® 284
7.10 SAP ARIBA CLOUD SOLUTIONS INTEGRATION WITH SAP ERP and SAP S/4HANA 285
7.11 Ariba Network and SAP ERP or SAP S/4HANA Integration Scenarios 286
7.11.1 Document Automation: 286
7.11.2 SAP Ariba Supply Chain Collaboration: 287
7.11.3 Contract Manufacturing Collaboration (Subcontract PO): 287
7.11.4 Consignment Collaboration (Consignment PO): 288
7.11.5 SAP Ariba Contracts 290
7.11.6 Master Data Integration 290
7.11.7 Direct Material Sourcing 290
7.11.8 Service Items Integration 290
7.12 THE FUTURE OF SAP: SAP S/4HANA AND THE INTEGRATED BUSINESS NETWORK 292
8 Literature sources 294
9 Additional Information 294
9.1 Links 294
9.2 How-to Guides 294
9.3 SAP Notes 295

1 Preface

SAP Master Data Governance is a Master Data Management solution and controls, harmonizes the master data process in the company. Master data is the beating heart of your organization. Without it, you're dead in the water.

This book is a description of the use of SAP-MDG. It is designed to provide project managers, IT managers and departmental managers with an overview of the application of MDG. MDG is a software tool that directly intervenes horizontally and vertically in business processes. The SAP MDG requires a system and business integration in the company. The book provides an overview of the MDG architecture, integration and implementation into the business process. Examples of implementation and project structures are presented. The book will serve the project manager as a compass. Due to the large number of different company structures and different handling of business processes, not all possible variants of business processes can be mapped. The company, which wants to introduce MDG must be clear that not only an SAP module is implemented, but rather established structures are changed. The MDG stands and falls with the degree of integration or integration. The audience of this document can be business analysts, techno-functional analysts, business process experts, master data experts, solution architects, technical developers, data scientists and customers.

2 Introduction SAP Master Data Governance (MDG)[1]

SAP Master Data Governance: the efficient way towards consistent and high quality master data – the foundation of your Digital Business

SAP MDG is the one single application that provides out-of-the-box, domain-specific master data applications as well as a flexible framework for custom data models and processes. This allows to de-centrally manage and consolidate, or centrally manage and distribute all master data for your company's complete system landscape.

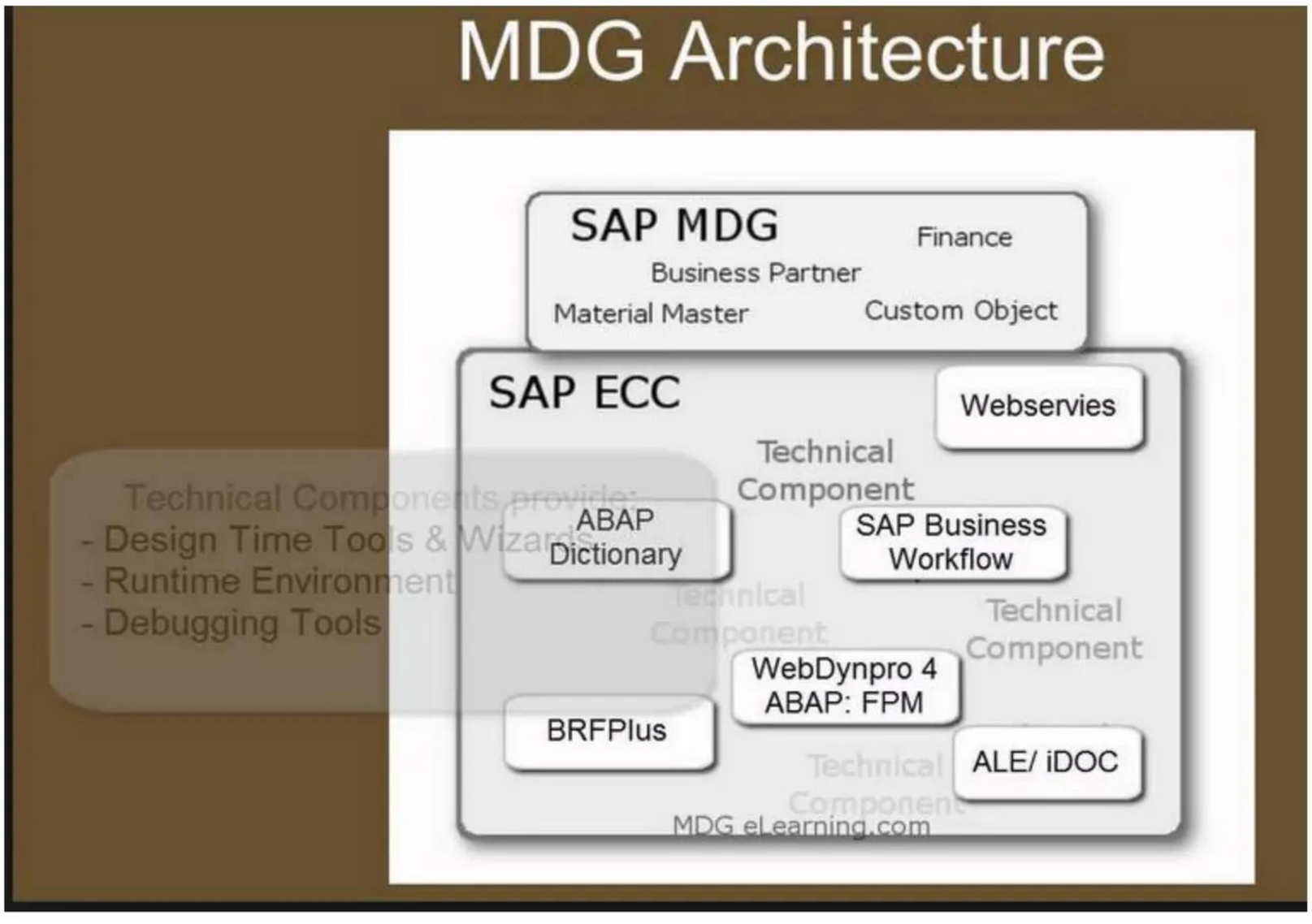
The clean master data from SAP MDG enables you to accelerate your enterprise software journey, and to enable new digital business models. For example…

- …SAP MDG supports your system landscape simplification and move to the cloud with synchronized master data across on-premise and on-demand applications
- …Leverage SAP MDG's harmonized master data and SAP's platform capabilities to connect transactional and IoT or other big data, and derive real business value
- …Mobilize your business with simplified master data access and efficient scenarios for request, approval, or to monitor KPIs and trends
- …MDG helps preparing a SAP S/4HANA move: get experience with a controlled subset, ensure master data quality before the implementation and maintain quality after go live

[1] SAP : Product overview 2018

SAP Master Data Governance (MDG) is a state-of-the-art master data management solution,
providing out-of-the-box, domain-specific master data governance to centrally create, change, and distribute, or to consolidate master data across your complete enterprise system landscape.
In this context, MDG can either run as a separate master data hub or be co-deployed on an operational system. In both cases MDG can run on:

- ECC on any DB
- ECC on HANA
- S/4HANA
- ARIBA
- Non - SAP



SAP Master Data Governance (MDG) is a **state-of-the-art master data management solution**, providing **out-of-the-box**, domain-specific master data governance to **centrally** create, change, and distribute, or to **consolidate** master data across your **complete enterprise system landscape**.
There are two versions of the product: **SAP MDG** (which technically runs on SAP ERP 6.0) and **SAP MDG on SAP S/4HANA**.

2.1 Central governance or consolidation for consistent master data

Enables centralized governance, compliance and transparency of master data during creation and change through integrated staging, approval and central audit trail

Delivers consistent definition, authorization and replication of key master data entities

Eliminates error-prone manual maintenance processes for master data in multiple systems

Supports consolidation of de-central master data in any enterprise system landscape in order to create best records and key mapping between duplicates

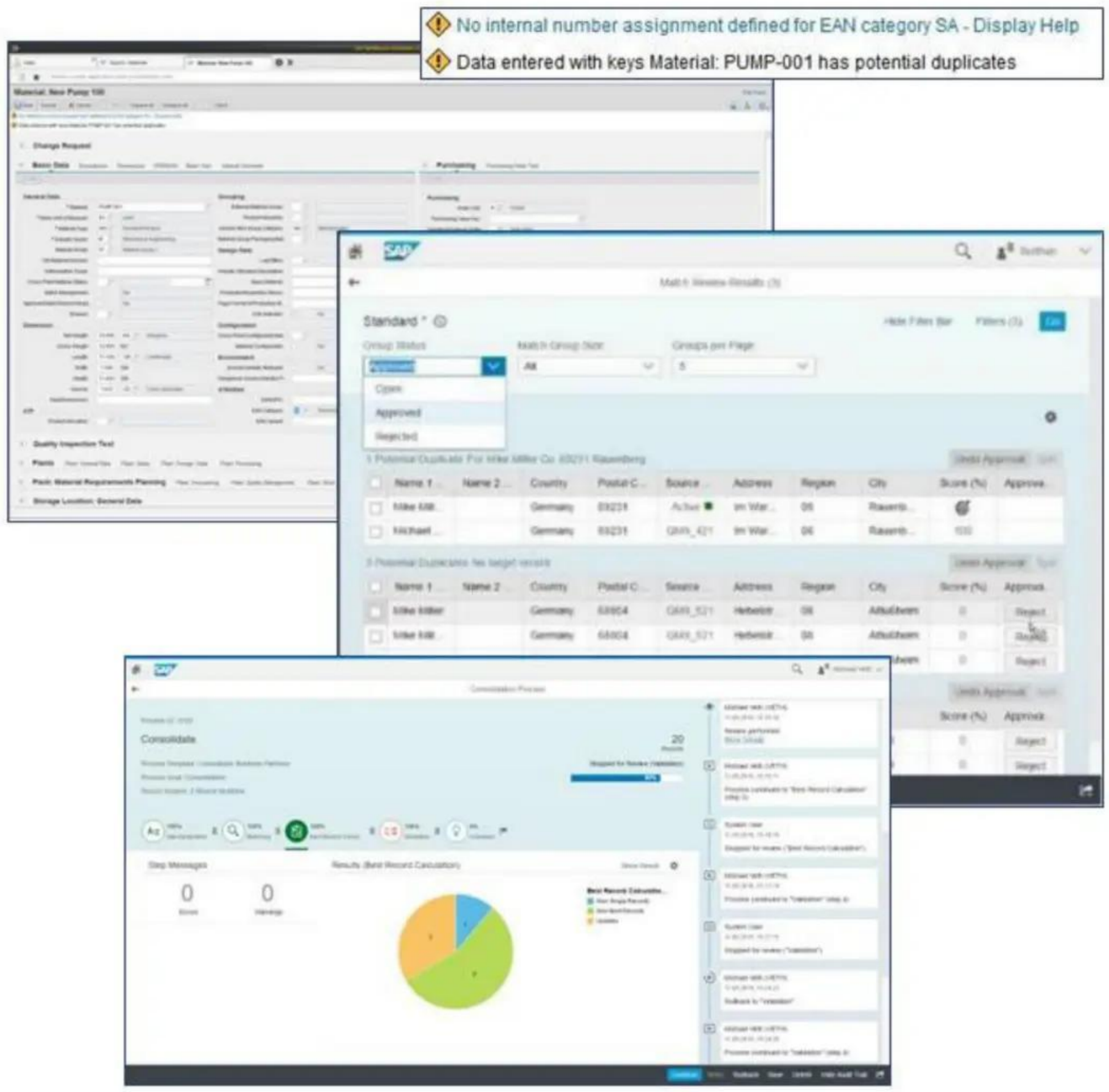
Optionally combines consolidation with central governance for sustained master data quality

2.2 Integration and re-use

Provides native integration and openness to integrate 3rd party services

Re-use of SAP data model, existing business logic and configuration for data validation

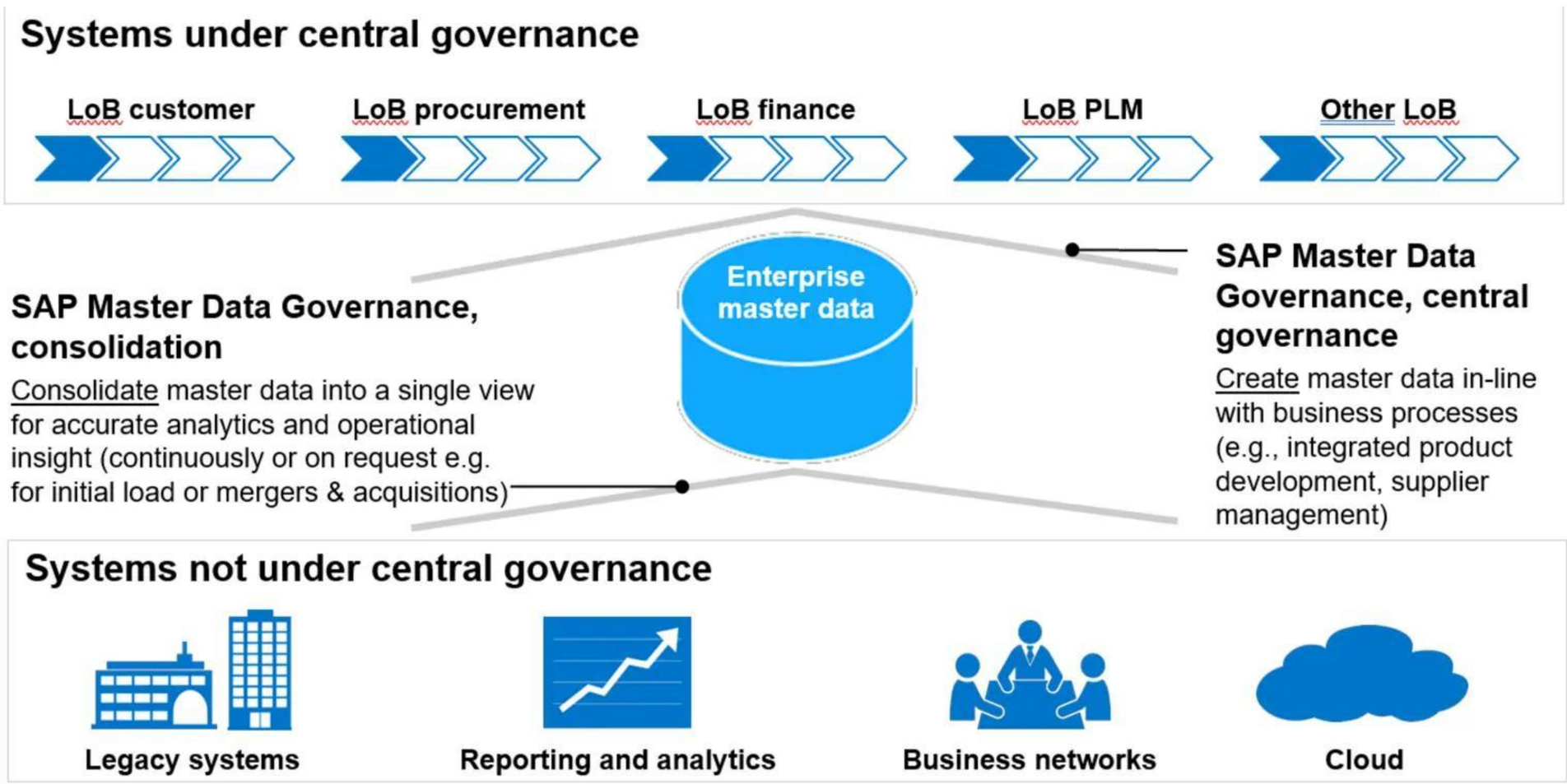
Open to manage your own master data flexibly also for non-SAP environments



2.3 Data quality

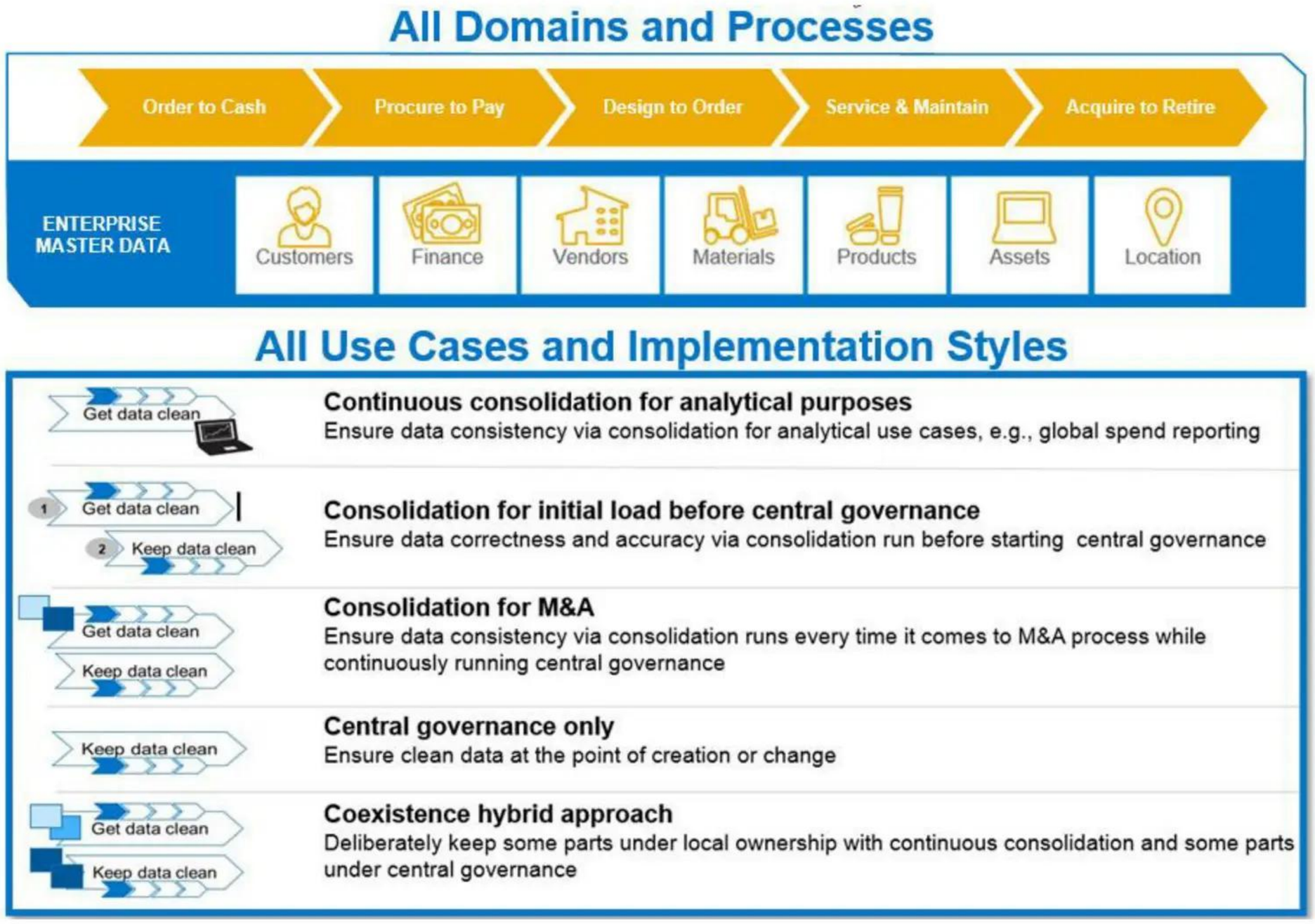
Analyze process quality and integrate with SAP HANA smart data quality and SAP Data Services / SAP Information Steward for quality, enrichment, and remediation

2.4 Central governance & distribution / de-central ownership & consolidation[2]

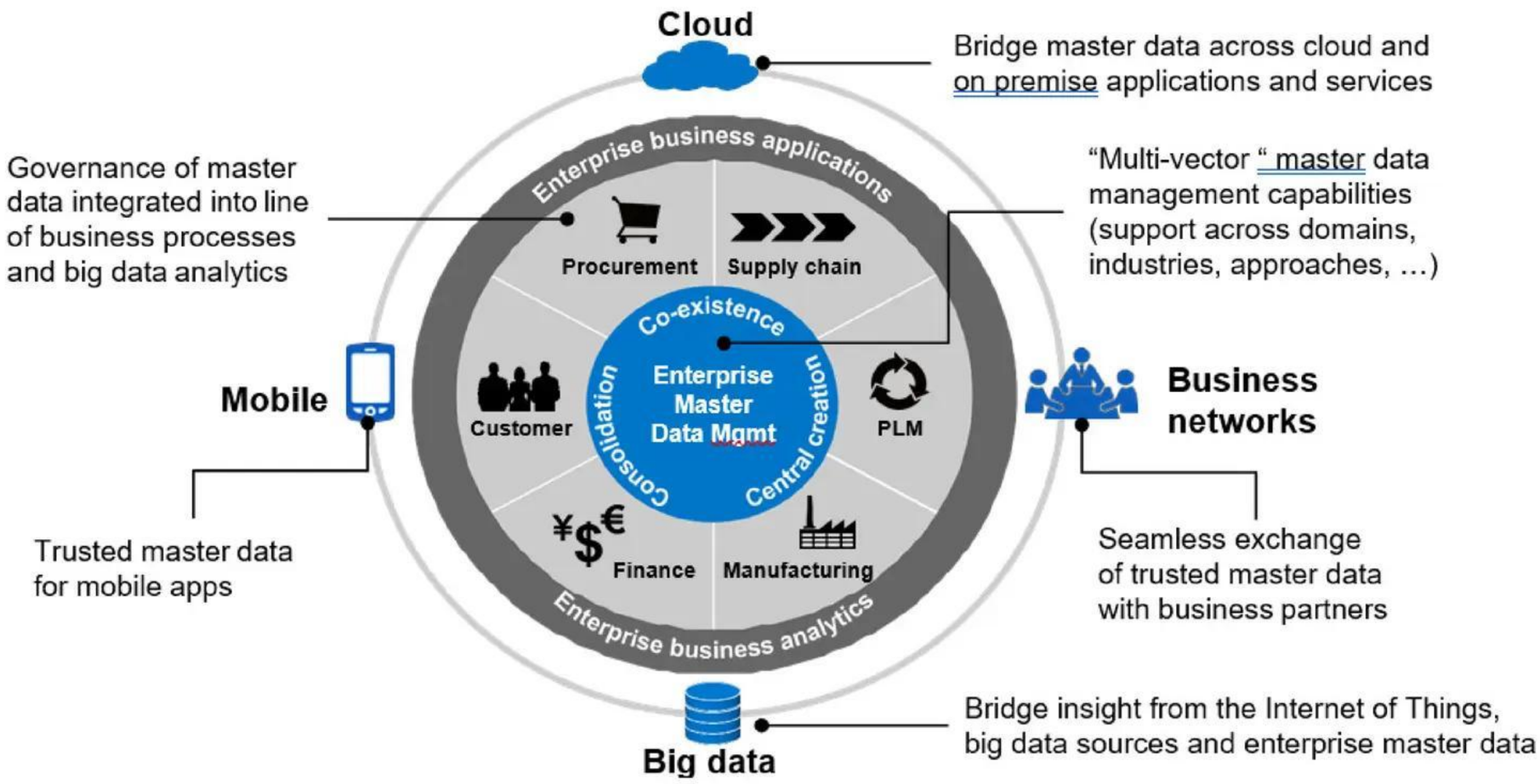


[2] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

2.5 One single application for Enterprise master data management



2.6 Key trends and impacts in business and technology



2.7 Master data governance capatilities[3]

Master Data Governance for all business data offers a central hub-based maintenance of material , customer, supplier , financial master data.

To address the key business issues related to such a governance solution for master data, MDG for material follows some basic principles that are shared across the domains of MDG

- Maintain once, use many
- One standardized process to be used by everybody
- One standardized user interface for everybody
- One standardized set of business logic

In order to fulfill these principles, SAP has built a Master Data Governance solution that offers

- Central based maintenance of master data
- Workflow-driven processes that are flexible while ensuring governance and standardization
- Harmonized user interface, based upon Enterprise Portal, Business Client or FIORI Launchpad
- Data model and business logic follow the best practices of SAP and reuse existing data structures, business validations and various services

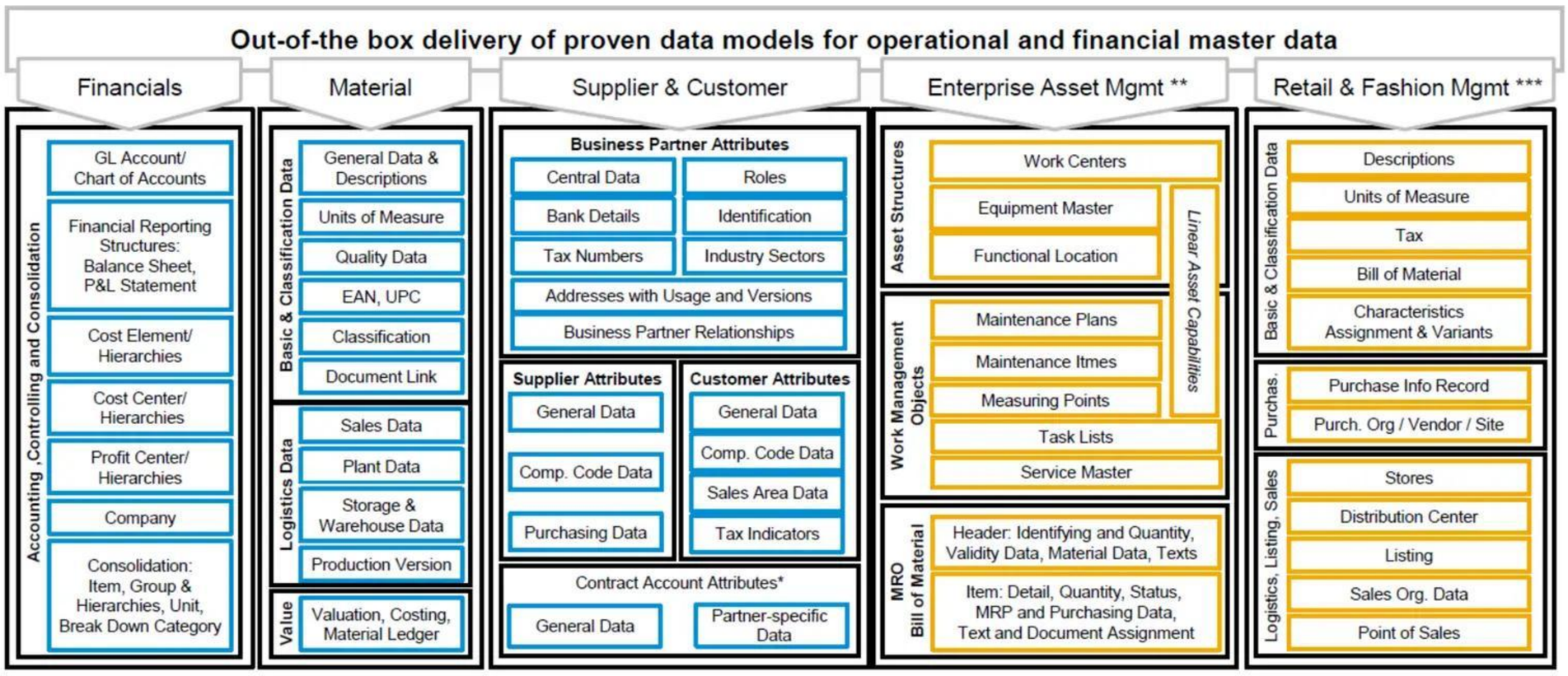
[3] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

2.7.1 Establish Best Quality Master Data to Optimize Business Processes

2.7.2 Deliver Out-of-the-Box

Low TCO	Full Integration to SAP systems as a client Offer integration services for non-SAP client systems
Flexibility	Workflow flexibility closely integrated to a framework for defining business rules
Ease of Consumption	Deliver a process-centric solution, with roles and a web-based user interface Deliver predefined data quality services
Best Practice	Deliver a data model proven by several applications & industries
Process Transparency	Reuse of existing processing logic
Integration	Out-of-the box data replication mechanisms, using stable mechanisms

2.7.3 Integrated Object Model[4]



Identifying Material Data ~ 10 fields

- Both SAP internal (material number) as well as external standards (GTIN)

Descriptive Material Data ~ 100 fields

- Help to understand what a material is but do not control processes
- Are as different as the products of our customers
- Typically realized via Classification (rather than Field Extension)

Process Controlling Material Data ~ 1.000 fields

- Control the behavior of SAP business processes
- Proprietary to SAP applications

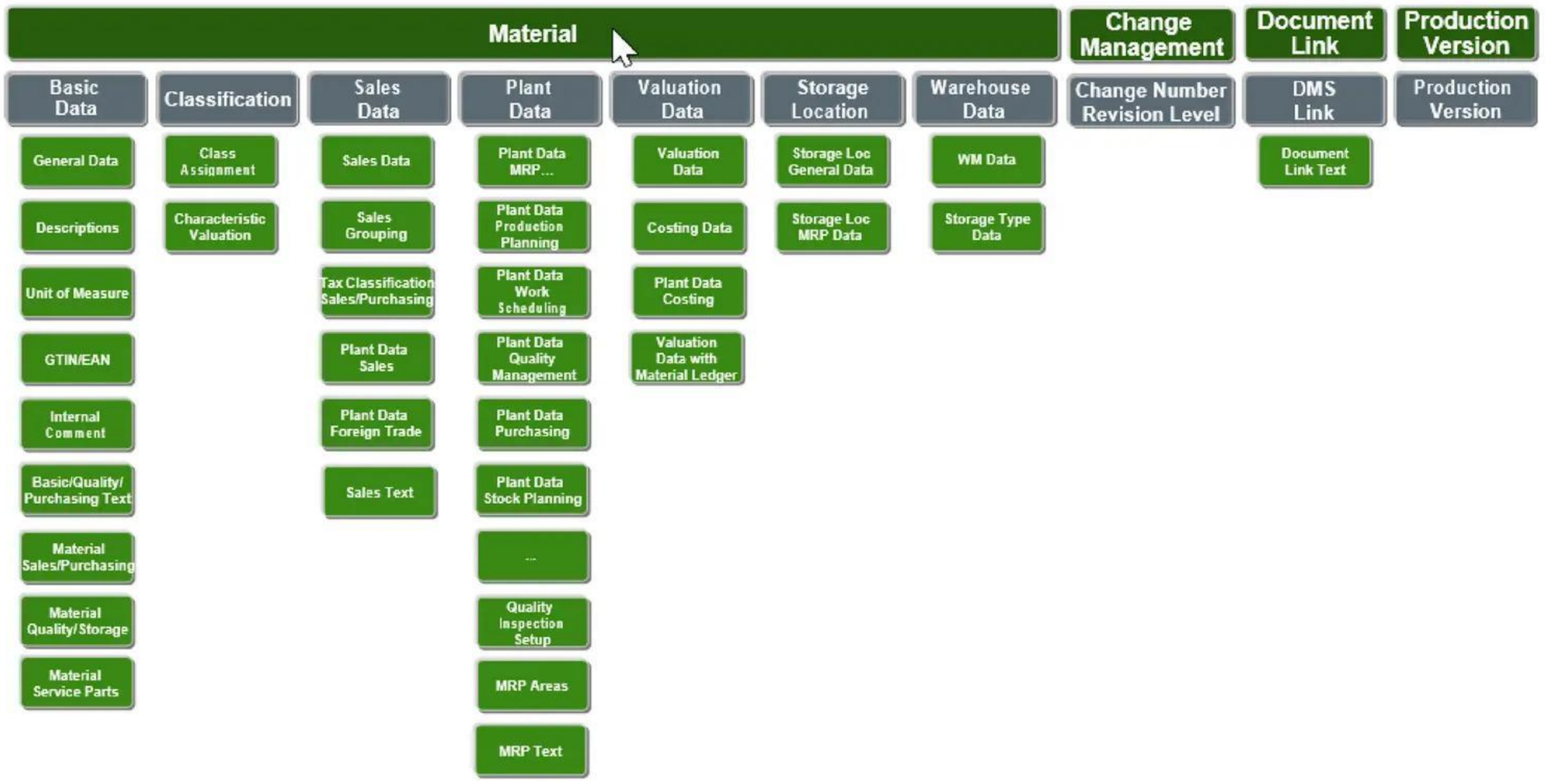
SAP MDG 9.1

SAP MDG Scope includes (~ 600 fields)

- Basic Data (Purchasing, Sales, Storage, Service Parts) and Long texts / notes
- Dimensions/ alternative UOMs / GTIN/EAN handling
- Classification / Lean Classification
- Sales Organization / Distribution Channel specific data / Tax Classification
- Plant specific data (with for Inspection Setup, MRP Area, Production Version)
- Storage Location Data
- Valuation data (with Material Ledger)
- Document Assignment
- Warehouse Data

Customers can enhance the data model

[4] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1



2.7.4 Master data governance in hybrid landscapes

All systems always in synch

- Master data creation can happen in on-demand systems, on-premise systems, or in SAP MDG on SAP S/4HANA
- De-central creation triggers a process in SAP MDG on SAP S/4HANA
- After approval, enriched high-quality master data is replicated to all relevant cloud and on-premise systems

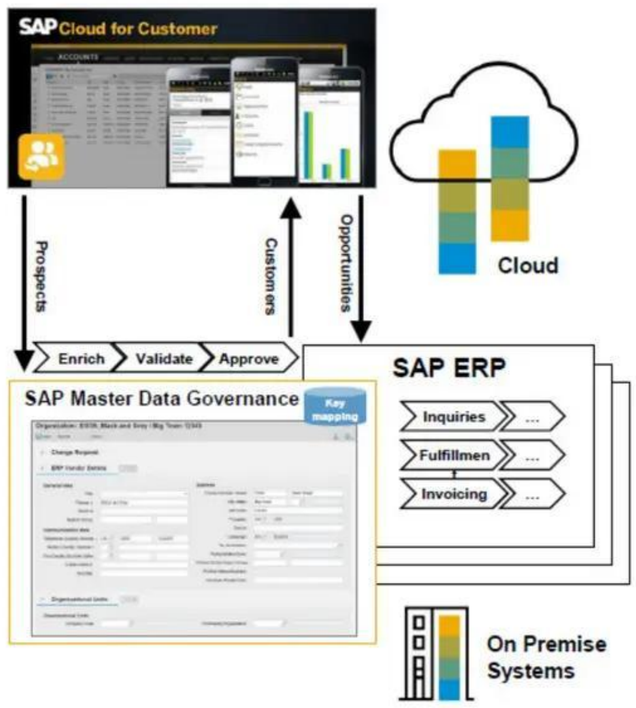
SAP MDG provides adaptable integration

- Integration scenarios can be implemented based on standard capabilities, like for example using extendable SOA services
- Any system can be involved as source and/or target
- Pre-built integration for selected SAP systems

Move to the cloud at your own speed

- A master data hub (like MDG) that synchronizes master data across on-premise and on-demand solutions is a prerequisite for your system landscape simplification and your step-by-step move to the cloud

Real-life customer example: SAP MDG managing prospect and customer data between cloud and on premise systems



2.7.5 End-to-end integration in heterogeneous landscapes [5]

Business-led integration

SAP Master Data Governance provides interfaces based on a process choreography for easy end-to-end integration…

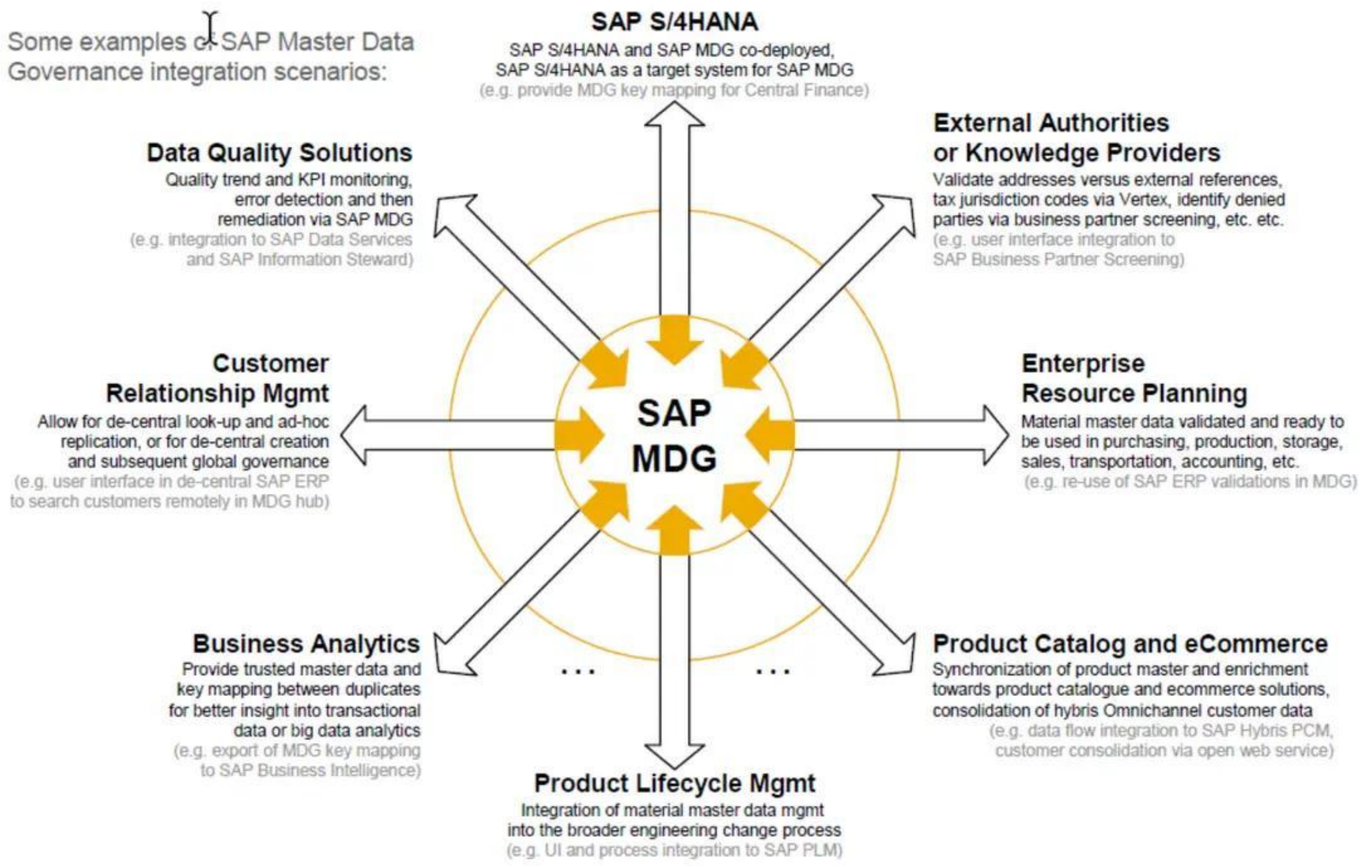
… not just technical interfaces

Not only for SAP

SAP Master Data Governance provides open interfaces to integrate with every solution as needed

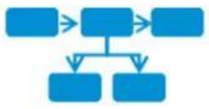
Out-of-the-box for SAP

In addition, for SAP to SAP integration, we provide out-of-the-box integration content

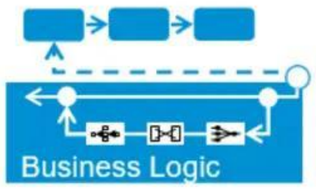


[5] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

2.7.6 Governance, Collaboration

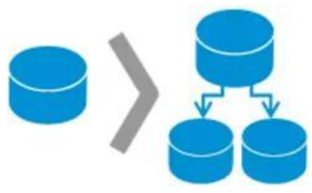


Linear or distributed and workflow based



Possibility to extend the process by adding business logic and thereby bringing in data quality

Has roles and responsibilities and also task authority
(e.g., create, change, approve)



Data in process stored in a separated repository, will be transferred to operational database after final approval

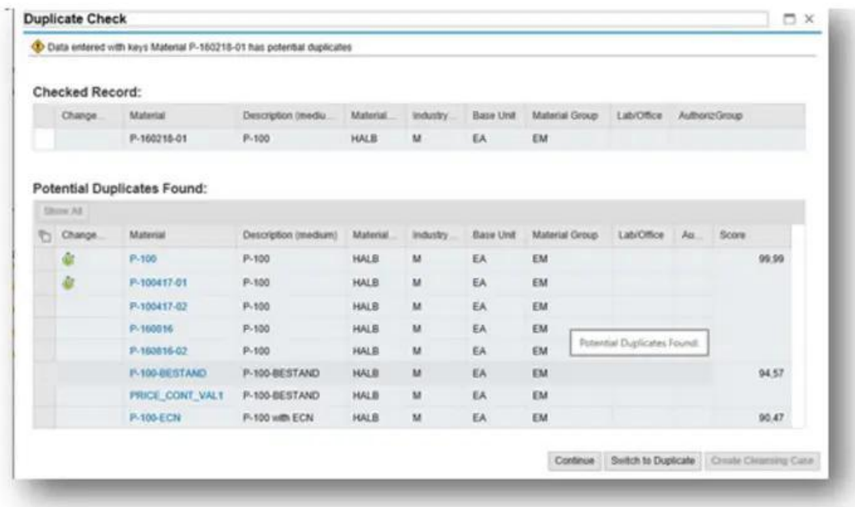
Can be adapted and tailored based on customer needs

Flexible enough to respect the distributed responsibilities existing in a company across various business units

2.7.7 Data quality

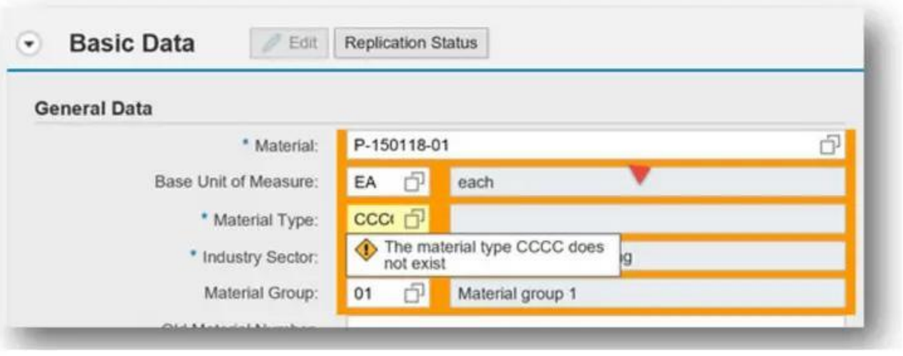
Prevent creation of duplicates for increased effectiveness and efficiency

- Checked early and embedded in the process
- High detection quality of matching using SAP HANA or Enterprise Search



Validations

- Re-use of existing validation logic
- Custom validations can be modeled and programmed (For example, code lists, simple checks, or rules using BRFplus)



Showing what was changed

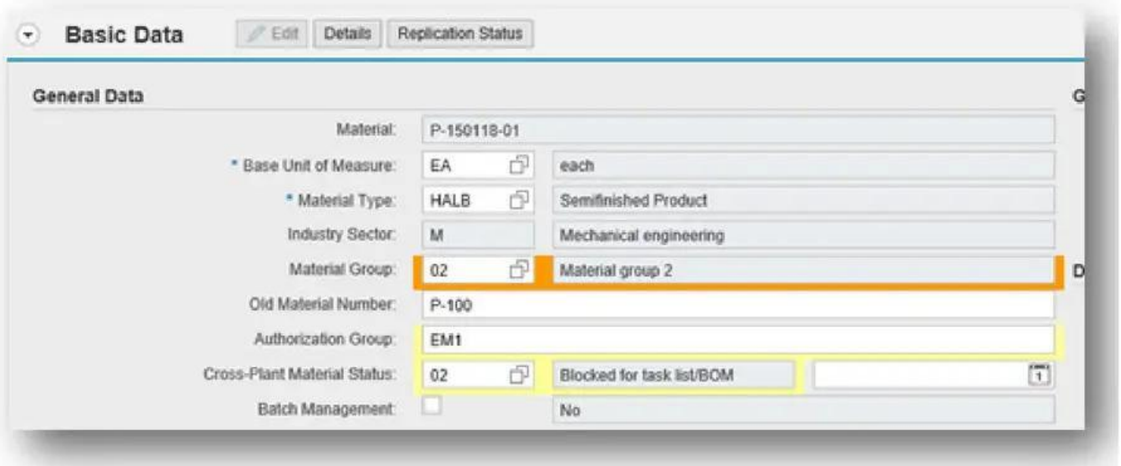
- Changes by earlier processors and own (unsaved) changes in two colors
- Previous, changed, and last saved value per attribute
- Available for all fields, including table cells, rows, navigation elements, etc.

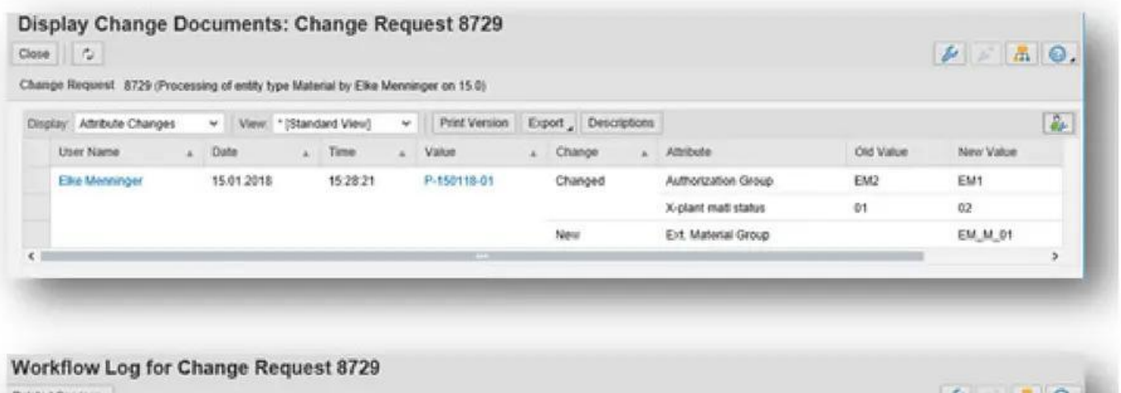
Helps people to do their job

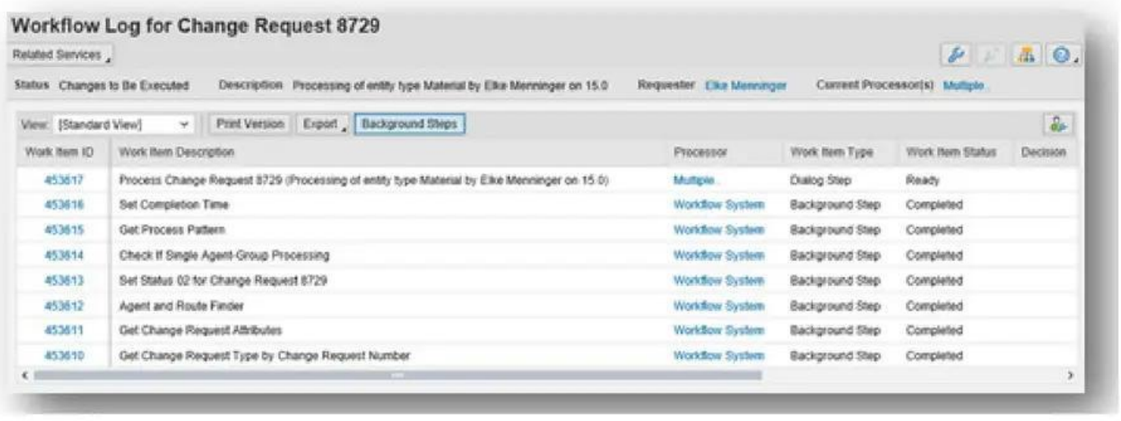
- Supports the processor to decide on their next action
- Supports approvers to quickly spot what they will approve
- Supports auditors to easily see what was changed

Transparency on what has happened earlier

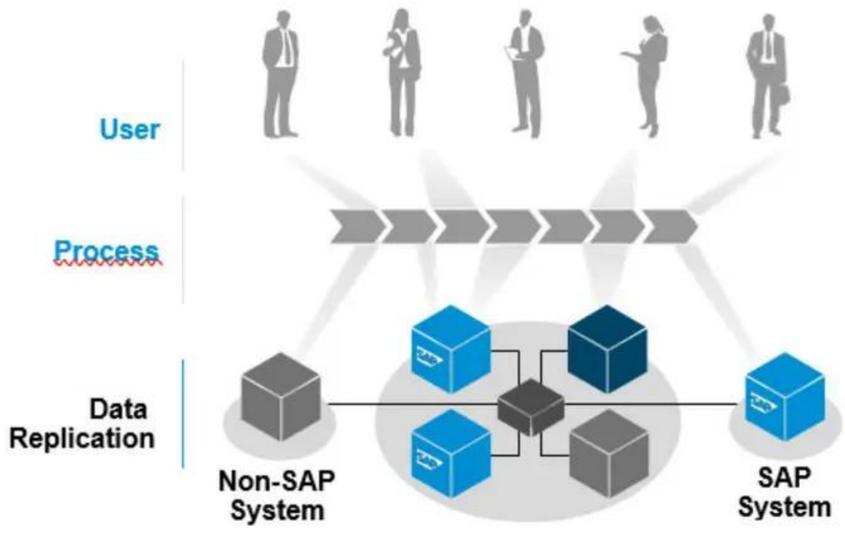
- Change documents provide information who changed which attribute from what value to what other value by when
- Workflow logs tell who was involved in the change request process and who approved which data change







2.7.8 Robust Data Replication



Replication

- Direct, pooled or manual replication to SAP and non-SAP systems
- Peer-to-peer or mediated

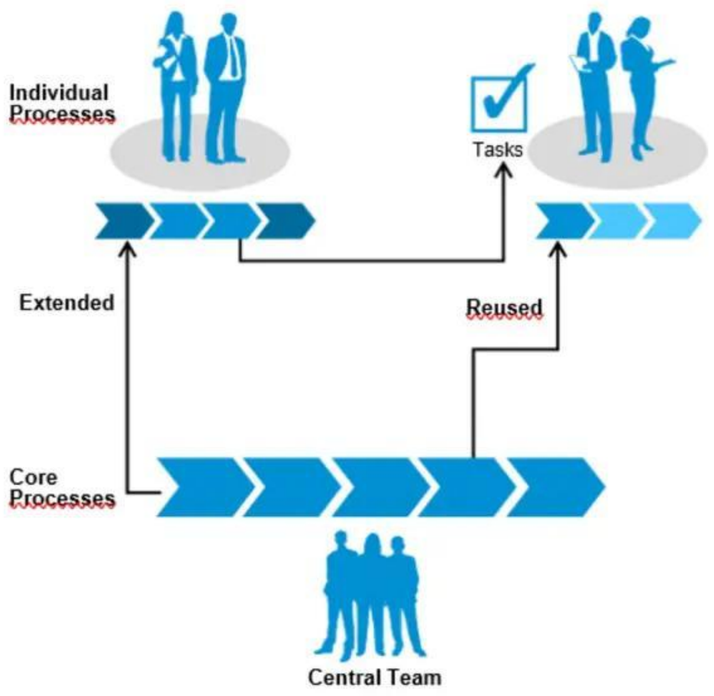
Replication Techniques

- DRF controlling replication mechanisms like
  - ALE for SAP S/HANA, SAP ECC, R/3 and non SAP systems
  - SOA service (only in S/4) for SAP and non SAP systems
- CIF for SAP SCM and MW for SAP CRM/SRM

Replication Features

- Flexibility to replicate selected data only to a specific client system by defining filters (controlled by DRF)
- Support of different object IDs or customizing keys between hub and client systems
  (via mapping in MDG or PI)
- Monitoring and error handling

2.7.9 Reusability & Extensibility



Reuse standard content

- Reuse existing data structures and functionality of SAP Business Suite

Enhance functionality

- Be able to enhance MDG functionality based on predefined content (e.g. plant or sales data) or customer-specific extensions

Extend data structures

- Provide easy-to-use extensibility capabilities for enhancing nodes and fields
  - data models
  - user interfaces
  - processes
  - data replication

2.7.10 MDG and S/4HANA[6]

As more customers move from the classic SAP ERP to SAP S/4HANA to transform their digital cores and make their enterprise landscape ready for the future, master data owners and IT teams responsible for master data must ask many tough questions: Should we also migrate our SAP Master Data Governance (SAP MDG) system to SAP S/4HANA? Should we upgrade our SAP MDG on SAP ERP or SAP MDG on the SAP Business Suite on SAP HANA to SAP MDG on SAP S/4HANA? What is the value of running SAP MDG on SAP S/4HANA? What functions and features are available in SAP MDG for SAP S/4HANA 1709 that aren't available in SAP MDG on SAP ERP?

The role of SAP MDG in the context of SAP S/4HANA

MDG can accelerate your S/4HANA roadmap

SAP S/4HANA does not change the role of SAP Master Data Governance

- Distribute or consolidate master data across all systems in the landscape: for example SAP Business Suite systems, SAP S/4HANA systems, other SAP and non-SAP systems…

Integration with SAP S/4HANA is available today

- SAP S/4HANA can be a distribution target for SAP Master Data Governance
- Central Finance can leverage MDG's centralized master data & key mapping for aligned postings
- SAP Master Data Governance can be co-deployed on SAP S/4HANA on premise
- Simplified versions of Consolidation, Mass Processing and Master Data Quality Remediation available for public cloud consumption in SAP S/4HANA Cloud

Preparing for a move to SAP S/4HANA

- SAP Master Data Governance is a good way to get experience with a controlled subset of SAP S/4HANA before a full blown implementation
- SAP Master Data Governance is a good way to ensure data quality and readiness before SAP S/4HANA implementation and reduce deployment risk.
- SAP Master Data Governance is a good way to establish strong governance processes that will help you maintain master data quality after SAP S/4HANA go live

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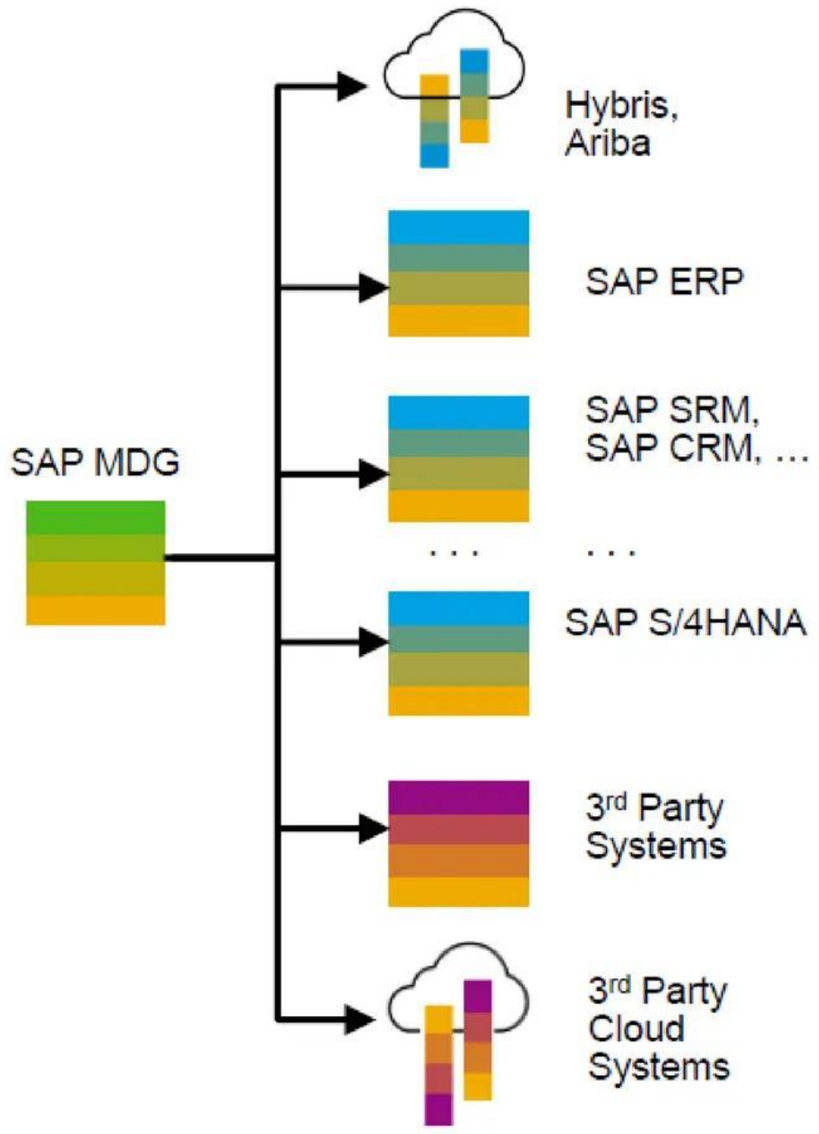
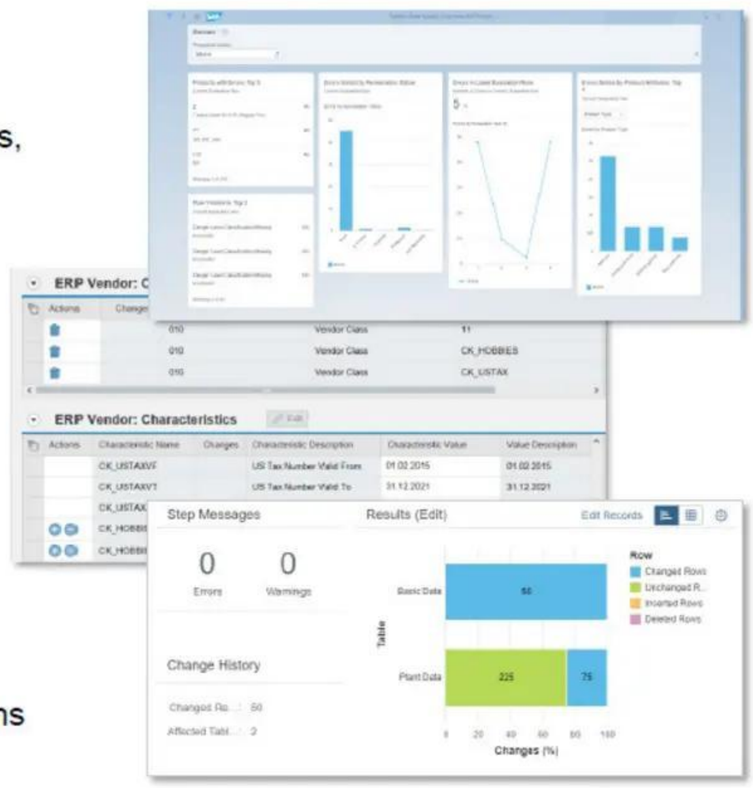
[6] Rheinwerk Verlag SAP Master Data Governance (SAP MDG) on SAP S/4HANA



Remark:
SAP expects innovations in SAP MDG on S/4HANA to outpace innovations in SAP MDG on ERP

2.8 Master data excellence for the digital business

Benefit in performing enterprise master data management with MDG innovations across key disciplines such as **master data consolidation, mass processing, master data remediation,** and **central governance.**



Master data consolidation and mass processing

- Consolidation of active data; handling of classification data, business partner relationships, mass creation or deletion scenarios, file upload, and transparency of changes – for business partner, material, and custom objects

Master data remediation

- Master data quality analytics for simple quality issue detection and correction of product data (only available on SAP MDG on SAP S/4HANA)

Central governance

- Improve the user experience for classification data handling for central governance of material data and enable classification data handling for central governance of customer and supplier data and within multiple records processing
- Capabilities related to the European Union General Data Protection Regulation (GDPR) like business partner blocking and deletion as well as read access logging
- Customer-driven optimization, simplification and round-offs across all master data domains

*) Innovations provided in SAP MDG on SAP S/4HANA 1709 and SAP MDG 9.1

End-to-end processes for the entire master data lifecycle, covering central governance and consolidation
Empowering business agility with data quality and process analytics and consumer-grade user experience
Strong foundation for improved business efficiency and process flexibility enabling simplification

Consolidation and mass processing

- Multiple vendor and customer assignments of business partners
- Classification data of materials, customers, vendors in mass processing

Central master data governance

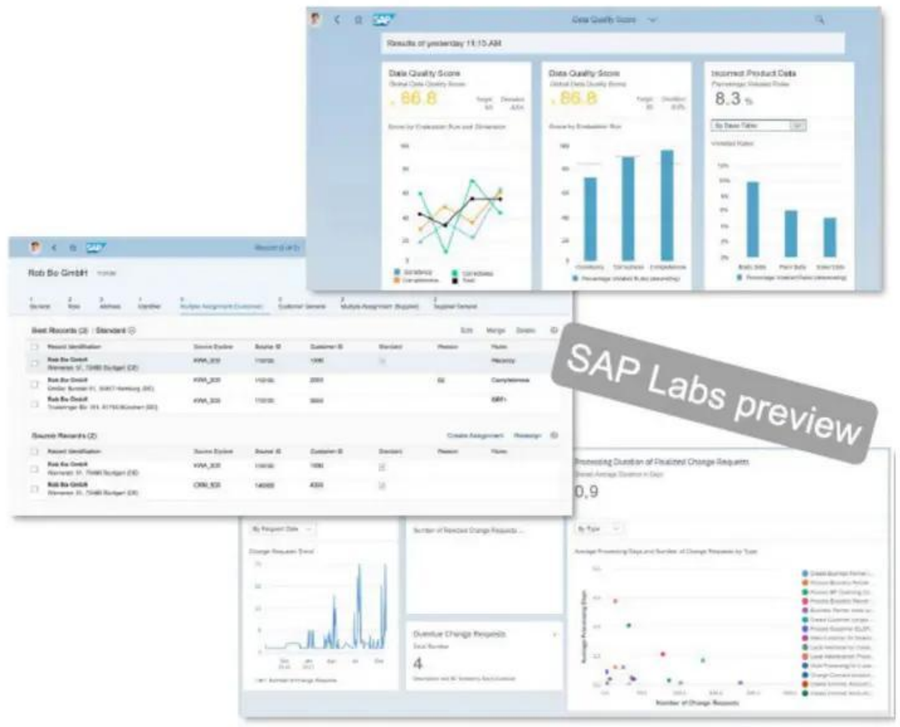
- Financial Data: Adopt S/4HANA universal journal, i.e. G/L accounts as basis for cost element; support for new domain: internal order
- Product: Adopt S/4HANA simplifications and industry-specific enhancements
- Business Partner: Multiple vendor and customer assignments; integration with SAP Credit Management

Integration scenarios

- e.g. SAP Ariba Supplier SLP, SAP Hybris C4C

Data quality and process quality

- Create business rules and analyze data quality
- Enhanced SAP Fiori user experience in process analytics



This is the current state of planning and may be changed by SAP at any time.

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2.9 MDG road map overview – key innovations[7]

Recent innovations[1]	2018 – Planned innovations[1,2]	2019 – Product direction[2]	2020+ – Product vision[2]
Master data consolidation ▪ Consolidation of active records ▪ Support of bus. partner relationships ▪ Classification **Master data mass processing** ▪ Support of bus. partner relationships **Central master data governance** ▪ Support for contract account ▪ Classification ▪ Framework enhancements like archiving of change requests, integrated object creation in hierarchy-processing UI, … ▪ Enhanced Material data model covering shelf-life data and service parts mgmt. ▪ Business partner enhancements like documents, payment cards, and more **Data quality management** ▪ Data quality control and remediation of product data	**Consolidation and mass processing** ▪ Multiple vendor and customer assignments of business partners ▪ Classification data of materials, customers, vendors in mass processing **Central master data governance** ▪ Financials: Adopt S/4HANA universal journal, i.e. G/L accounts as basis for cost element. Support internal order ▪ Product: Adopt S/4HANA simplifications and industry-specific enhancements ▪ BP: Multiple vendor/customer assign-ments. Integration /w SAP Credit Mgmt. **Integration scenarios** ▪ e.g. SAP Ariba Supplier SLP, SAP Hybris C4C **Data quality and process quality** ▪ Create bus. rules & analyze data quality ▪ Enhanced SAP Fiori user experience in process analytics	**Expand the foundation…** ▪ Support all critical MDM capabilities in one single packaged application ▪ Flexibly support heterogeneous environments, with faster time to value than just a toolbox ▪ Leverage SAP's footprint to simplify MDM and integration across on premise and cloud applications **…and support new master data challenges** ▪ Increasing velocity and growing amounts of master data, demand for a higher degree of automation ▪ Leverage new technologies, like machine learning, to enable the new promises of the digital transformation	**Support strategic trends** ▪ Enterprise-wide master data mgmt across on premise & cloud apps ▪ Seamless master data exchange across the entire business network ▪ Master data for end-to-end IoT solutions spanning devices and apps **Simplify creation, management and use of master data** ▪ One platform for all domains and use cases natively using EIM and SAP HANA ▪ Intuitive user experience and high business user efficiency ▪ Embedded analytics, stewardship, monitoring, and remediation ▪ Further automation of routine tasks like error correction or data completion

1. Potential data protection and privacy features include simplified deletion of personal data, reporting of personal data to an identified data subject, restricted access to personal data, masking of personal data, read access logging to special categories of personal data, change logging of personal data, and consent management mechanisms. 2. This is the current state of planning and may be changed by SAP at any time without notice.

2.10 New MDG functions

Integration scenarios

- SAP Ariba Supplier SLP
- SAP Hybris C4C

Central Master Data Governance for Customer / Supplier

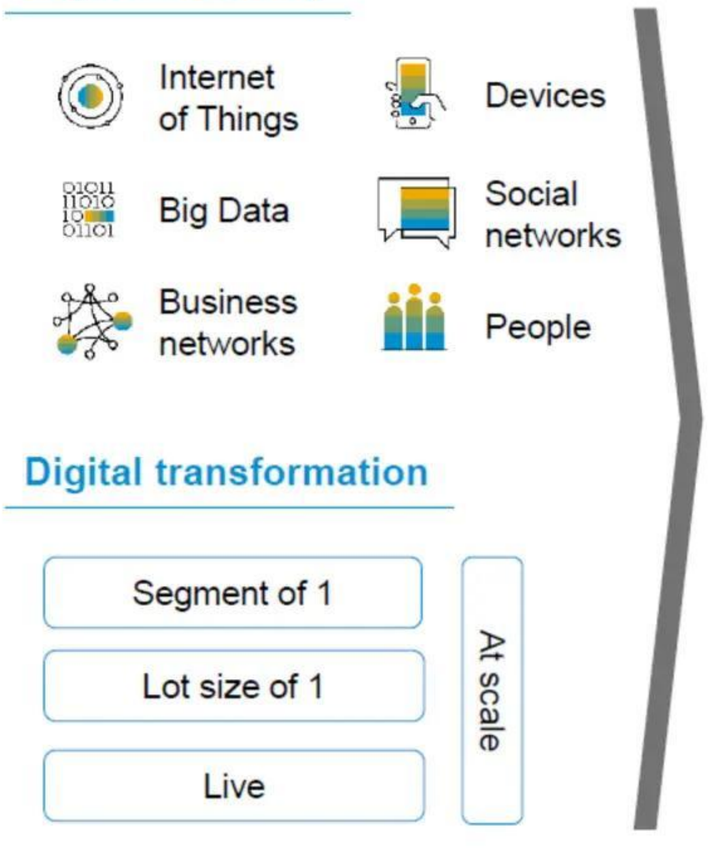
- Roundoff Multiple vendor and customer assignments of business partners (creation process within one Change Request)
- Global search variants (MDG on S/4HANA only)
- Enhancement by SAP Credit Management data

Central Master Data Governance for FI Contract Account

- Data Load
- Data Replication via SOA-Service
- Enable Client Maintenance
- Time Dependency

[7] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

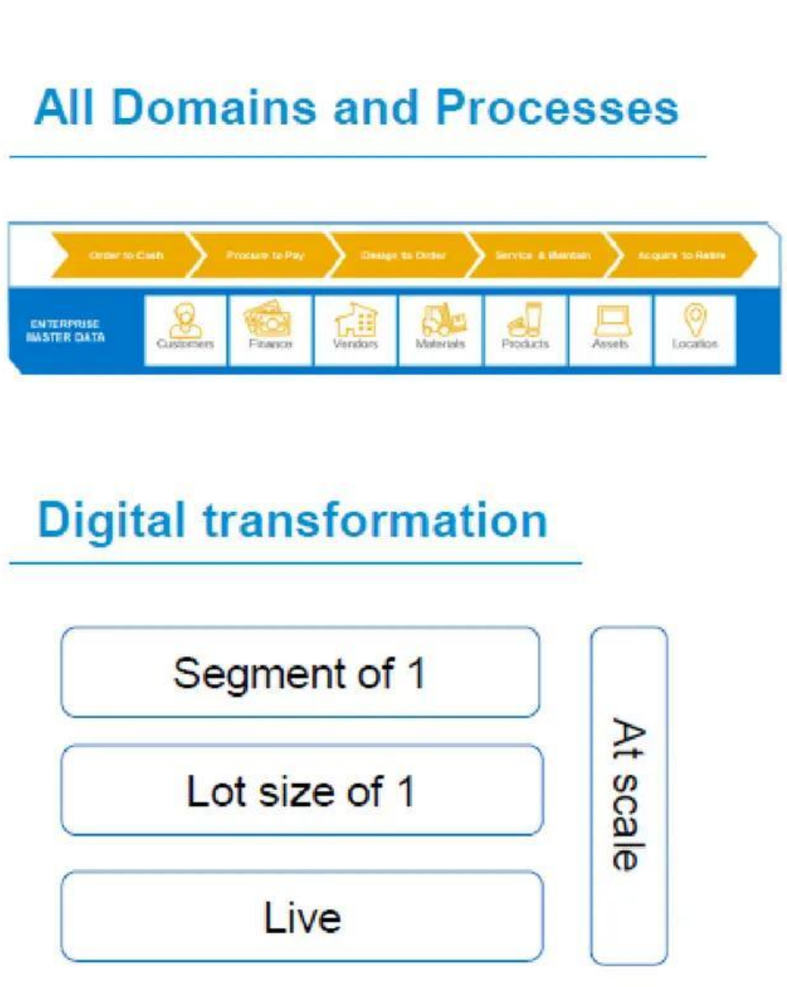
2.10.1 The impact of digitization on master data



Key impact on master data (management)

- Scalable, efficient processes and higher degree of automatization **require increased quality of master data** across the enterprise, supported by **machine learning**
- **Combined products & services, blending of industries** and 'smart' capabilities integrated into physical products
- **Individualized products/services** to cover the individual needs of the customer/consumer, often based on modularization
- **Data is everywhere**: not everything can be centralized, leave data where it lives – **active and passive governance** at the same time

Expand the foundation …
Further strengthen SAP MDG's ability to support all critical capabilities for master data management in one single application. Continue to deliver a packaged solution and out of the box content, with all flexibility to support heterogeneous environments, but with a faster time to value than just a toolbox
Leverage SAP's footprint in applications, analytics, and business networks; and simplify master data management and integration across on premise and cloud applications
… and support new master data challenges . Increasing velocity and growing amounts of master data, demand for a higher degree of automation in handling of master data
Start to leverage new technologies, like machine learning, to enable the new promises of the digital transformation, such as customized offers to the individual customer in real time.



2.10.2 MDG > 2020+ Product vision - High quality master data for every use case

Support current and future strategic trends

Enterprise-wide master data management across on premise and cloud deployed applications

Seamless exchange of master data across the entire business network value chain

Support master data for end-to-end IoT solutions spanning devices, platforms & applications

Mobile capabilities that are fully integrated into backend systems

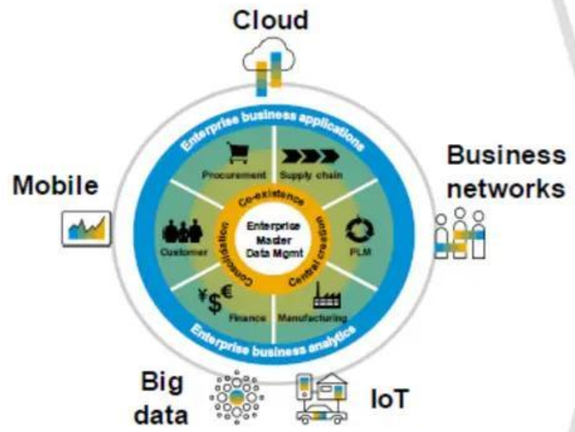
Simplify the creation, management and use of master data

One platform supporting all domains, styles and use cases available on premise or in the cloud, natively using SAP's Information Management portfolio and SAP HANA

Intuitive user experience and high efficiency through business oriented interfaces, flexible workflow, and integration of content and context

Embedded analytics and stewardship capabilities for monitoring, root cause analysis and remediation

Automation using machine learning for things like error detection and correction, completion of missing attributes, entity resolution, rule detection, and more



2.10.3 SAP Master Data Governance and cloud consumption models

Enterprise master data is part of companies' Digital Core

SAP MDG is offered on premise and in the private cloud, today

In the future, we plan to leverage SAP S/4HANA Cloud and SAP Cloud Platform as the platform to extend MDM scenarios for consumption in the public cloud

Master Data Management support for cloud applications

SAP MDG customers synchronize master data between all cloud and on premise applications, today

SAP provides cloud-based data integration and data quality capabilities

Advanced local master data maintenance in SAP S/4HANA Cloud, like e.g. business partner data enrichment, or re-use of MDG code for mass maintenance…

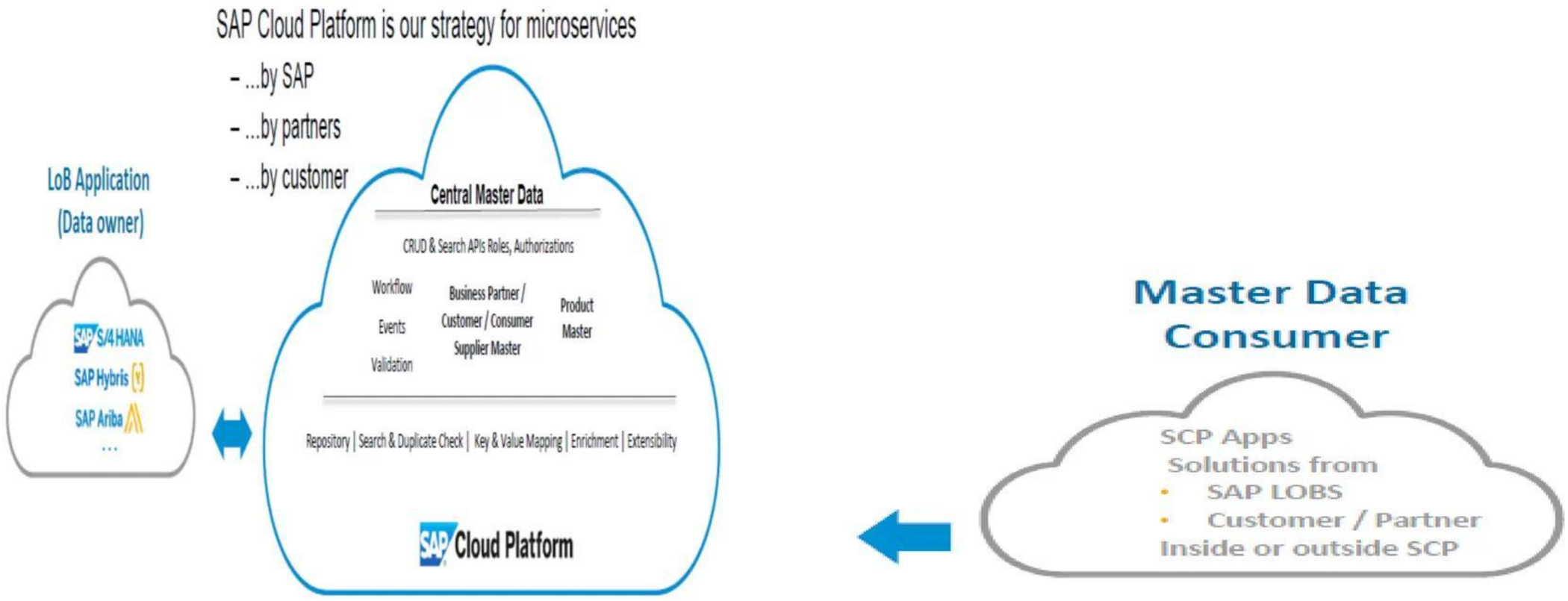
Leveraging Micro Services and Business Networks

SAP Data Quality Management, microservice, Identity-as-a-Service, …

SAP Asset Intelligence Network, SAP Ariba, SAP Hybris, …

SCP-based microservices planned as an alternative way to synchronize between SAP MDG and any public cloud application (SOAP & ODATA)

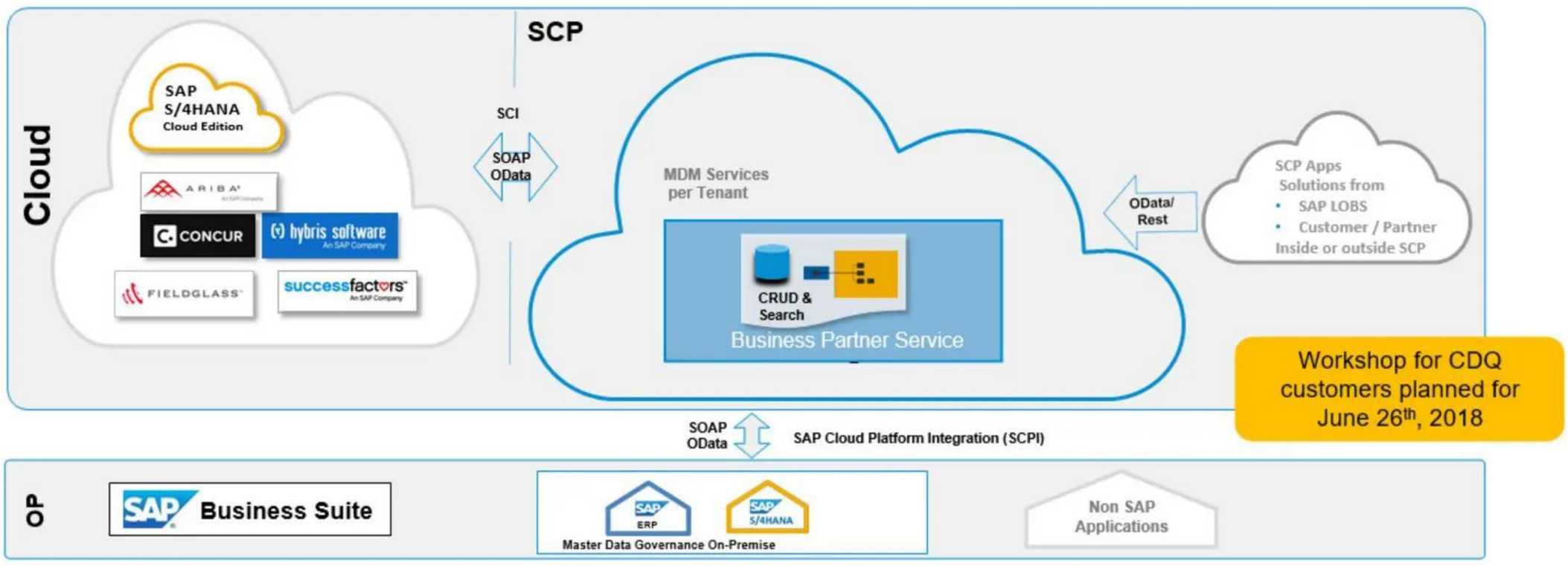
2.10.4 Master data services on SAP Cloud Platform



Guiding Principles

1. Shared core data model between applications
2. Local enhancements by specific attribute-sets in local apps
3. Start with Business partner/customer re-use service, extend to Supplier and Product Master
4. Built on SAP Cloud Platform, services published in API Hub

2.10.5 MDG – Business Partner Service Consumption



2.11 Strategic direction, using connectivity in business networks

Connect to suppliers with SAP Ariba

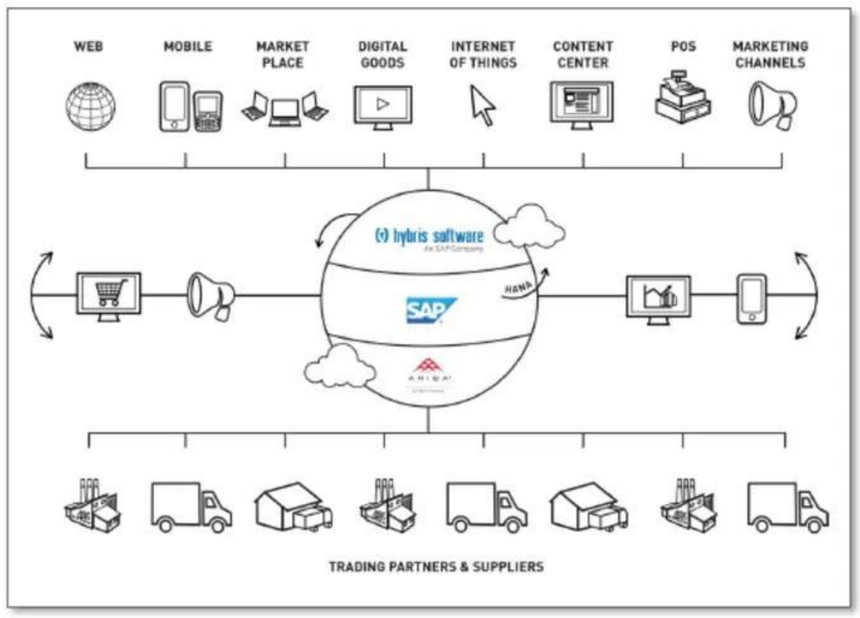
Exchange master data between business partners via the SAP Ariba network

Manage data internally in SAP MDG

Governed provisioning of master data with SAP Master Data Governance, ready to be used in all internal business applications

Provide product data via SAP Hybris PCM

Enrich product data for use in catalog, eCommerce and other customer-facing channels



3 SAP MDG for Financials (MDG-F)[8]

Master Data Governance for Financials (MDG-F) is an application to centrally govern the maintenance of financial master data.

•MDG-F provides central steering and control mechanisms independently of local or centralized execution.

•In MDG-F, master data records can be created, completed or changed in a staging area, following a distributed step-by-step way, before making the record available to connected business applications.

SAP Master Data Governance provides out-of-the box solutions for the central management of master data objects. Domain-specific solutions include business partner (MDG-BP), customer (MDG-C), supplier (MDG-S) governance, material governance (MDG-M), and financials governance (MDG-F).
This guide provides you with the foundation knowledge you need to know about financial data and its related governance solution financial governance (MDG-F).

Inconsistencies in financial master data arise from local, non-standardized maintenance processes (e.g. different charts of accounts) with, in some cases, manual workflow steps in a heterogeneous global system landscape. This not only affects the operational efficiency, but also the input required for the respective financial statements when preparing evaluations and the subsequent manual adjustments.

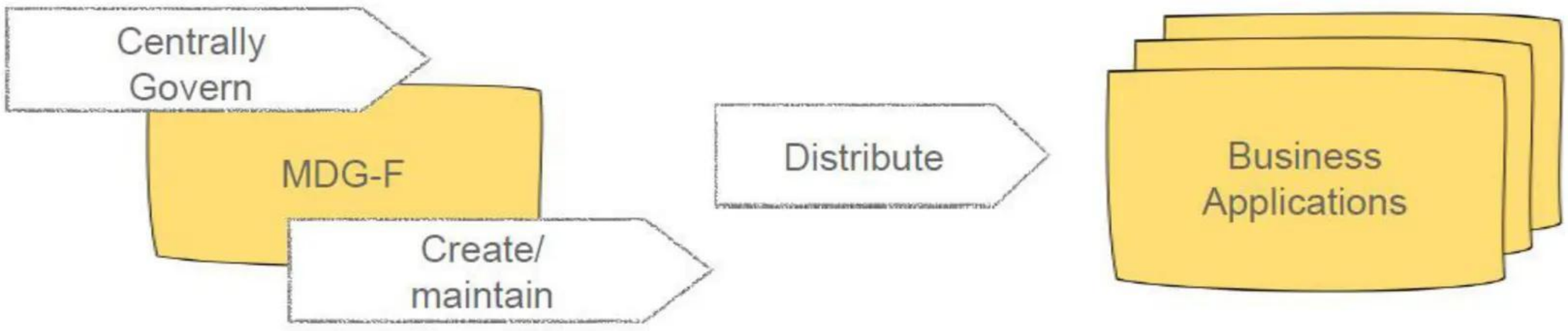
In addition to this, increasing complex regulations require organizations to have change transparency and audit trails.

Master Data Governance for Financials (MDG-F) is an application to centrally govern the maintenance of financial master data.

•MDG-F provides central steering and control mechanisms independently of local or centralized execution.

[8] SAP Master Data Governance on SAP S/4HANA 1709 / SAP Master Data Governance 9.1

•In MDG-F, master data records can be created, completed or changed in a staging area, following a distributed step-by-step way, before making the record available to connected business applications.



SAP MDG for Financials (MDG-F) offers central maintenance and governance solutions for a wide range of financial master data. These include, for example, accounting objects such as standard and group charts of accounts, general ledger accounts, companies and Controlling objects such as profit centers, cost centers and their hierarchies. In addition, customer-specific objects can also be implemented with MDG-F.

3.1 Specifics of SAP MDG-F:[9]

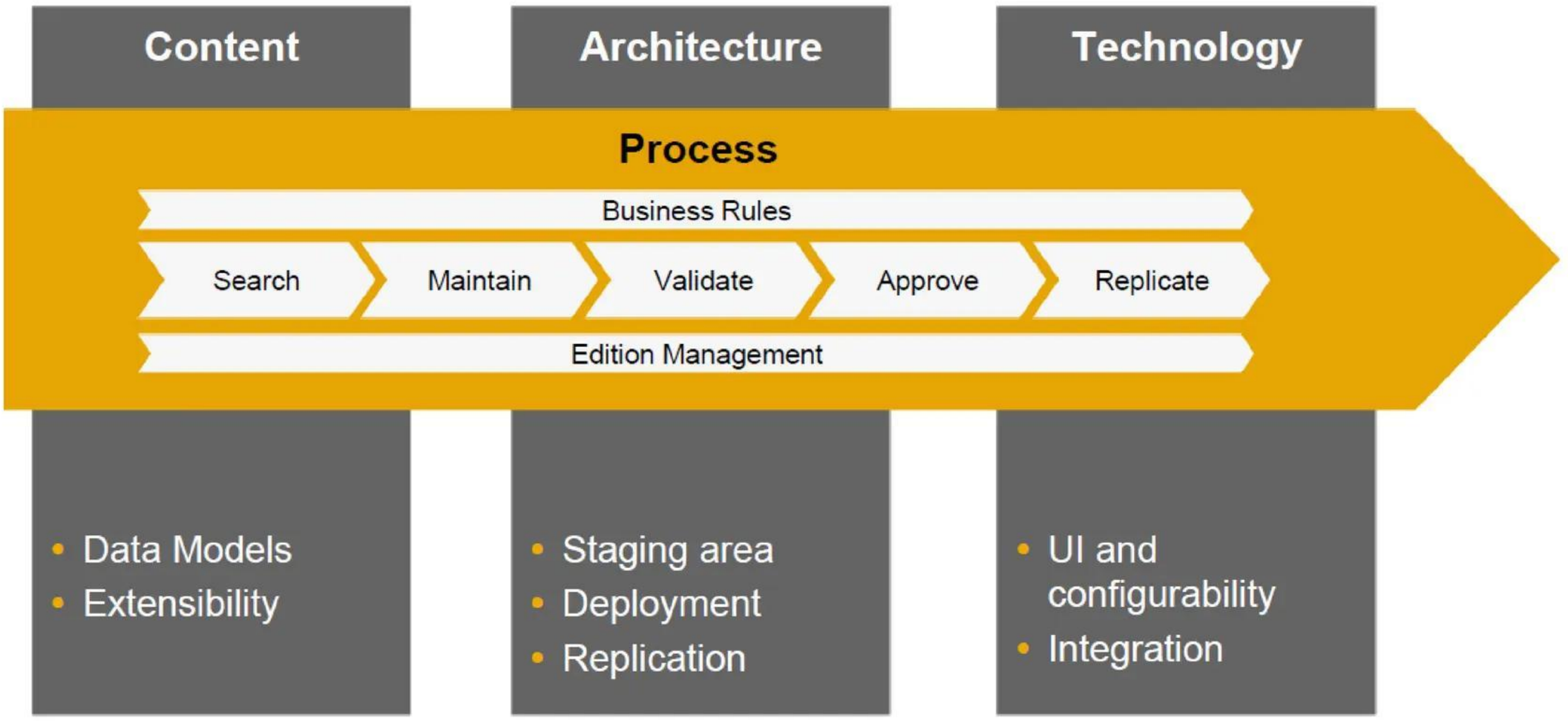
- Consolidation of company financial data with the financial data of the subsidiaries
- Mapping between corporate and operative standard charts of accounts
- Tracking of the change processes and process setup possible within the scope of IFRS, Sarbanes-Oxley, GAAP, etc.
- Time-based and period-based versioning: Transparency, traceability, edition (summary of multiple work steps for the final release)

The centralized MDG-F system permits the user-friendly management of your financial data and specific customizing options in order to comply with legal requirements with the aid of holistic integration into your SAP ERP system. In particular, SAP MDG-F is notable for allowing existing SAP ERP data structures and objects to be reused, which in turn allows checks and validations to be used analogously. The user interface is based on ABAP WebDynpro programming to respond flexibly to changes in requirements.

[9] SAP SE

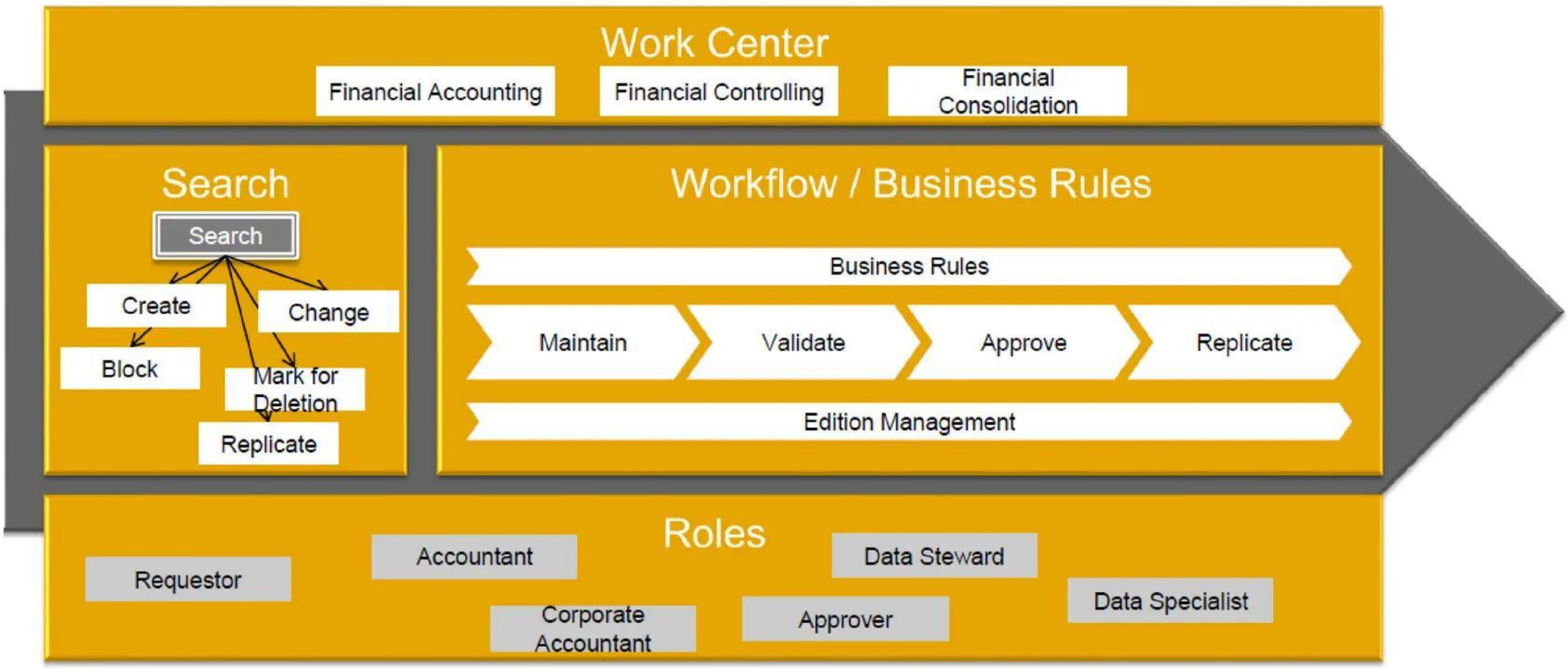
3.2 MDG for Financials Addresses Main Action Areas[10]

3.3 The Solution Building Block Overview



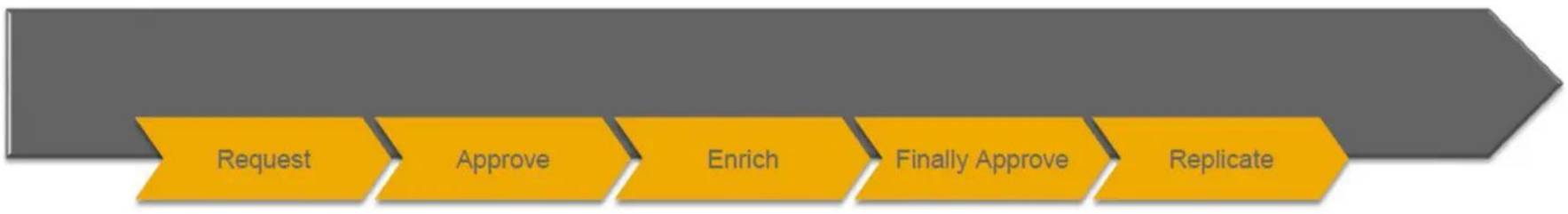
[10] SAP SE

3.4 MDG-F Process Framework



3.5 The Change Request

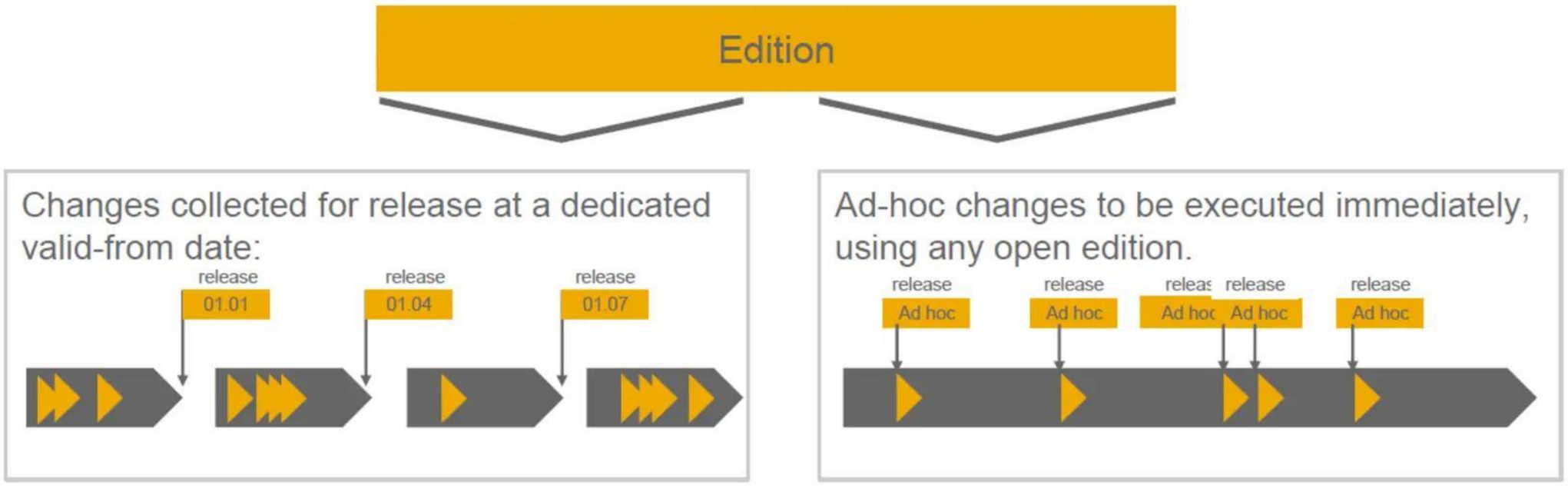
- **The Change Request is an MDG-specific object that:**
- **acts as the container for new and changed data**
  - Master data content
  - Comments
  - Approval history
- **defines and controls the process flow and processors according to defined Change Request type**
- **optionally holds multiple changes**



3.6 Editions

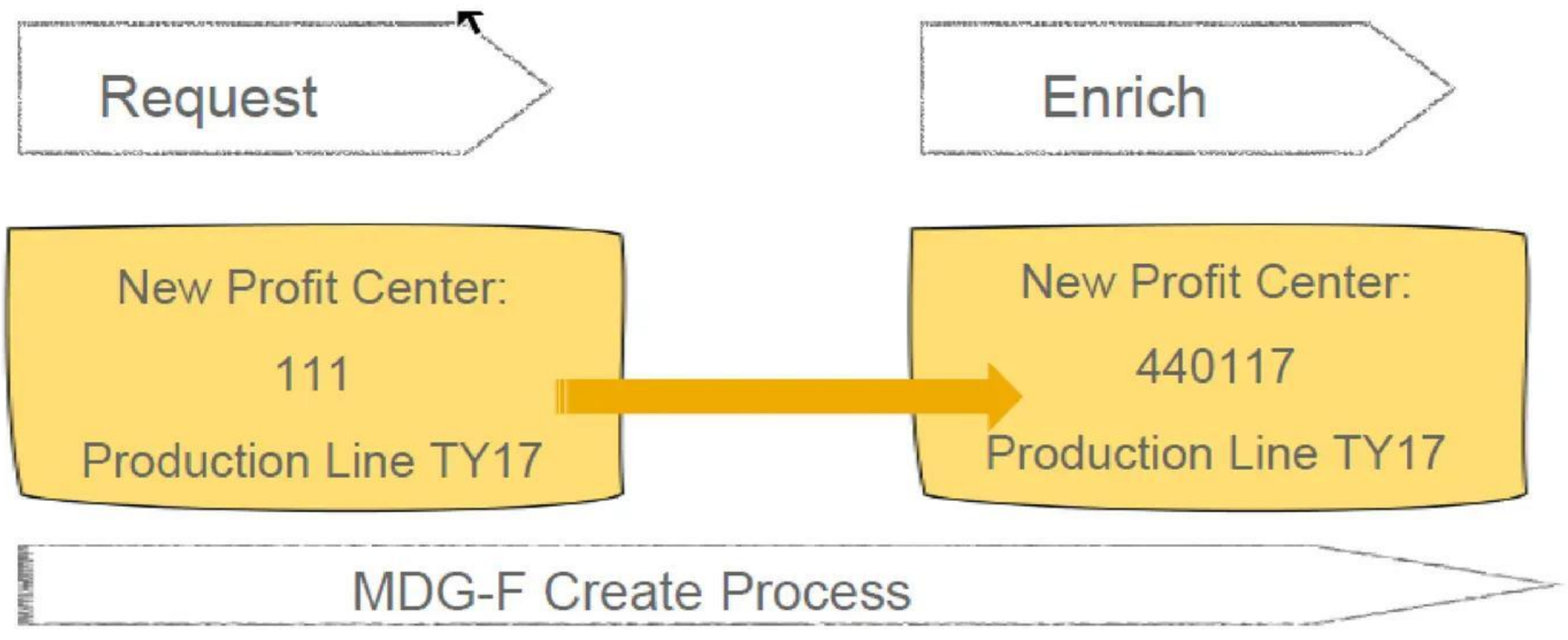
Editions are used to control the validity of a change. In the edition, planned change requests for the creation of new data or for changes to existing data are collected and released collectively at a certain

date. Time-dependent objects (Profit/Cost Center) automatically inherit the valid-from date from the edition's valid-from date.



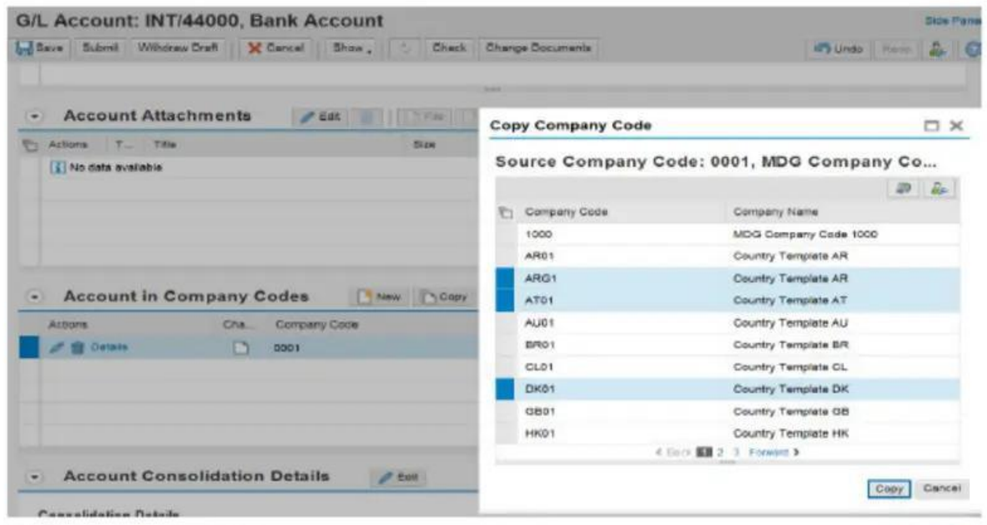
3.7 Work with Changeable Keys

When requesting a new financial master data item, often the requesting user does not yet know the number to be used in Business Suite Applications. To provide flexibility to the requestor whilst adhering to corporate numbering standards, the single-object maintenance in SAP MDG-F offers the feature to use a generated key first, and changing it to the desired final number in a later step of the creation process.

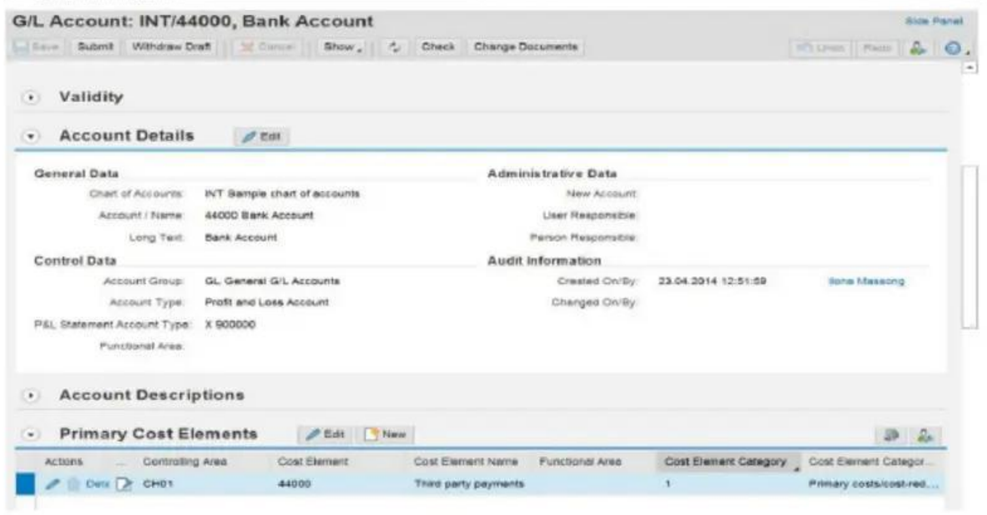


3.8 Process Features

- Copy multiple company codes: create multiple company code-specific G/L accounts by simple copying



- One-step creation of primary cost element: automatically create the primary cost element connected to a new account in the same maintenance step



3.9 Hierarchy Assignment in Single-Object Processing

Master Data and Hierarchy Assignment in the same Main Financial Objects UI

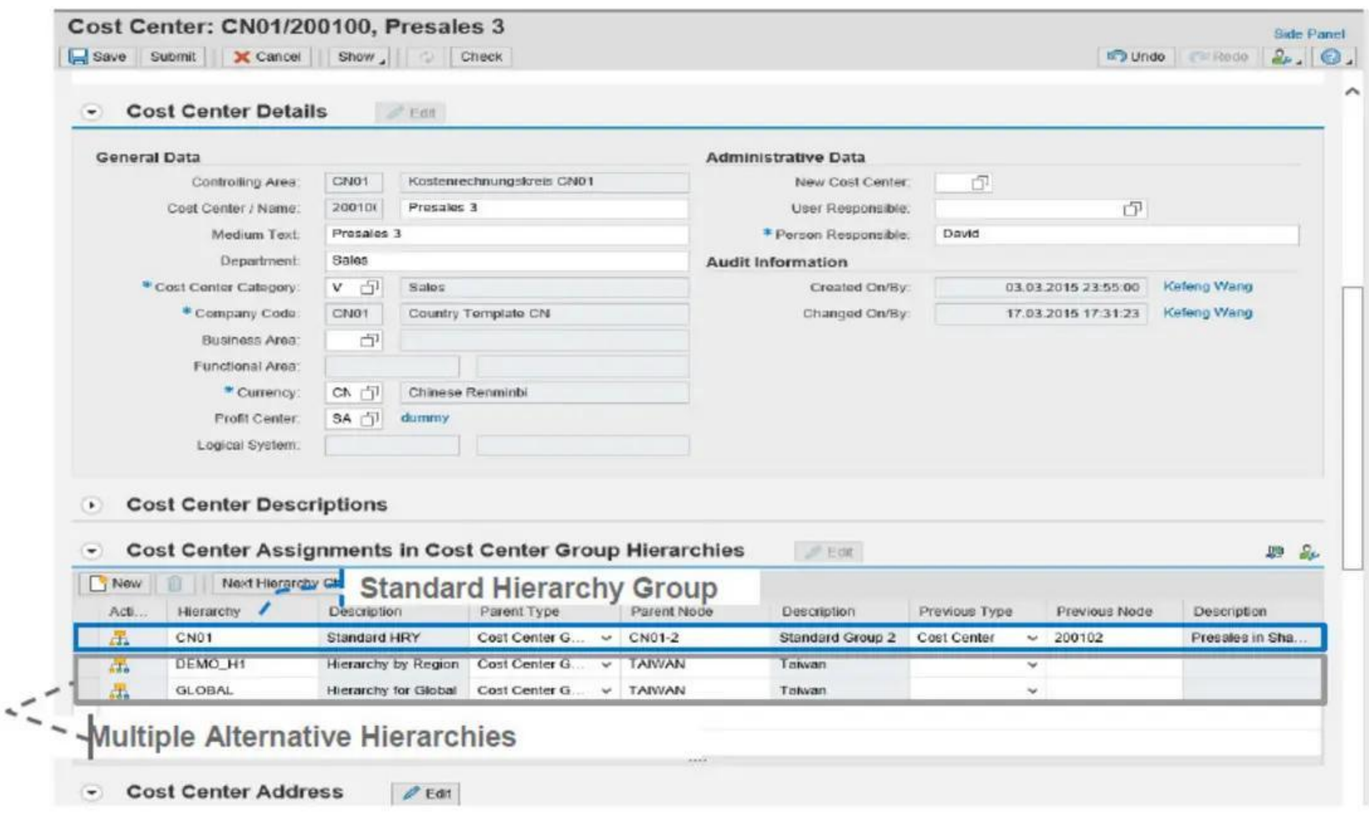
- Cost Center
- Profit Center
- Cost Element
- Cost Center Group
- Profit Center Group
- Cost Element Group

And also:

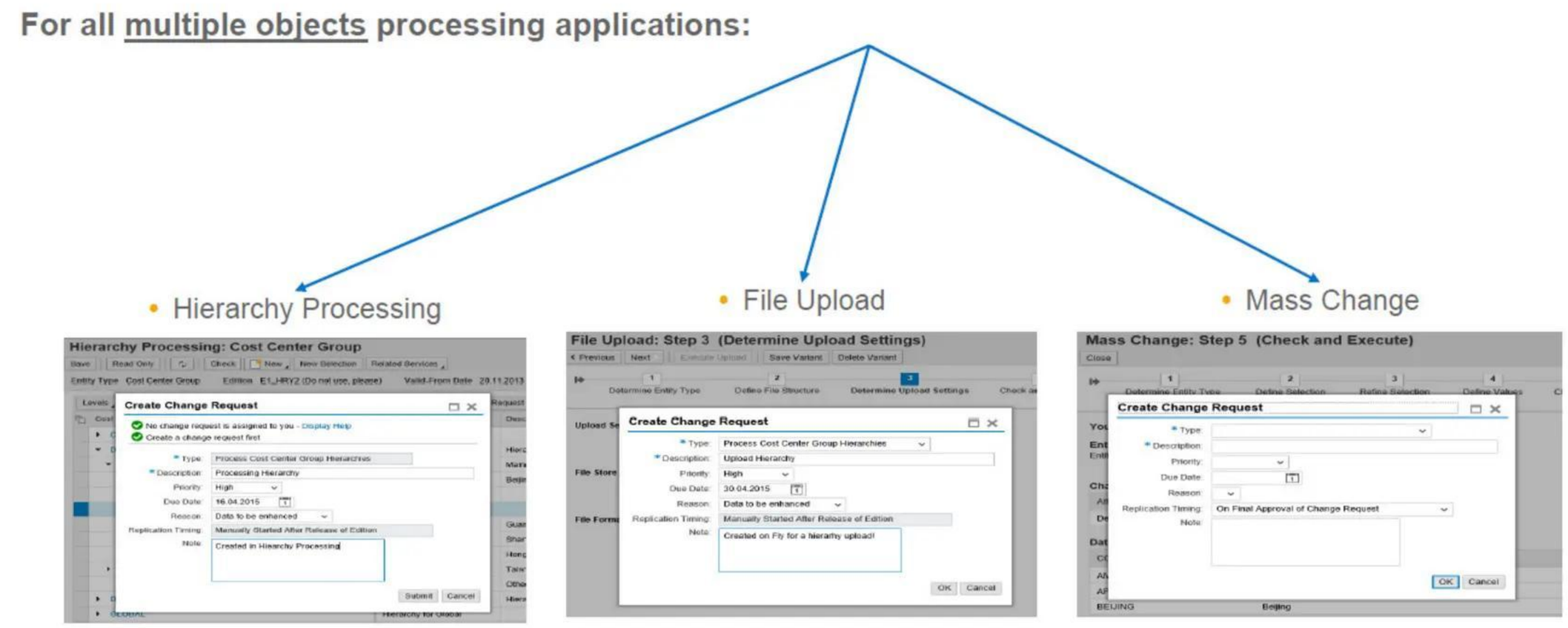
- G/L Account
- Financial Reporting Structure Item

In addition to ERP:

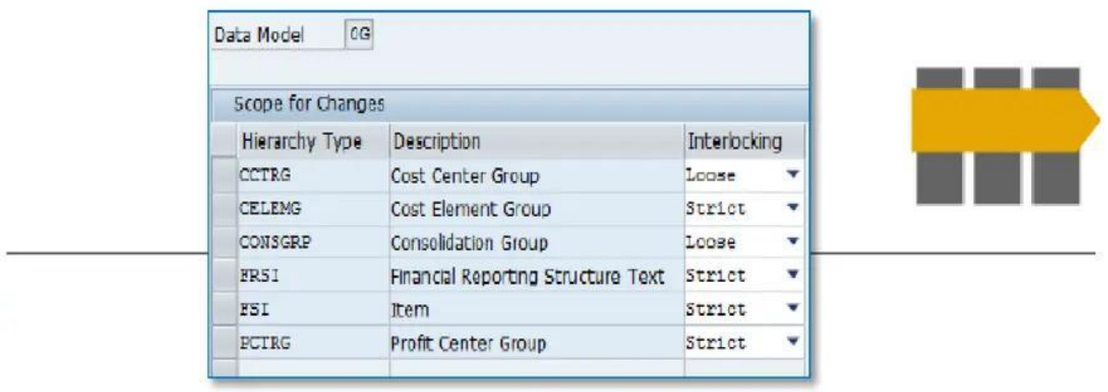
- Overview of all assignments in all hierarchies
- Display and maintain not only standard group, but also alternative hierarchies on master data screen

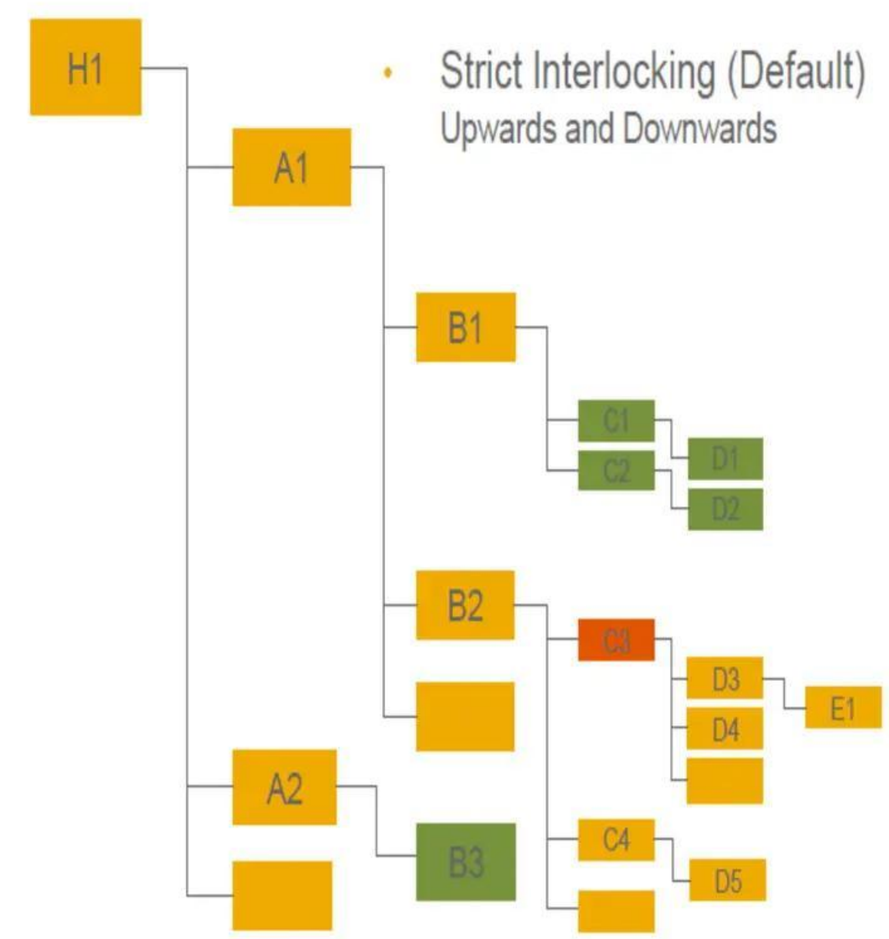


3.10 Integrated Change Request



3.11 Hierarchy Interlocking





Node currently maintained in a pending change request

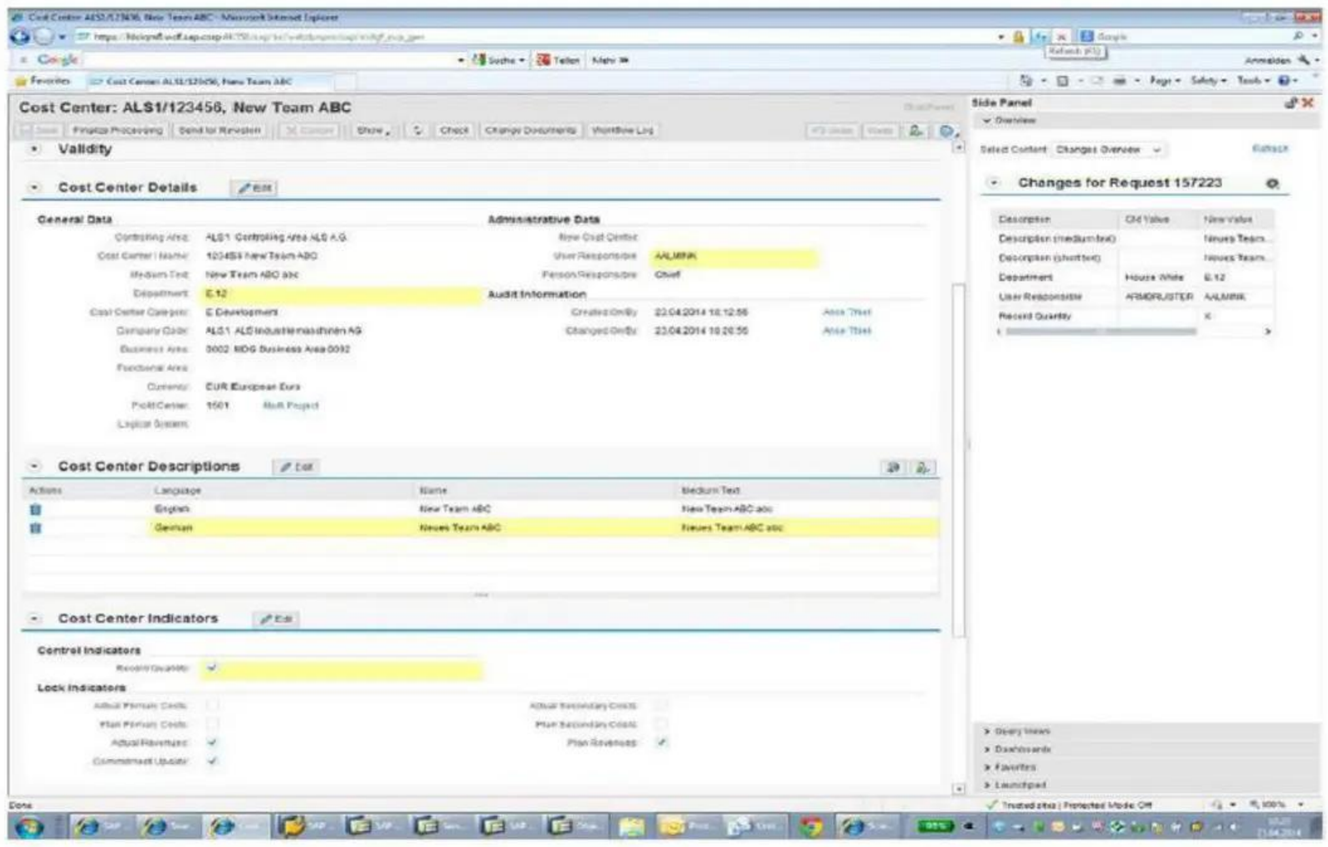
Interlocked: You must use the pending change request

Not Interlocked: You can use a separate change request

3.12 Usability Features in Single-Object Maintenance UI

Side panel / Business Context Viewer:
Context information can be displayed in the side panel to provide further related information about the current data record, e.g. the change history

Highlight Changes: Fields that were changed in the previous step are highlighted in color

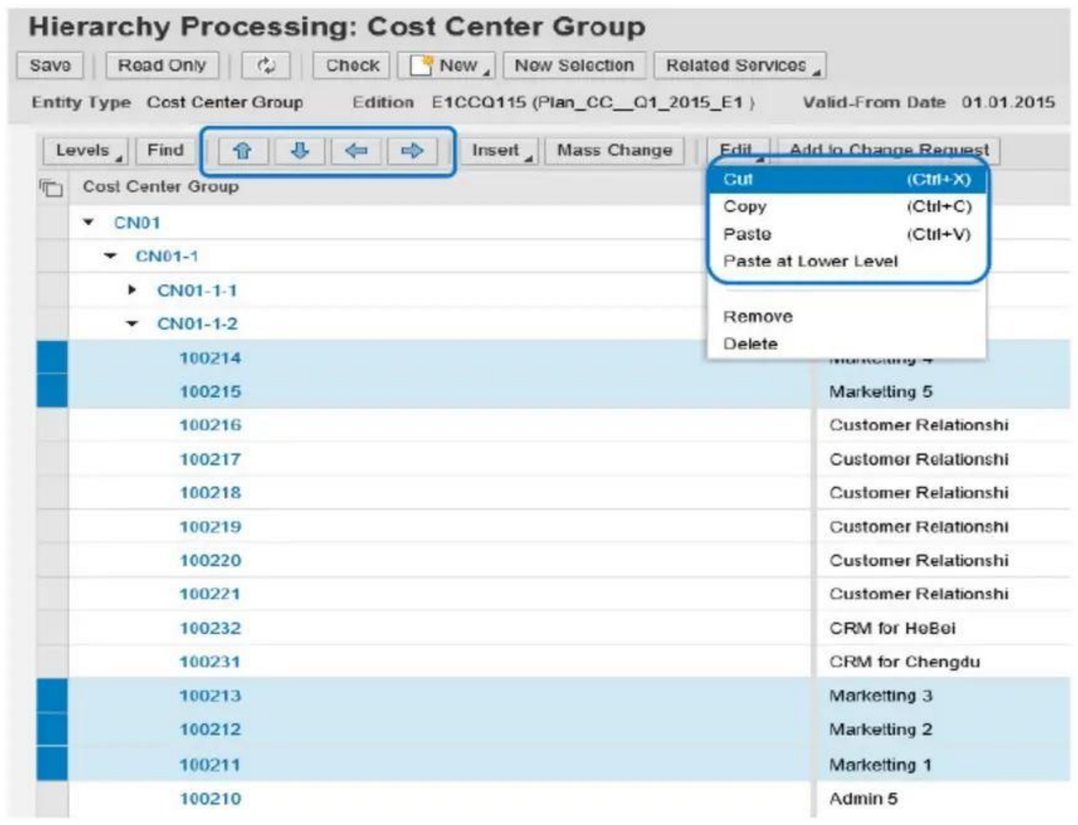


3.13 Usability Features in Hierarchy Processing

Easy Actions supported for multiple objects:

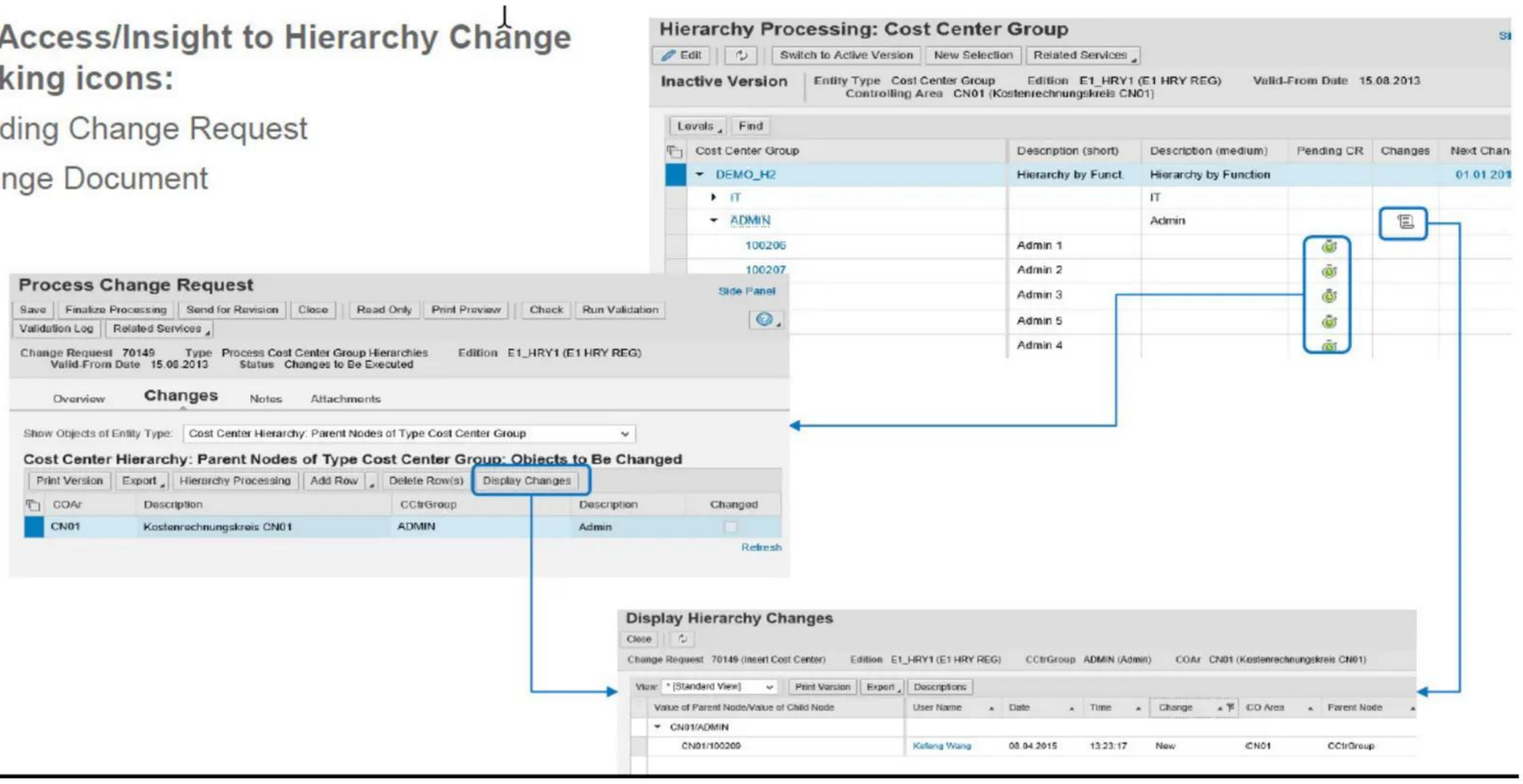
- Drag & Drop
- Move Up, Down
- Rank Higher, Rank Lower
- Copy & Paste or Paste at Lower Level
- Cut & Paste or Paste at Lower Level

Easy to use:
Select multiple subnodes of one node and carry out any of the actions above.

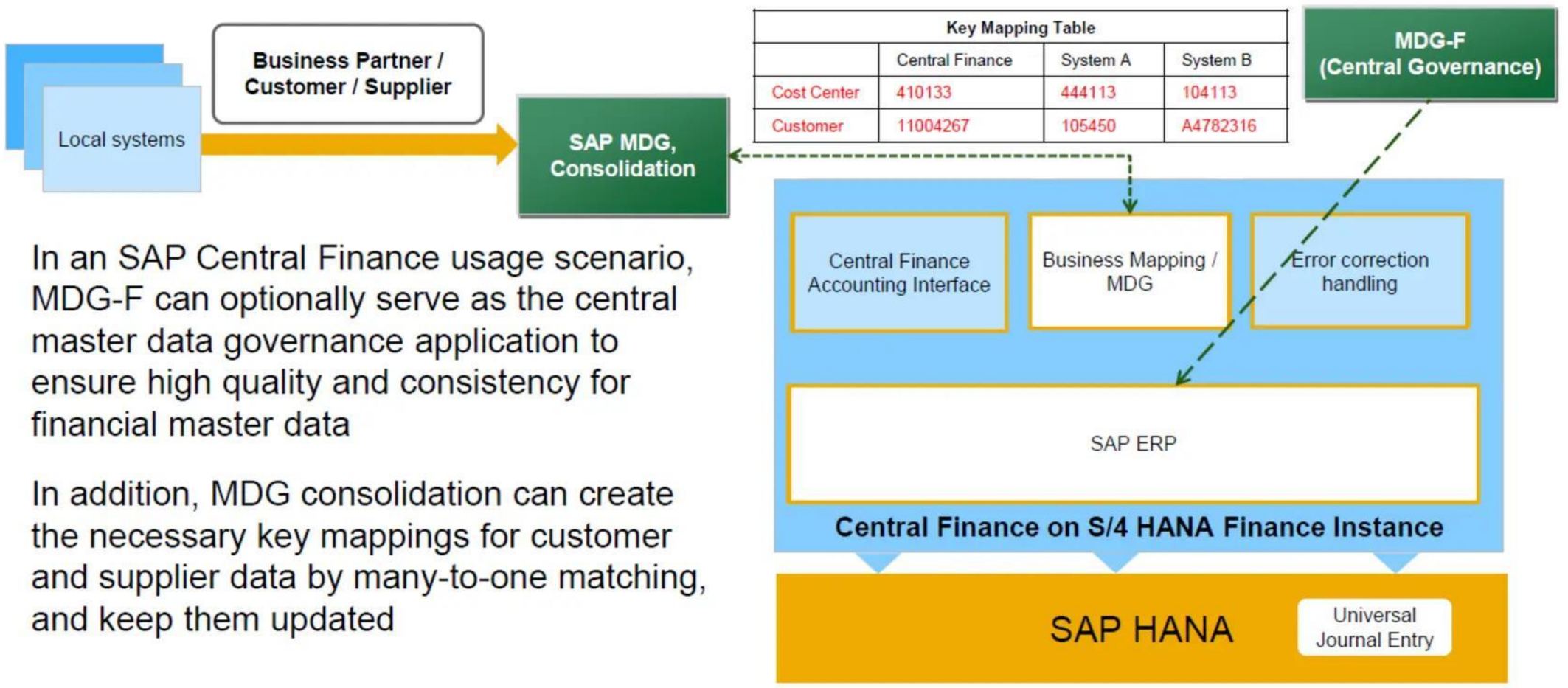


Direct Access/Insight to Hierarchy Change by clicking icons:

- Pending Change Request
- Change Document



3.14 SAP MDG Supporting Central Finance (S/4 HANA Finance)[11]

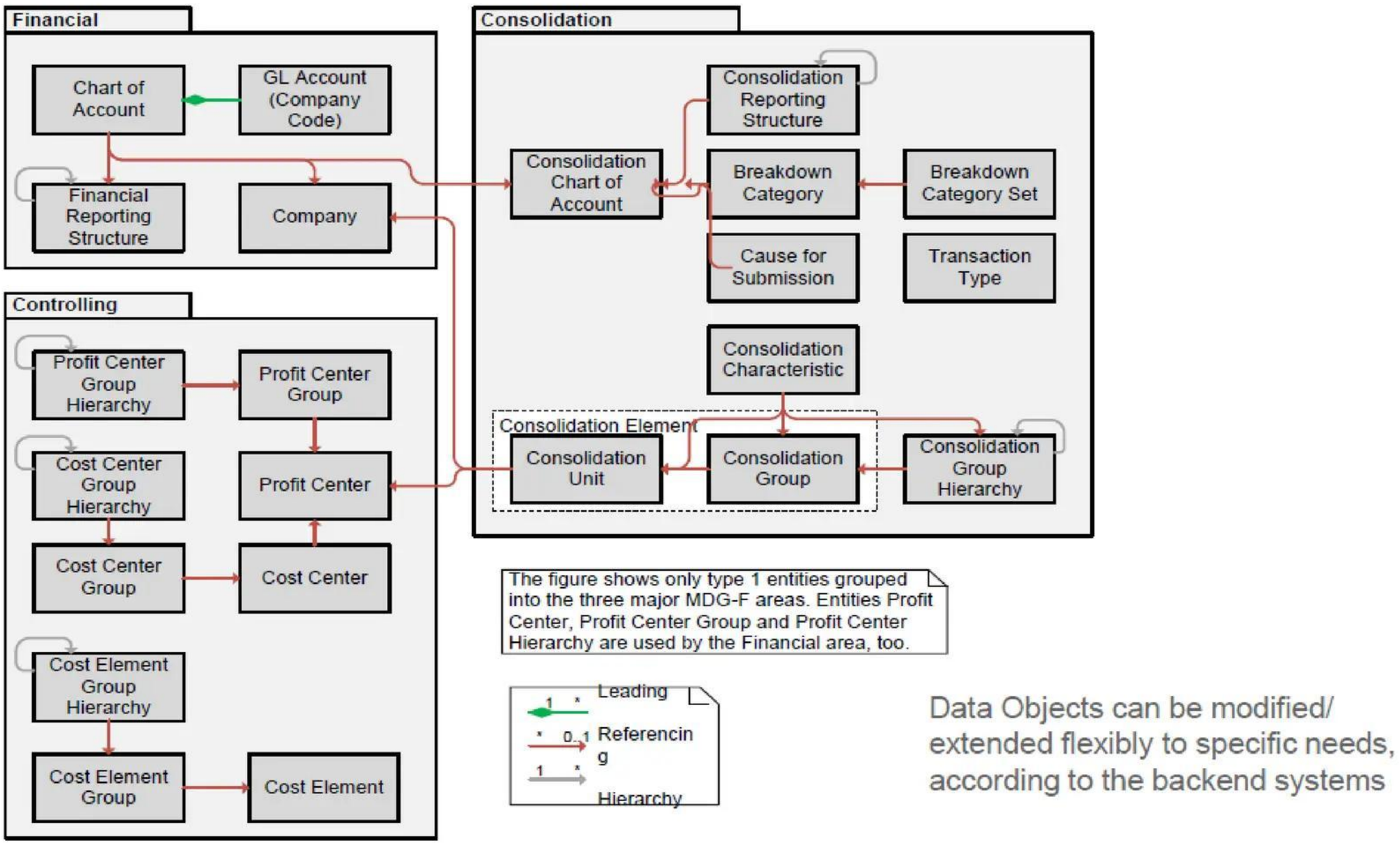


In an SAP Central Finance usage scenario, MDG-F can optionally serve as the central master data governance application to ensure high quality and consistency for financial master data

In addition, MDG consolidation can create the necessary key mappings for customer and supplier data by many-to-one matching, and keep them updated

[11] SAP SE

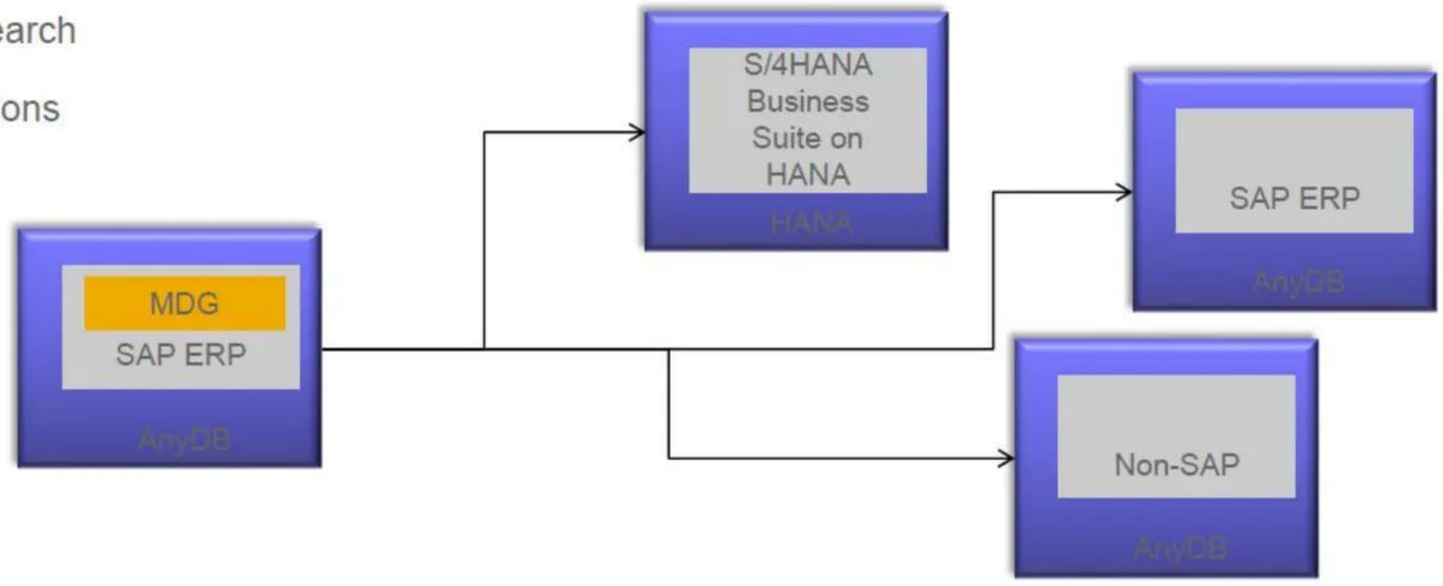
3.15 Standard Objects in Data Model 0G



3.16 Deployment

- MDG is delivered as a business function within SAP ECC
- Deployment as standalone hub or on top of existing ERP instance
- Optional embedding of HANA search
- Replication into various applications including S/4HANA

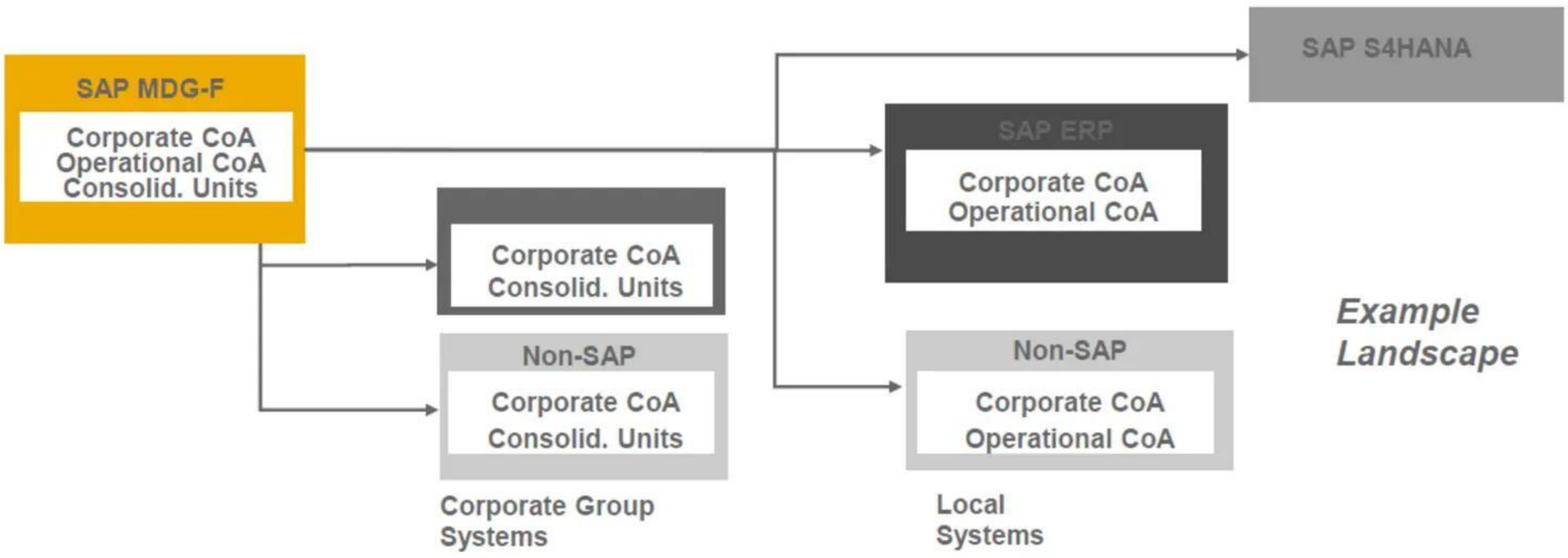
Example



> MDG is pre-integrated with SAP ECC, and provides publish/subscribe and master/slave implementation modes with automatic distribution to SAP and 3rd party applications.

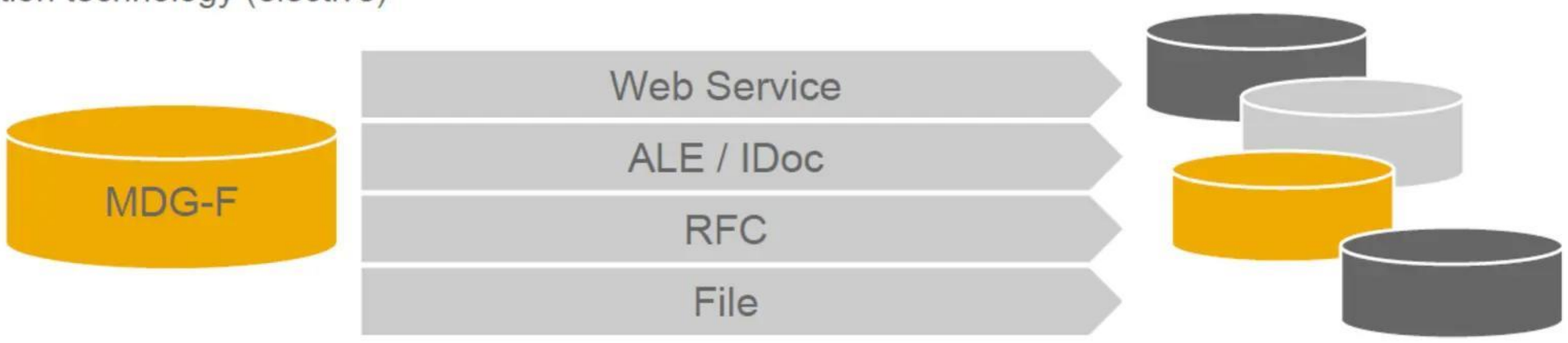
3.17 Replication

MDG-F replicates the data according to the needs of the connected business systems. This is independent of the deployment option, if MDG is used on top of one of the operational ERP systems, or as a standalone MDG data hub.



3.18 Integration

Integration technology (elective)



SOA

Replication Model	Outbound Impl.	Description
SOA_ACC	1010	Replication model for Account (S
SOA_CCTRH	1110	Replication model for Cost Center Group
SOA_CELE	1180	Replication model for Cost Element (SOA)
SOA_CELEH	1190	Replication model for Cost Element Group Hierarchy (SOA)
SOA_COA	1000_V1	Replication model for Chart of Account (SOA)
SOA_COMP	1140	Replication model for Company (SOA)
SOA_COST	1100	Replication model for Cost Centre (SOA)
SOA_FRS	1020	Replication model for Financial Reporting Structure (SOA)
SOA_PCTH	1130	Replication model for Profit Center Group Hierarchy (SOA)
SOA_PCTR	1120	Replication model for Profit Center (SOA)

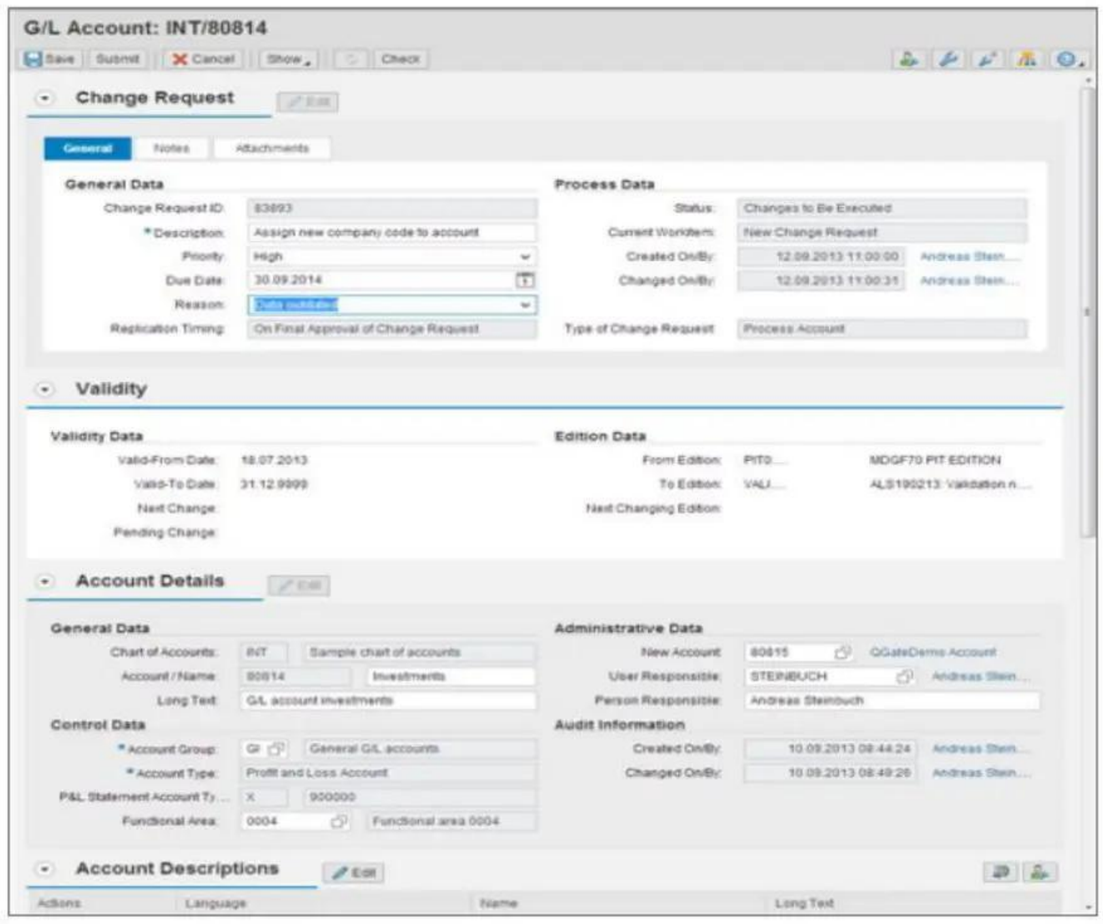
IDoc

Replication Model	Outbound Impl.	Description
IDOC_ACC	1012	Replication model for Account (I
IDOC_COST	1102	Replication model for Cost Center (IDoc)
IDOC_CCTRH	1112	Replication model for Cost Center Group Hierarchy (I
IDOC_CELE	1182	Replication model for Cost Element (IDoc)
IDOC_CELEH	1192	Replication model for Cost Element Group Hierarchy (IDoc)
IDOC_PCTR	1122	Replication model for Profit Center (IDoc)
IDOC_PCTH	1132	Replication model for Profit Center Group Hierarchy (IDoc)

3.19 User Interface

Technology

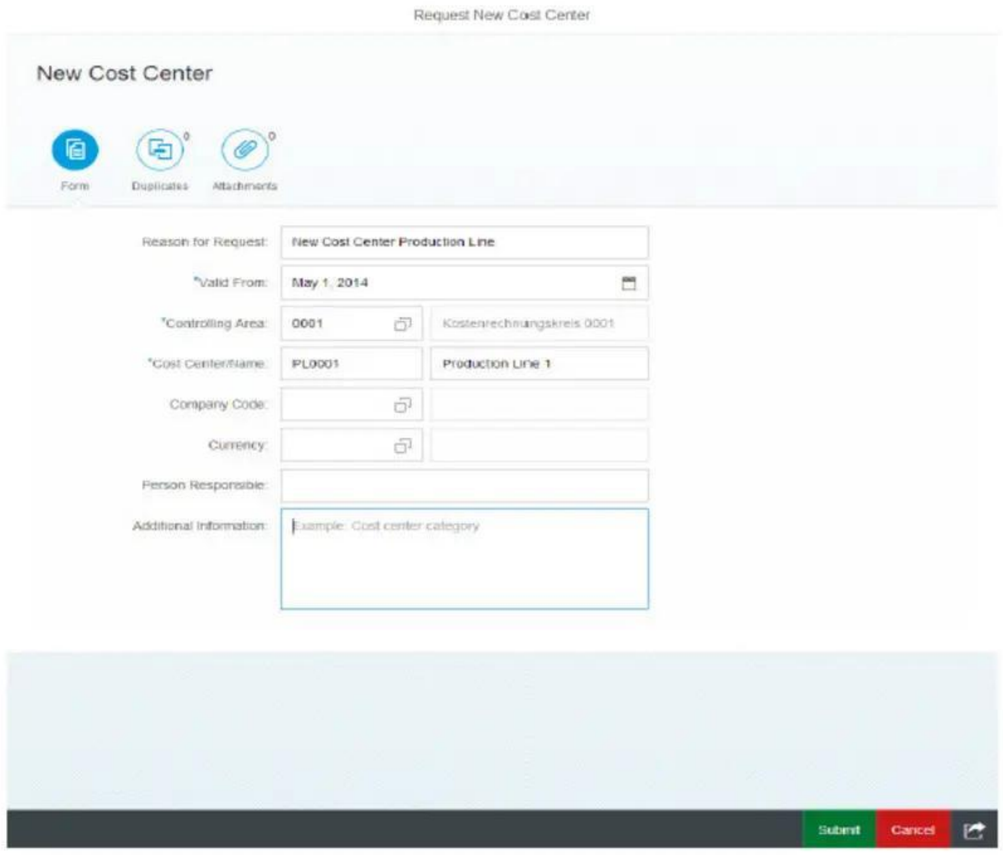
- Browser-based UI (Web Dynpro ABAP) to allow decentral task execution
- Use of Floorplan Manager for easy screen configuration and personalization
- Pre-delivered workcenters provide a structured entry point for individual roles



3.20 Request Cost Center or Profit Center With FioriApp[12]

You can also use Fiori apps to request a new cost center or a new profit center.

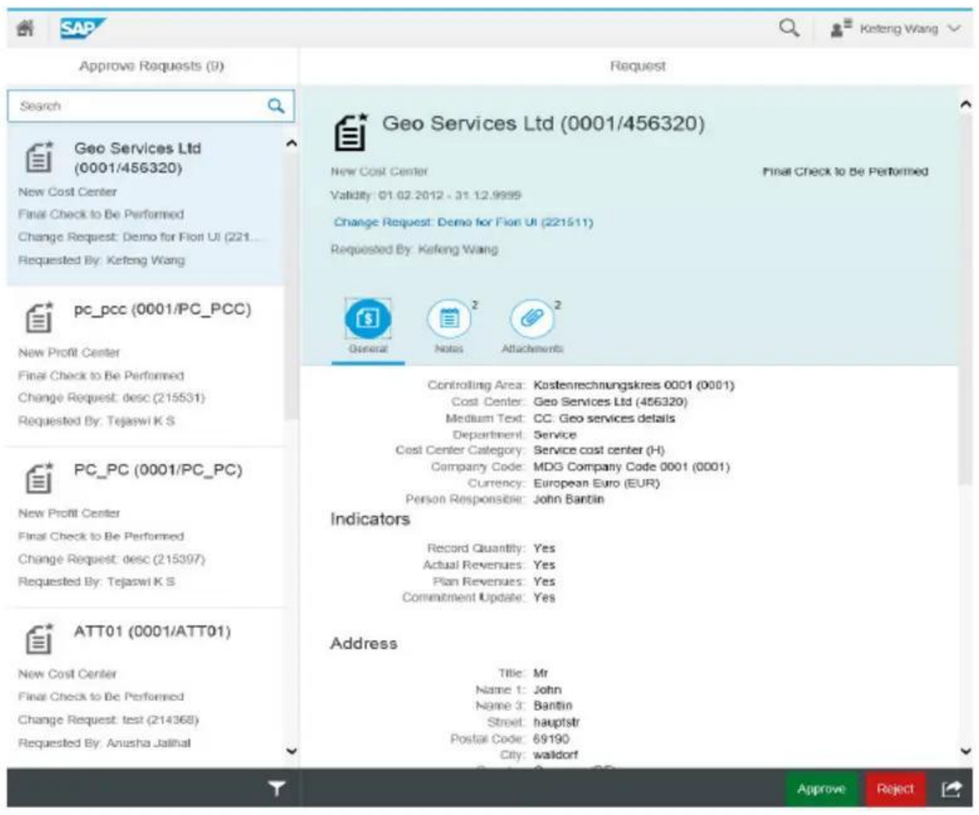
These apps can be flexibly used on a variety of devices including tablets and smartphones.



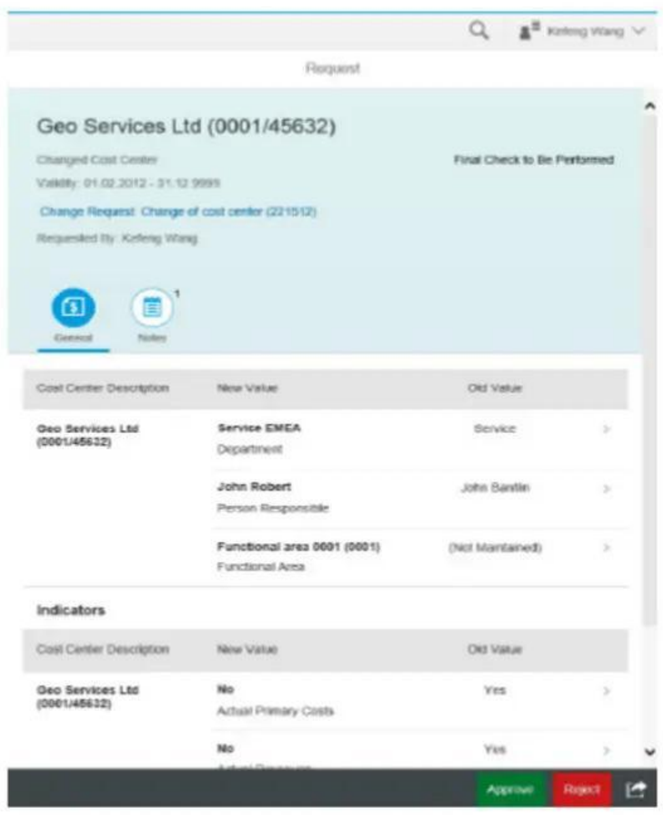
[12] SAP SE

3.21 Approve Cost Center, Profit Center or G/L Account With Fiori App

Approve new object: Only entered fields are displayed



Approve change: Only changed fields (with old values) are displayed



3.22 Copying of accounts

When creating a G/L account, you can now save time and effort by copying the

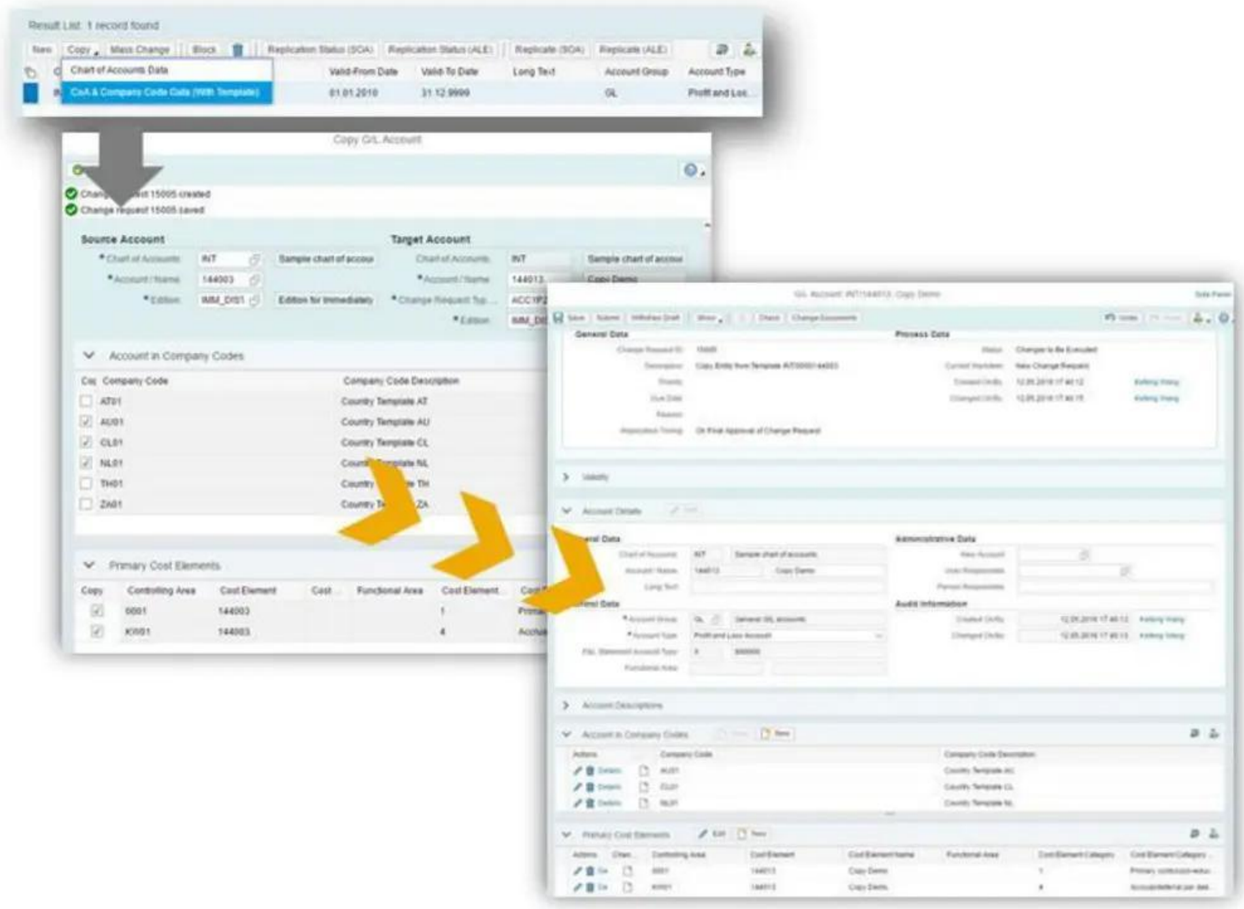
Chart of Accounts and Company Code Data (With Template)

from an existing G/L account using the new Copy function on the Search screen. Alternatively, you can copy just the Chart of Accounts Data.

Copy relevant G/L Account data in one step, including:

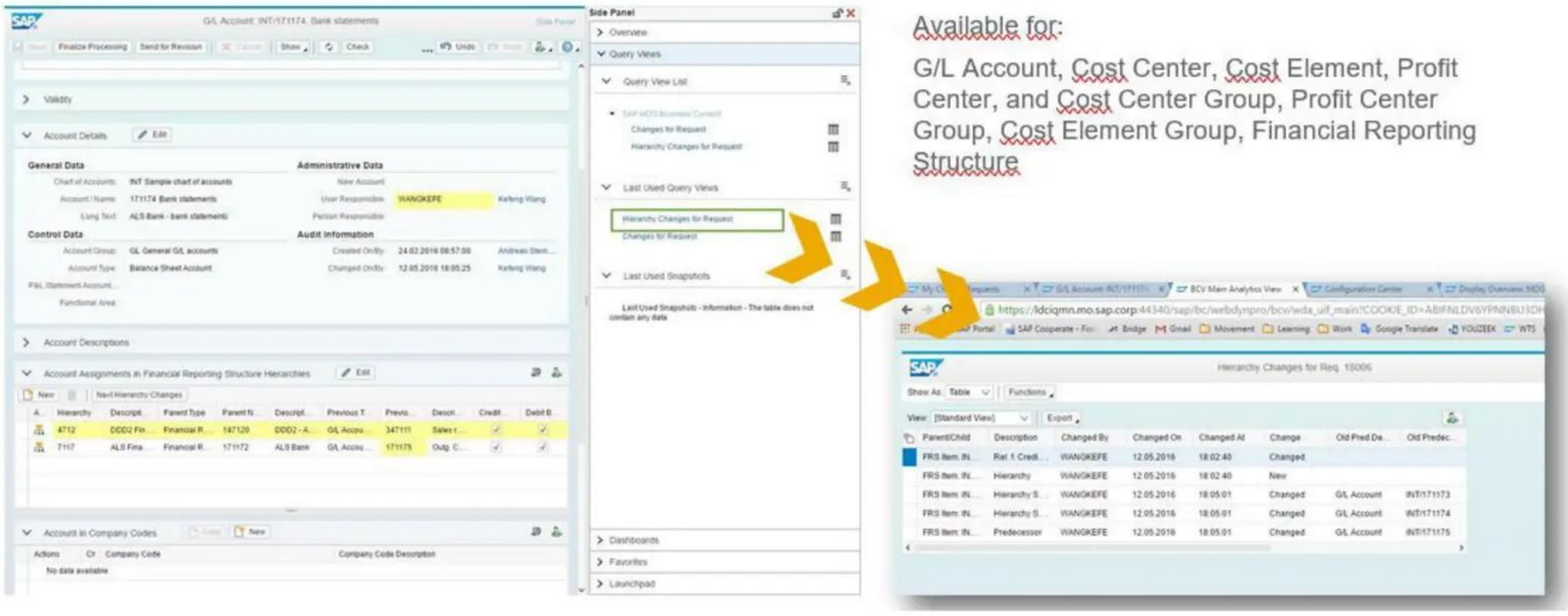
- Chart of Accounts data
- Company Code data
  (selectable on Company Code level)
- Primary Cost Element
  (selectable on Controlling Area).

- New G/L Account can be changed on the populated change request drafts and submitted directly from there.



3.23 Enhanced BCV content to display hierarchy changes

A new display option was added to the Query View. In addition to displaying changes per change request, you can now also display only hierarchy-specific changes for a change request for a detailed overview.



Available for:

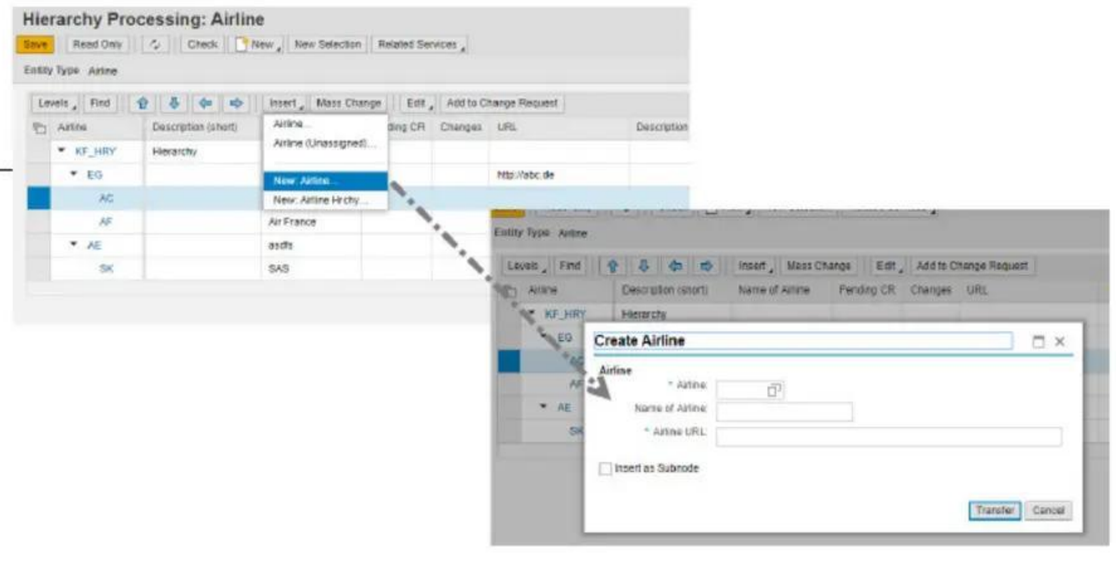
G/L Account, Cost Center, Cost Element, Profit Center, and Cost Center Group, Profit Center Group, Cost Element Group, Financial Reporting Structure

3.24 New functions (planned Enhancements in MDG Application Framework)

3.24.1 Integrated Object creation in Hierarchy Processing UI

Feature description

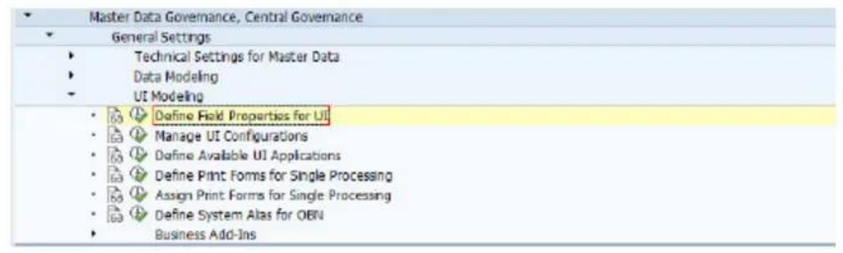
- In hierarchy processing UI, create and insert object into hierarchy within one step.



Business value

- Simplified maintenance process for hierarchy management
- Improved user usability for hierarchy management

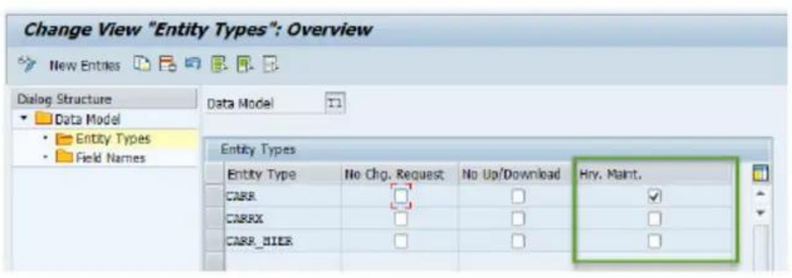
Valid in MDG-F hierarchy

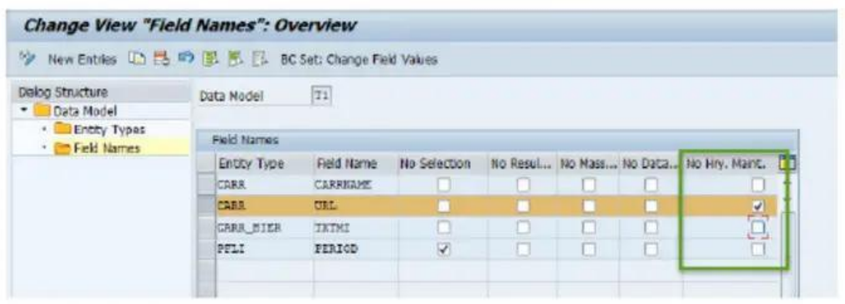


Integrated Object Creation feature is configured under MDG Customizing -> UI Modelling.

You can configure whether Integrated Object Creation in Hierarchy Processing UI is allowed for hierarchy related entity type.

You can configure entity attribute level whether it should NOT be maintained on the creation Popup in Hierarchy Process UI.





Valid in MDG-F hierarchies

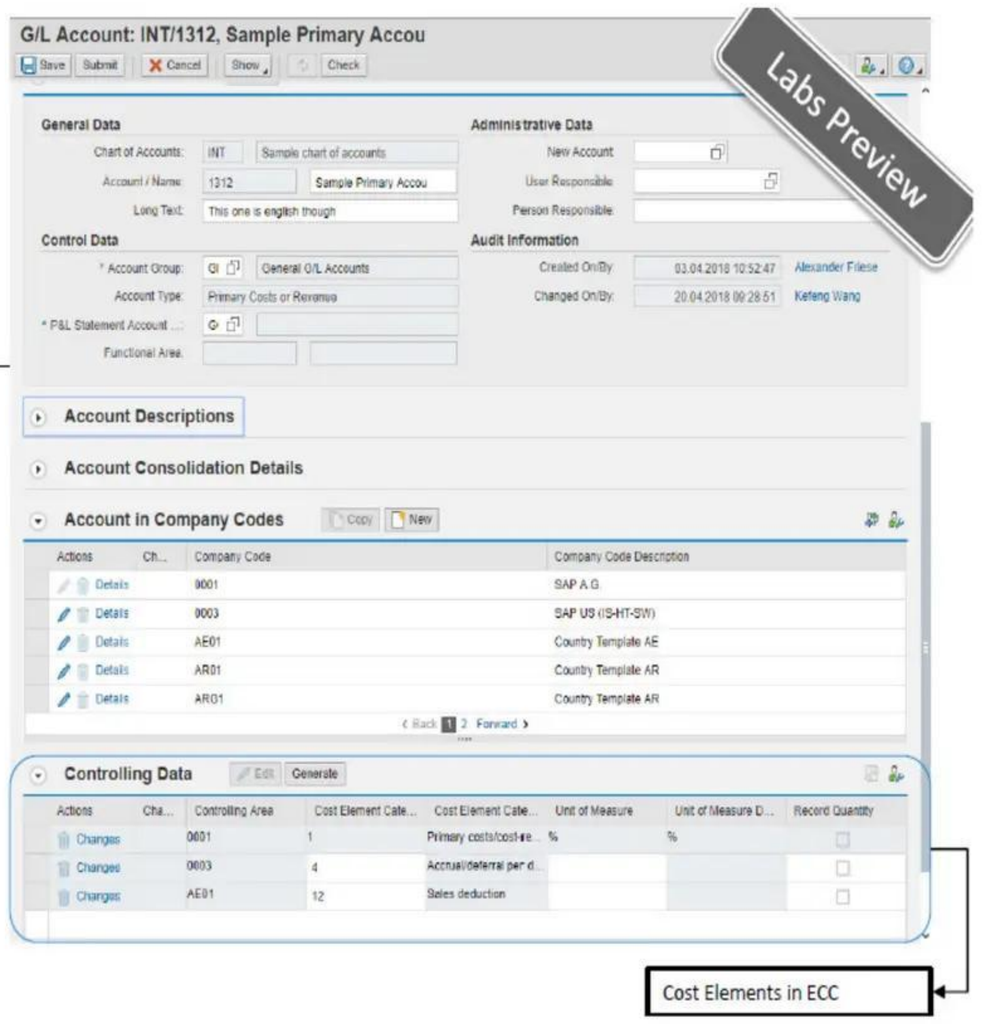
3.24.2 Merge Cost Element into G/L Account

Feature description

- Maintain cost element data in G/L Account UI
- Only for MDG on S/4HANA

Business value

- Simplified financial master data maintenance
- Harmonized behavior with S/4 HANA environment



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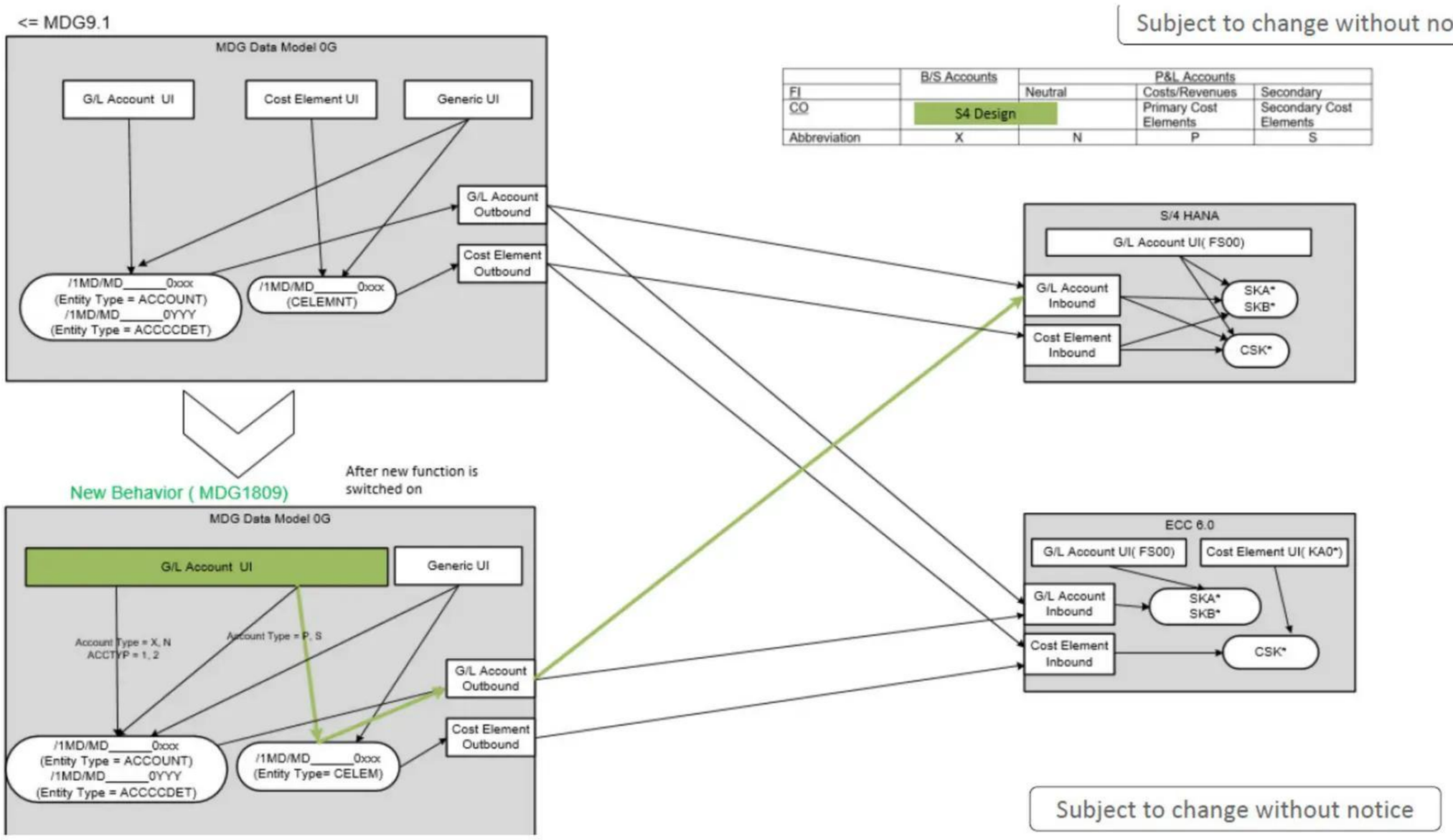


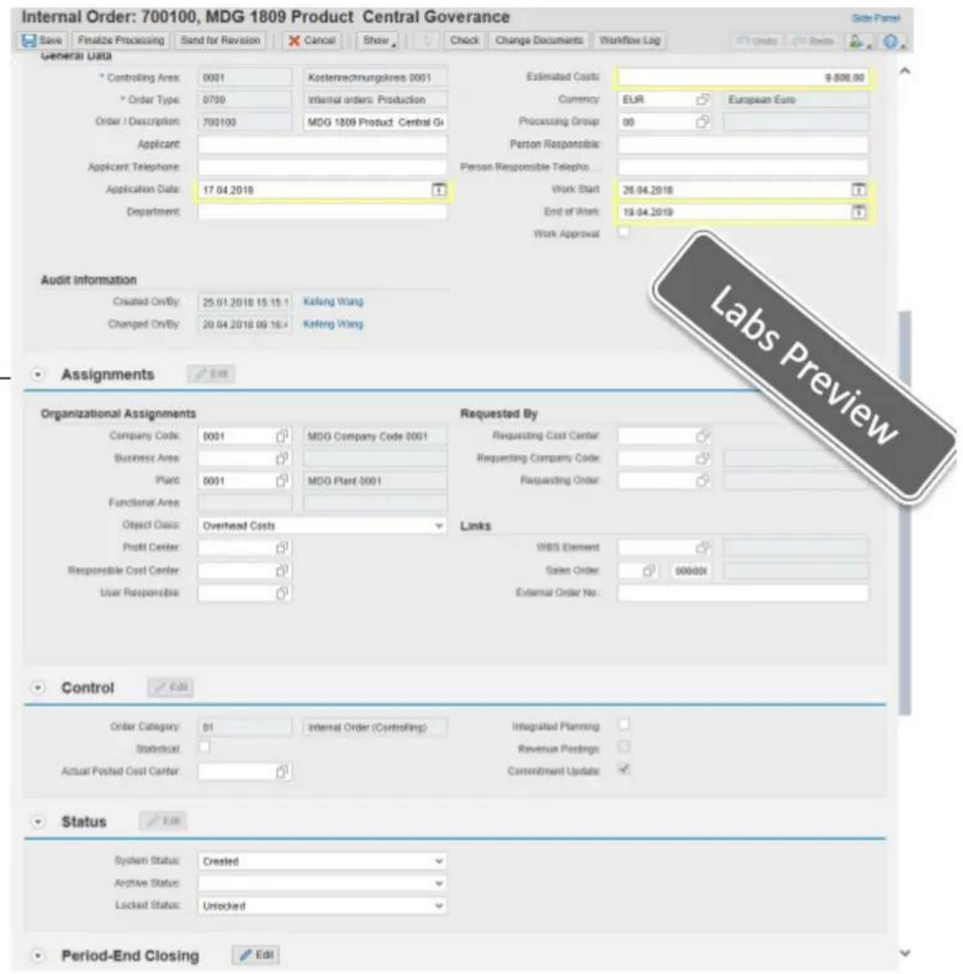
3.24.3 Internal Order Central Governance

Feature description

- Central maintenance and governance of Internal Order
- Replicate Internal Order to ECC or S/4HANA system (Both SOA and Idoc)
- Initial Load Internal Order from ECC or S/4HANA into MDG

Business value

- Centralized governance, compliance and transparency of master data during creation and change through integrated staging, approval and central audit trail
- Delivers consistent definition, authorization and replication of internal order master data, keeps data consistent cross systems
- Eliminates error-prone manual maintenance processes for internal order master data in multiple systems



3.25 MDG-F road map overview – key innovations[13]

Recent innovations[1]

Master data consolidation
- Consolidation of active records
- Support of bus. partner relationships
- Classification

Master data mass processing
- Support of bus. partner relationships

Central master data governance
- Support for contract account
- Classification
- Framework enhancements like archiving of change requests, integrated object creation in hierarchy-processing UI, …
- Enhanced Material data model covering shelf-life data and service parts mgmt.
- Business partner enhancements like documents, payment cards, and more

Data quality management
- Data quality control and remediation of product data

2018 – Planned innovations[1,2]

Consolidation and mass processing
- Multiple vendor and customer assignments of business partners
- Classification data of materials, customers, vendors in mass processing

Central master data governance
- Financials: Adopt S/4HANA universal journal, i.e. G/L accounts as basis for cost element. Support internal order
- Product: Adopt S/4HANA simplifications and industry-specific enhancements
- BP: Multiple vendor/customer assign-ments. Integration /w SAP Credit Mgmt.

Integration scenarios
- e.g. SAP Ariba Supplier SLP, SAP Hybris C4C

Data quality and process quality
- Create bus. rules & analyze data quality
- Enhanced SAP Fiori user experience in process analytics

2019 – Product direction[2]

Expand the foundation…
- Support all critical MDM capabilities in one single packaged application
- Flexibly support heterogeneous environments, with faster time to value than just a toolbox
- Leverage SAP's footprint to simplify MDM and integration across on premise and cloud applications

…and support new master data challenges
- Increasing velocity and growing amounts of master data, demand for a higher degree of automation
- Leverage new technologies, like machine learning, to enable the new promises of the digital transformation

2020+ – Product vision[2]

Support strategic trends
- Enterprise-wide master data mgmt across on premise & cloud apps
- Seamless master data exchange across the entire business network
- Master data for end-to-end IoT solutions spanning devices and apps

Simplify creation, management and use of master data
- One platform for all domains and use cases natively using EIM and SAP HANA
- Intuitive user experience and high business user efficiency
- Embedded analytics, stewardship, monitoring, and remediation
- Further automation of routine tasks like error correction or data completion

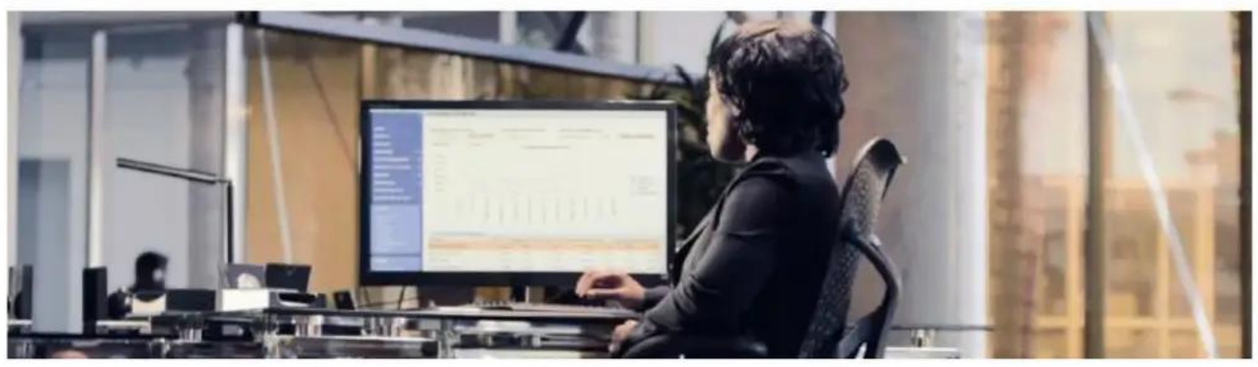
1. Potential data protection and privacy features include simplified deletion of personal data, reporting of personal data to an identified data subject, restricted access to personal data, masking of personal data, read access logging to special categories of personal data, change logging of personal data, and consent management mechanisms. 2. This is the current state of planning and may be changed by SAP at any time without notice.

13 SAP SE

3.26 Benefits of SAP Master Data Governance for Financials

What can be improved?	How can SAP Master Data Governance help?	What benefits can be achieved?
Process efficiency		
• Intercompany reconciliations • Period closing • Corporate controlling • Financial consolidation	**Greater governance across local and corporate systems** • Automation of data maintenance through integrated validations, derivations and role-based transactional guidance • Reduction of manual correction work due to "First Time Right" data entry • More automation in reconciliation of charts of accounts across multiple ledgers	• Faster financial close • Increased coordination across operating units • Increased trust in reporting data • Reduced cost of finance as a percentage of revenues

What can be improved?	How can SAP Master Data Governance help?	What benefits can be achieved?
Compliance Management		
• Transparency • Traceability • Risk exposure	**Greater governance across local and corporate systems** • Separation of duties between change and approval authority for financial master data • Standardized workflow and automated routing • Complete history and audit trail of changes to financial master data	• Increased visibility into change impact on financial statements • Reduced audit workloads and costs • Greater collaboration in support of internal policies and government/industry regulations • Fewer non-compliance penalties
Quality and Consistency		
• Master data • Business rules • Governance processes	**Greater governance across local and corporate systems** • Single Source of Truth allowing for creating and storing the right data at the right place, even in distributed and changing landscapes • Standardized data models, validation rules and workflows • Easier Cross-System Data Synchronization through configurable and selective distribution logic	• Reduced effort to find information • Increased trust in data by business users • Greater consistency of financial statements and less need for restatements • Improved modeling, planning and analysis



SAP Master Data Governance (MDG) is used for embedded Master Data Management (MDM), that is, centralized, out-of-the-box, domain-specific creation, modification, and distribution of master data with a focus on SAP Business Suite.

For MDG-F, however, customers might want to run the MDG hub in the same physical client as the ERP client due to the nature of the MDG-F data model. Since all MDG-F entity types use the flexible mode their data is stored only in the MDG generated tables and not within ERP tables.

This document explains how-to setup an ALE scenario for data replication where MDG hub and ERP client share the same physical client of the same system.

We recommend that you study the following how-to guide before working with the current guide:

Extensibility Options for SAP Master Data Governance for Financial Data Overview about MDG-F

Strong ALE configuration expertise is required, too.

SAP Master Data Governance (MDG) is used for embedded Master Data Management (MDM), that is, centralized, out-of-the-box, domain-specific creation, modification, and distribution of master data with a focus on SAP Business Suite.

Domain-specific content (data models, user interfaces, workflows) is provided as part of the standard for several application areas. It is a common requirement from customers to adapt the MDG data models to their specific needs.

This document explains the SAP-delivered solution for financial master data within MDG (MDG-F). It describes the key concepts and implementation details and links to guides that provide real-life use cases of MDG-F. All software artifacts of the MDG hub that are provided or used by MDG-F belong to the **MDG_FND** software layer.

If you use either the accounting components or the controlling components of MDG-F, we recommend that the MDG hub includes the **SAP_APPL** layer. This layer owns customizing tables that the relevant accounting entity types or controlling entity types use.

If you use the consolidation components of MDG-F only, the **MDG_APPL** and **SAP_APPL** software layers are optional.

Master Data Governance for Financials (MDG-F) is an application to centrally govern the maintenance of financial master data.

•MDG-F provides central steering and control mechanisms independently of local or centralized execution.

•In MDG-F, master data records can be created, completed or changed in a staging area, following a distributed step-by-step way, before making the record available to connected business applications.

3.27 Financial Master Data[14]

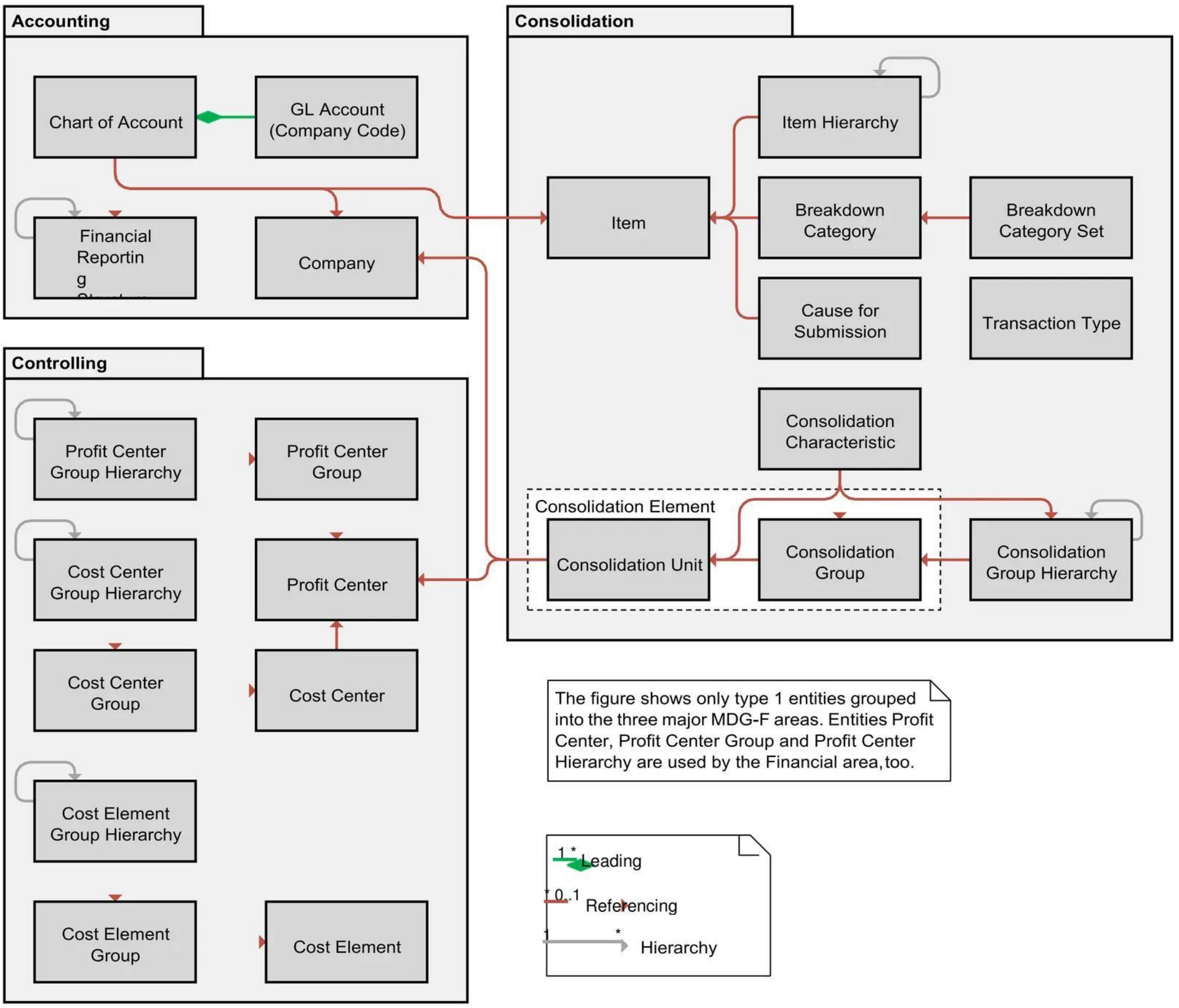
3.27.1 MDG for Financials

MDG offers a domain specific solution for financial governance (MDG-F). The current MDG-F data model is called **0G**. It covers entity types of the accounting, controlling and consolidation components of financial master data as indicated by the examples below:

- Accounting: G/L Account **(ACCOUNT & ACCCCDET)**, Company **(COMPANY)**
- Controlling: Cost Center **(CCTR**), Cost Element **(CELEM)** and Profit Center **(PCTR)**
- Consolidation: Consolidation Unit **(CONSUNIT)**, Item **(FSI)**

MDG offers a single data model combining several entity types of financial master data. The entity types can be separated roughly in three areas: accounting, controlling and consolidation. Nevertheless most of the entities are connected with each other. The figure below shows the currently supported entity types and their relations.

[14] SAP SE



For an excel file providing a detailed overview of all fields in MDG-F, see MDG Financials – (Un-) Supported Segments and Fields.

3.28 Data Model 0G

The MDG-F entity types, their attributes, and relations are shipped within the data model **0G**.

The data model uses only **flexible entity types**. Both the actual (active) data as well as the staging (inactive) data that exists only within a governance process, is stored in generated MDG tables. Although some entity types refer to existing objects of SAP systems, the active master data is not stored directly in the related backend tables. It requires a data replication to send the MDG active data into the related SAP system tables.

Some of the entity types with *SU Type* 1 **reference** other entity types with *SU Type* 1 as attributes (for example the entity type “account” uses the entity type “company” as an attribute “trading partner”). Using a reference requires that the record to be referenced already exists as an active one within MDG-F:

- Keep the validity of an object according to the chosen edition in mind. If you have created a company that is valid from the 1st of January 2015, the same company cannot be used for an account that is valid from 1st of January 2013 since the company does not exist on the 1st of January 2013.
- Keep the sequence of objects to be created in mind. You must create the company that an account uses before creating the account. It is not sufficient to create the change request for the company only. The change request must be processed completely so that the company record is activated.
- Keep the edition and change request type in mind. If you want to create dependent objects at the same time within a single change request, you have to define both the edition and change request type accordingly. They must contain both entity types.

Each of the MDG-F entity types with *SU Type* 1 uses the **MDG Edition functionality**. Editions enable the grouping of different entity types to ensure the data consistency across all entity types. Furthermore, the edition enables the time-dependent maintenance of master data. This is required for some of the related SAP ECC objects like cost or profit centers.

Using editions for the master data maintenance requires that you configure **Edition Types** to indicate which of the entity types with *SU Type* 1 can be grouped within an edition. It is recommended to use the SAP pre- defined edition type **0G_ALL**. The edition type uses all entity types with *SU Type* 1. If you want to create your own edition types, keep in mind that:

- Many entity types with *SU Type* 1 reference other entity types with *SU Type* 1.
- You cannot assign the same entity type with *SU Type* 1 to multiple edition types as long as the edition types are not released.

- If objects are not part of the same edition, you will not be able to create cross references.

Additionally **Change Request Types** are needed for master data maintenance.

SAP pre-defines both Edition Types and **Change Request Types** in release dependent BC-Sets:

Release	BC-Sets	SAP Note
MDG 7.0 SP00	**CA-MDG-APP-FIN_EDITION_CR_04**	1908432
MDG 7.0 SP02	**CA-MDG-APP-FIN_EDITION_05** **CA-MDG-APP-FIN_CR_TYPES_05**	2003344
MDG 8.0 SP00 MDG 9.0 SP00	**CA-MDG-APP-FIN_EDITION_05** **CA-MDG-APP-FIN_CR_TYPES_06**	2138778

Each MDG-F entity type with *SU Type* 1 uses **external and non-changeable keys**. The entity's key must be defined manually during the creation of the entity. It cannot be changed. With MDG 7.0 SP02 MDG-F offers more flexibility with regard to changeable keys during the creation process of an object (see the related how- to guide for more details).

Most of the MDG-F entity types with *SU Type* 1 support **attachments**. This enables adding documents and/or links to the master data. The attachments are a MDG specific functionality. They are not related to the SAP ECC document features and thus not part of data replication.

All **hierarchies** that are defined in data model **0G**use the **MDG Edition functionality**. It is possible to create the same hierarchy in different, multiple points in time. This allows a different structure, for example one for the active business year and a second one for planning the next business year. Note that each hierarchy is an independent object. Following the example above, if you change something in the hierarchy for the active business year, the same change is not applied automatically to the planning hierarchy of the next business year. You have to repeat the change in the planning hierarchy.

In **MDG 7.0** it is only possible to maintain hierarchies using the hierarchy processing user interface.

MDG 8.0 introduces the hierarchy maintenance within the single object maintenance user interfaces for accounting-related and controlling-related entity types.

MDG 9.0 does not introduce any new change request types.

3.29 Accounting

Entity types of the accounting area relate to common SAP ECC objects:

MDG Entity Type	Description
Chart of Accounts (**ACCOUNT**)	Reflects the so called "***A-Segment***" of the SAP ECC ***Account*** (tables **SKA1** and **SKAT**). It is the chart of accounts dependent data of an account.
G/L Account in Company Code (**ACCCCDET**)	Reflects the so called ***B-Segment*** of the SAP ECC ***Account*** (table **SKB1**). It is the company code dependent data of an account.
Entity types **ACCOUNT** *and* **ACCCCDET** *are usually combined within MDG-F to the term* ***G/L Account****.*	
Company (**COMPANY**)	Reflects the SAP ECC ***Company*** customizing (table **T880**).
Financial Reporting Structure (**FRS**)	Reflects the SAP ECC ***Financial Statement Version / Financial Reporting Structure*** (table **T011**).
FRS Item (**FRSI**)	Reflects the SAP ECC ***Financial Statement Version / Financial Reporting Structure Items*** (tables **FAGL_011***).
Entity types **FRS** *and* **FRSI** *are usually combined within MDG-F to the term Financial Reporting Structure. The financial reporting structure supports the creation of a hierarchy, too. You can assign accounts to this hierarchy.*	

3.30 Controlling

Entity types of the controlling area relate to common SAP ECC objects:

MDG Entity Type	Description
Cost Center (**CCTR**)	Reflects the SAP ECC ***Cost Center*** (tables **CSKS**and **CSKT**).
Cost Center Group (**CCTRG**) ***Cost Center Group Hierarchy*** (**CCTRH**)	Reflect the SAP ECC ***Cost Center Group Hierarchy***. In MDG-F the entity ***Cost Center Group Hierarchy*** defines the root node of the hierarchy. Both entities are required to build the full hierarchy. Cost Centers are added as leafs.
Cost Element (**CELEM**)	Reflects the SAP ECC ***Cost Element*** (tables **CSKA**, **CSKB**, and **CSKU**). In MDG-F a ***Cost Element*** does not differentiate between common and controlling area dependent data as done in SAP ECC.
Cost Element Group (**CELEMG**) ***Cost Element Group Hierarchy*** (**CELEMH**)	Reflect the ***Cost Element Group Hierarchy***. In MDG-F the entity ***Cost Element Group Hierarchy*** defines the root node of the hierarchy. Both entities are required to build the full hierarchy. Cost Elements are added as leafs.
Profit Center (**PCTR**)	Reflects the SAP ECC ***Profit Center*** (tables **CEPC**, **CEPC_BUKRS**and **CEPCT**).
Profit Center Group (**PCTRG**) ***Profit Center Group Hierarchy*** (**PCTRH**)	Reflect the ***Profit Center Group Hierarchy***. In MDG-F the entity ***Profit Center Group Hierarchy*** defines the root node of the hierarchy. Both entities are required to build the full hierarchy. ***Profit Centers*** are added as leafs.

3.31 Consolidation

Entity types of the consolidation area relate to the SAP solution for Business Consolidation Services of Strategic Enterprise Management (SAP SEM BCS):

MDG Entity Type	Description
Consolidation Characteristic (**CONSCHAR**)	Defines the basis for consolidation groups and consolidation elements. The characteristic's scope either relates to a company or a profit center.
Consolidation Unit (**CONSUNIT**)	Defines the consolidation unit of a consolidation element. Depending on the used consolidation characteristics, either company or profit center specific data can be maintained.
Consolidation Group (**CONSGRP**)	Defines the consolidation group of a consolidation element.
Entity types **CONSUNIT***and* **CONSGRP***are usually combined within SAP SEM BCS Consolidation Element. In MDG-F both are treated as separated entity types.*	
Consolidation Group Hierarchy (**CONSGRPH**)	Defines the root node of the consolidation group hierarchy. Consolidation groups and units can be added to the hierarchy.
Item (**FSI**)	Defines the item. MDG-F uses Items for the maintenance of Group Accounts, too. Therefore entity type **ACCOUNT**has a cross reference to **FSI**.
Item Hierarchy (**FSIH**& **FSIT**)	Reflect the item hierarchy resp. consolidation reporting structure of SAP SEM BCS. Entity type **FSIH** defines the root node of the hierarchy. Entity type **FSIT**defines text items usable as sub-nodes of the hierarchy. Item can be assigned as leafs.
Breakdown Category (**BDC**)	Defines the breakdown category. Each category uses sets. A category is usually assigned to an item.
Breakdown Category Set (**BDCSET**)	Defines the breakdown category sets.
Cause for Submission (**SUBMPACK**)	The cause for submission does not exist in SAP SEM BCS. The entity type serves as an example how-to extend MDG-F with additional information.

Transaction Type (**TRANSTYPE**)	Defines the transaction type.

All entity types are created according to the **SOFEX** example model that is shipped as a template model within SAP SEM BCS. Any changes performed in the SAP SEM BCS model have to be applied to MDG-F, too.

3.32 User Interface

The user interfaces for MDG-F are based on ABAP Web Dynpro. They are built with the floor plan manager (FPM) using its specific Business Object Layer (BOL) / Generic Integration Layer (genIL) technology. Some advantages to be mentioned are:

- Loose coupling of the user interfaces to the MDG-specific processes.
- High flexibility for the creation of the user interfaces. The huge amount of fields to be displayed is split into small User Interface Building Blocks (UIBBs). UIBBs support lists, forms and special kinds like pop-ups, search input and search results.
- Possibility to create object-specific user interfaces to create a common look and feel and/or a similarity of the MDG user interfaces compared to the SAPGUI maintenance transactions.
- Reuse of the MDG-F generated tables, structures and fields (including naming) during the user interface creation.

3.33 Generic Interaction Layer (genIL)

The generic interaction layer (genIL) is required for the MDG-F user interfaces. It consists of a genIL model and one or more genIL implementation classes for the specific model. Transaction **GENIL_MODEL_BROWSER** can be called in the SAP backend to view the model. A genIL model basically consists of objects and relations.

- Objects consist of attributes. Each attribute reflects a usable field for the user interface.
- Relations connect one object to another. They define the cardinality of objects in a relation, too. Relations are reflected in the user interface by the wires (connections) from one UIBB to another. It is mandatory that the UIBB hierarchy in the overview page is consistent to the genIL object hierarchy as defined by the relations.

SAP provides the following genIL models for MDG-F:

- **MDG 7.0**: Model **MDGF**and its implementation class **CL_MDGF_GENIL_ADAPTER.**
- **MDG 8.0**: Model **MDGFHP**and its implementation class **CL_MDGF_GENIL_ADAPTER_HRY**.

Both models are dynamic genIL models. It is strictly forbidden to change the models manually.
The models are generated by their implementation classes according to the given run-time information of the MDG-F data model **0G**.

Model **MDGF**is used by the single object maintenance user interface for the common entity types and their attributes.

- Each entity type with *SU Type* 1 of the data model is transferred to a genIL root object.
- Each entity type with *SU Type* 4 of the data model is transferred to a genIL dependent object.

- Relations (for example, between entity types with *SU Type* 1 and other entity types with *SU Type* 1; or between entity types with *SU Type* 1 and entity types with *SU Type* 4) are determined and transferred into genIL relations.
- Each entity type with *SU Type* 1 retrieves additional genIL query and query result objects to support the search.
- If an entity type with *SU Type* 1 supports multi-lingual texts, a dependent object is created in genIL to enable the text maintenance within a table.
- If an entity type with *SU Type* 1 supports attachments, two dependent objects are created in genIL to enable the attachment maintenance within a table and related pop-ups.
- The generated structures belonging to an entity are used for the genIL key and attribute structures. This ensures that all fields of the MDG data model **0G** are available for the creation of the related user interfaces. Attribute structures are used by FPM to build the field catalog being available during the user interface creation

Model **MDGFHP** is used by the single object maintenance user interface for the entity types that are used within hierarchies.

- Each entity type with *SU Type* 1 that is used by hierarchies as a root node is transferred into a genIL root object. Those objects are used to build the hierarchy tree in the user interface.
- Each entity type with *SU Type* 1 that defines a hierarchy is transferred into a genIL dependent object. Those objects are used to build the hierarchy tree in the user interface.
- Each entity type with *SU Type* 1 that is used by hierarchies as a root node, a grouping node, or leaf node is transferred into a genIL access object. Those objects are used to build the hierarchy assignment lists in the user interface.
- Each entity type with *SU Type* 1 that is used by hierarchies as a root node, a grouping node, or a leaf node retrieves an additional genIL query objects to support the search. The search result is displayed as hierarchy tree.
- If a hierarchy defines hierarchy attributes for a specific entity type with *SU Type* 1, a corresponding genIL dependent object is created additionally.

3.34 Caution

Enhancements of the MDG-F data model 0G are reflected immediately after activation of the data model in the genIL components. A manual change or enhancement of the genIL components is strictly forbidden.

If enhancements in genIL are required, all changes have to be implemented in a related genIL implementation class. It is mandatory that this class inherits from the SAP classes **CL_MDGF_GENIL_ADAPTER**respectively **CL_MDGF_GENIL_ADAPTER_HRY**.

3.35 Context Based Adaptations

A context based adaptation (CBA) is an FPM concept that allows changing the user interface in a flexible way based upon given values (for example application parameters, user input, and others). A CBA consists of an **Adaptation Schema** that consists of one or more **Adaptation Dimensions.**

- The pre-defined adaptation schema for MDG-F is **MDG_FIN**.
- The adaptation schema includes the following adaptation dimensions:
  - **USMD_OTC**: usable for adaptations according to the current business object type code.
  - **ACTION**: usable for adaptations according to the current logical action like “create” or “mark for deletion”.
  - **CRTYPE**: usable for adaptations according to the current change request type.
  - **WFSTEP**: usable for adaptations according to the current workflow step.

Using both the adaptation schema and its dimensions, you can create various adaptations of the user interface (for example changing the layout of an overview page (OVP), or either adding rows to or removing rows from a list UIBB, and so on). You can combine several dimensions to create a very specific adaptation.

The CBA concept is based on the common FPM event handling. You can trigger one or more CBA events that are handled by FPM's event loop processing. Unfortunately FPM handles multiple CBA events one after the other. It does not cumulate the dimension information given within each event (which basically means that the final CBA event wins).

3.36 Work Centers, Roles and Landing Pages

Work centers and roles are used within MDG for the creation of so called landing pages. A landing page is considered as a single point of entry for an end user to the MDG-F specific user interface.

Work centers define a grouping of the different MDG-F entity types into a single menu role. Additionally each role uses the common MDG components such as data replication, change request, editions, and so on.

Three work center menu roles are pre-defined according to the MDG-F data model setup:

- **SAP_MDGF_ACC_MENU_04**: reflects all accounting entity types.
- **SAP_MDGF_CTR_MENU_04**: reflects all controlling entity types.
- **SAP_MDGF_CO_MENU_04**: reflects all consolidation entity types.

Next to the menu roles as mentioned above, authorization roles are pre-defined, too. The authorization roles follow the general role setup of all MDG based applications:

- The **Display Role** contains all authorizations to display master data, change requests, change documents, replication status, and other information for the respective work center data.
  The role is assigned to an auditor, who is then authorized to display all information, but not to change anything.

- The **Requester Role** contains all authorizations to create and display change requests within the respective work center. The role is assigned to a business user, who is then able to request changes for master data.
- The **Specialist Role** contains all authorizations to process and approve change requests in the respective work center.
- The **Steward Role** contains all authorizations needed for special tasks that involve the processing and replication of master data within the respective work center. The role is assigned to the user responsible for the quality of the master data. It is recommended to combine this role with the specialist.

The combination of the work center and the explained general authorization roles results in the following pre- defined roles:

Release	Role	Description
MDG 7.0	SAP_MDGF_ACC_DISP_04	MDG-F: Accounting Display MDG-F:
	SAP_MDGF_ACC_REQ_04	Accounting Requester MDG-F: Accounting
	SAP_MDGF_ACC_SPEC_04	Specialist MDG-F: Accounting Data
	SAP_MDGF_ACC_STEW_04	Steward MDG-F: Controlling Display
	SAP_MDGF_CTR_DISP_04	MDG-F: Controlling Requester MDG-F:
	SAP_MDGF_CTR_REQ_04	Controlling Specialist MDG-F: Controlling Data Steward
	SAP_MDGF_CTR_SPEC_04	MDG-F: Consolidation Display MDG-F:
	SAP_MDGF_CTR_STEW_04	Consolidation Requester MDG-F: Consolidation
	SAP_MDGF_CO_DISP_	Specialist MDG-F: Consolidation Data Steward

	04 SAP_MDGF_CO_REQ_ 04 SAP_MDGF_CO_SPEC _04 SAP_MDGF_CO_STEW_0 4	
MDG 8.0[1]	SAP_MDGF_ACC_REQ_0 6 SAP_MDGF_CO_REQ_06 SAP_MDGF_CTR_REQ_06	MDG-F: Accounting Requester MDG-F: Consolidation Requester MDG-F: Controlling Requester

Each pre-defined role must be considered as a template. The roles must not be assigned directly to user. An SAP template role must be copied and adjusted according to the customer project requirements.

3.37 MDG-F User Interfaces

All software artifacts related to the new MDG-F user interfaces for single object maintenance are stored in package **USMDZ10**.

A complete user interface consists of the following elements:

- The application configuration that defines the general settings of a user interface including application parameters.
- The communicator settings configuration that defines the search of each overview page. The communicator settings configuration must have the same name as the application configuration.
- The Overview Page (OVP) that combines one or more form UIBBs and/or list UIBBs for the actual data maintenance. Forms, lists and their related feeder classes can be identified by their naming.
  - Web Dynpro Component **MDGF_0G_CCTR**belongs to the cost center.
  - Feeder class **CL_MDGF_GUIBB_FI_ACCOUNT**belongs to the account.

The new user interface uses the business object type code (OTC) as a key parameter. The OTC links the data model and the entity type with the user interface application. It is mandatory that each user interface uses an OTC. Pre-defined combinations are listed in the next chapters.

3.37.1 Accounting

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_FI_ACCOUNT **MDGF_0G_FI_ACCOUNT_OVP 892**	**ACCOUNT** **ACCCCDET**	The OVP combines two entity types with SU Type 1 for account master data maintenance in a single user interface. Accounts can be assigned to financial reporting structure hierarchies in collective processing.
MDGF_0G_OVP_COMPANY **MDGF_0G_COMPANY_OVP 154**	**COMPANY**	The OVP represents the company master data.

MDGF_0G_OVP_FI_REPORT **MDGF_0G_FI_REPOR** **T_OVP 901**	**FRS** **FRSI**	The OVP combines two entity types with SU Type 1 for financial reporting structure maintenance in a single user interface. The user interface creates single entities that can be used to define the complete financial reporting structure hierarchy in collective processing.

3.37.2 Controlling

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_CCTR **MDGF_0G_CCTR_** **OVP 158**	**CCTR**	The OVP represents the cost center master data. The assignment of the cost center to its (standard) hierarchy must be done in collective processing.
MDGF_0G_OVP_CCTRG **MDGF_0G_CCTRG** **_OVP 895**	**CCTRG**	The OVP represents cost center groups. The assignment of the cost center groups to hierarchy nodes must be done in collective processing.
MDGF_0G_OVP_CCTRH **MDGF_0G_CCTRH** **_OVP 897**	**CCTRH**	The OVP represents the root node of a cost center group hierarchy. Hierarchy maintenance must be done in collective processing.
MDGF_0G_OVP_CELEM **MDGF_0G_CELEM** **_OVP 983**	**CELEM**	The OVP represents the cost element master data. The assignment of the cost center to its hierarchy must be done in collective processing.
MDGF_0G_OVP_CELEMG **MDGF_0G_CELEM** **G_OVP 984**	**CELEMG**	The OVP represents cost element groups. The assignment of the cost element groups to hierarchy nodes must be done in collective processing.
MDGF_0G_OVP_CELEMH **MDGF_0G_CELEM** **H_OVP 985**	**CELEMH**	The OVP represents the root node of a cost element group hierarchy. Hierarchy maintenance must be done in collective processing.
MDGF_0G_OVP_PCTR **MDGF_0G_PCTR_** **OVP 229**	**PCTR**	The OVP represents the profit center master data. The assignment of the profit center to its (standard) hierarchy must be done in collective processing.
MDGF_0G_OVP_PCTRG **MDGF_0G_PCTRG** **_OVP 896**	**PCTRG**	The OVP represents profit center groups. The assignment of the profit center groups to hierarchy nodes must be done in collective

processing.

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_PCTRH **MDGF_0G_PCTRH_** **OVP 898**	**PCTRH**	The OVP represents the root node of a profit center group hierarchy. Hierarchy maintenance must be done in collective processing.

3.37.3 Consolidation

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_BDC **MDGF_0G_BDC_OVP** **MDGF_BCG**	**BDC**	The OVP represents the breakdown category master data. Breakdown categories include breakdown category sets. They can be used to refine items.
MDGF_0G_OVP_BDCSET **MDGF_0G_BDCSET_O** **VP MDGF_BCS**	**BDCSET**	The OVP represents breakdown category sets.
MDGF_0G_OVP_CONSCH AR **MDGF_0G_CONSCHAR_OV** **P MDGF_CCC**	**CONSCHA R**	The OVP represents the consolidation characteristic master data.
MDGF_0G_OVP_CONSGR P **MDGF_0G_CONSGRP** **_OVP 904**	**CONSGRP**	The OVP represents the consolidation group. The assignment of the consolidation groups to hierarchy nodes must be done in collective processing.
MDGF_0G_OVP_CONSGR P **MDGF_0G_CONSGRP** **_OVP 894**	**CONSGRP H**	The OVP represents the root node of a consolidation group hierarchy. Hierarchy maintenance must be done in collective processing.
MDGF_0G_OVP_CONSUNI T **MDGF_0G_CONSUNIT** **_OVP 905**	**CONSUNIT**	The OVP represents the consolidation unit master data. Consolidation units can be assigned to consolidation group hierarchies in collective processing.

MDGF_0G_OVP_CO_ACCOUNT **MDGF_0G_CO_ACCOUNT_OVP 900**	**FSI**	The OVP represent the item. Items can be assigned to item hierarchies in collective processing.

Application / OVP / OTC	Entity Type(s)	Description
MDGF_0G_OVP_CO_REPORT **MDGF_0G_CO_REPORT_OVP 902**	**FSIH** **FSIT**	The OVP combines two entity types with SU Type 1 for item hierarchy maintenance in a single user interface. The user interface creates single entities that can be used to define the complete item hierarchy in collective processing.
MDGF_0G_OVP_SUBMPACK **MDGF_0G_SUBMPACK_OVP MDGF_SMP**	**SUBMPACK**	The OVP represents the cause for submission. A cause for submission is used to refine items.
MDGF_0G_OVP_TRANSTYPE **MDGF_0G_TRANSTYPE_OVP MDGF_TTP**	**TRANSTYPE**	The OVP represents the transaction types.

3.37.4 User Interfaces for Collective Processing

The previous chapters Accounting, Controlling and Consolidation listed user interfaces for single object maintenance.

Collective processing summarizes the user interfaces for both mass change and for hierarchy maintenance. The related user interfaces are not refactored. They are accessible through links in the related work centers.

3.37.5 Search and Value Helps for Single Object Maintenance User Interfaces

MDG-F uses the **generic text helper** functionality provided by the MDG framework with class **CL_USMD_GENERIC_GENIL_TEXT**. The text helper is able to provide values of domains and check tables that have only a single key field. This functionality is combined with the **OVS search** technology of the user interface:

- Fields having domain fixed values are usually displayed as drop-down list boxes in the user interface. The values of the drop-down list box are determined by the text helper.
- Fields having simple check tables are usually displayed as input fields including a related read-only field for the description. MDG-F uses the generic OVS search implementation

provided by FPM (see class **CL_MDGF_GUIBB_FORM**). The text helper is called indirectly by the MDG-F specific genIL implementation.

- Fields having complex check tables or depending on other values are usually displayed as input fields including a related read-only field for the description. MDG-F implements specific OVS search helps in the related feeder classes since this cannot be done with a generic approach.

There are two options for filtering search help results:

- The text helper includes the enhancement spot **USMD_GENERIC_GENIL_TEXT** respectively BAdI **USMD_GENERIC_GENIL_TEXT_BADI**. They are usable to restrict the values being shown on the user interface for domains or simple check tables.
- The generic form feeder class **CL_MDGF_GUIBB_FORM**provides a rather generic OVS filter method called **OVS_OUTPUT_FILTER**. You can enhance this logic (for example, using an implicit enhancement at the end of the method) with your custom filters.

3.38 Data Transfer

Data Transfer in MDG-F covers both the initial load of master data into and MDG system and the replication of created or modified data to client systems.

The initial load is used for preparing the MDG system with the currently existing data of client systems. The initial load can be accomplished using a file-based approach or using the MDM Generic Extractor. When setting up the system it is important to load all objects that shall be put under governance completely. For time-dependent objects such as Profit Centers or Cost Centers, it is ensured that all existing time slices of a single object are loaded to MDG.

Data replication is possible using either SOA Services or ALE IDocs. Combining both for the same receiving system is not recommended. IDocs are only available for accounting and controlling entity types.

Consolidation entity types do not exist within SAP ECC systems and thus do not offer IDocs.

SAP Note (Un-) Supported Segments & Fields offers a detailed overview about all fields covered by MDG-F and how they are used in the different replication technologies.

Data replication supports different modes that are explained in the next chapters.

3.39 Object-based Replication

This is a new feature provided by MDG 7.0. It enables the immediate replication of an existing object, for example by selecting the same in the search result list and choosing the **Replicate** button. It is possible to select one or more target systems for the current replication run. The edition cannot be selected. The outbound implementation always uses the currently valid data for non-time-dependent objects respectively all valid time slice for time-dependent objects. It is not possible to replicate outdated objects.

3.40 Change-Request based Replication

The change-request based replication has not been changed. It is triggered once the change request has been activated. If the changed object is non-time-dependent and the change happened for a time slice that is not valid (for example, in an outdated or future edition), the object is not replicated. This prevents overwriting non-time-dependent objects with invalid data in the target systems.

3.41 Edition-based Replication

The edition-based replication has not been changed. It still requires a released edition to replicate all objects of the same edition. For non-time-dependent objects the same rules as defined by the change-request based replication apply. The system prevents the replication of invalid time slices.

3.42 Replication of Deletions

MDG-F supports the replication of deletion only using SOA messages. It is not possible to use ALE IDocs. A “Deletion” means both the actual deletion of a complete object and the deletion of a single time slice to set

the “valid to date” to a specific value.

3.43 Replication of Accounts

MDG-F provides two entity types with **SUT1** for the account.

- The entity type **ACCOUNT** represents the so-called A-Segment of an account. It relates to the SAP ECC table **SKA1** ***Accounts***.
- The entity type **ACCCCDET** represents the so-called B-Segment of an account. It relates to the SAP ECC table **SKB1** ***Accounts in Company Code***.

The predefined replication models for the account reflect this setup.

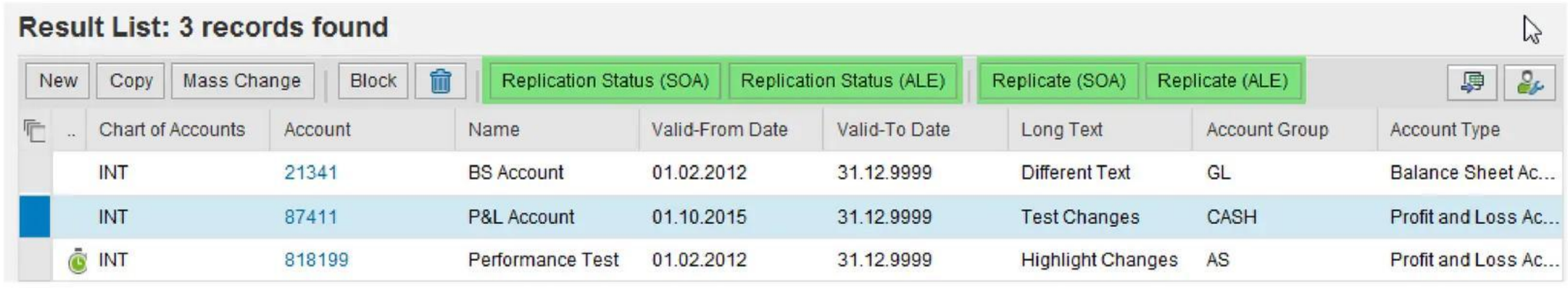
The IDoc based replication uses the common **GLMAST** IDoc type that combines both the A and B Segments. The replication is triggered using the business object type for the general ledger accounts master (892). It is possible to replicate A segments without B segments. If the object list to be replicated contains only B segments, related A segments are added automatically.

The SOA service-based replication is more complex since there are two separate service messages used for the replication.

- Service **ChartOfAccountsReplicationRequest_V1** contains only the A segment without any attributes of the B segment. The replication is triggered using the object type code for the financial accounting chart of accounts (899).
- Service **GeneralLedgerAccountMasterReplicationBulkRequest** contains additional attributes of the A Segment and the complete list of attributes of the B segment. The replication is triggered using the object type code for the general ledger accounts master (892).

The service definition requires that any replication of A segments has to send both of the abovementioned services to ensure data consistency in the receiving system. This is implemented by the corresponding outbound implementations. B segment replication is possible without the chart of accounts service.

A side-effect of the different outbound implementations that are required to ensure data consistency is the design of the search result list for accounts as shown in the screenshot below.



The button **Replication Status (SOA)** shows the status for any replication of accounts using the chart of accounts replication request. The replication itself is triggered with the button **Replicate (SOA)**.

The button **Replication Status (ALE)** shows both the status for any IDoc-based replication of accounts and the status for the SOA-based replication using the general ledger account master service. The replication itself is triggered with the button **Replicate (ALE)**. You choose this button for IDoc based replication only.

3.44 Replication Monitoring

Data replication offers several monitors to check the current replication status.

- The search result list of MDG-F user interfaces offers a link to the DRF replication status. The button

  Replication Status is active for all entity types that support data replication in general.
- ALE IDoc replication offers a specific monitor. You can use backend transaction **WE02** to display both outbound IDocs (on the MDG hub) as well as inbound IDocs (on the MDG client). You can use the monitor for detailed troubleshooting. Post-processing of IDocs is possible using transaction **BD87**.
- SOA Service replication offers a specific monitor. You can use backend transaction **SXMB_MONI** (available until SAP Basis 7.32) to display outbound services (on the MDG hub). You can use backend transaction and **SRT_MONI** (available in SAP Basis 7.40 and newer) to display inbound services (on the MDG client). You can use the monitor for detailed troubleshooting. In the forward error handling, you can post-process services, using backend transaction /**SAPPO**/**PPO2**.

3.45 Validations and Data Derivation

As mentioned before, the data model **0G**uses only **flexible entity types**. This means that there is no access class available that could be used either for validations or for data derivation, or for both. Nevertheless validations and data derivation are needed by the MDG-F entities, too.

MDG-F implements the common MDG rule service BAdI **USMD_RULE_SERVICE**for both validations and derivations. All software artifacts are stored in package **USMDZ7**.

You can enhance the existing validations with custom checks, too. The recommended approach is a custom class that inherits from the pre-defined SAP class to ensure, that all SAP pre-defined validations are executed. SAP Note 2337685 - MDG-F: Standard Validations of Data Model 0G provides a detailed overview of all pre-defined validations in MDG-F.

MDG supports different types of derivation:

- BRF+ based derivation
- Single entity derivation of the rule service BAdI **USMD_RULE_SERVICE**
- Cross entity derivation of the rule service BADI **USMD_RULE_SERVICE_CROSS_ET**

The default implementations provided by SAP use only the cross entity derivation. There is a specific how to guide “Cross Entity Derivation in MDG-F” available that explains all details.

3.46 Data Modeling

3.46.1 Q: Why is there an active data model 0F in the MDG hub?

Data model 0F has been introduced with the very first shipment of MDG-F for SAP ECC 6.0 EhP4. As of today, the data model is outdated and must not be used anymore. Delete the data model in your system.

3.46.2 Q: Is it recommended to copy data model 0G?

It is **not** recommended to create a copy of the data model

0G. The existing data model can be easily enhanced in

various ways:

- Adding new fields to an existing entity type.
- Adding new tables as type 4 entity to an existing entity type.
- Adding new entity types with *SU Type* 1 to the data model.

The generic implementation of the genIL layer simplifies the integration of the enhancements in the user interface.

3.46.3 Q: How do I add custom Data Dictionary search helps to attributes of data model 0G?

The new user interfaces for MDG-F consider the following data dictionary-based search helps:

1. Search helps that are defined within the related data element or domain.
2. Search helps that are defined within the UI configuration, for example using a re-definition of the feeder class method GET_DEFINITION. The UI configuration overrules a definition on data element or domain level.

The new user interfaces **ignore** all data dictionary-based search help that are maintained in the MDG data model customizing views. It is recommended to use the UI configuration instead.

3.47 User Interface Design

3.47.1 Q: Is it recommended to copy existing user interfaces?

It is **not** recommended to copy existing user interfaces.

The existing user interfaces can be easily enhanced in various ways without the need of a copy or a modification of the pre-defined user interfaces:

- You can customize the existing user interface by adding fields or removing fields.
- You can adapt the user interface adapted dynamically using context-based adaptations.
- You can enhance the user interface with custom UIBBs using the enhancement functionality of FPM.
- How-to guide “Extending the Data Model by New Fields in MDG-F” provides a real-life example how the user interface can be enhanced without copying.

A copy might destroy the binding to the improvements and corrections made by SAP.

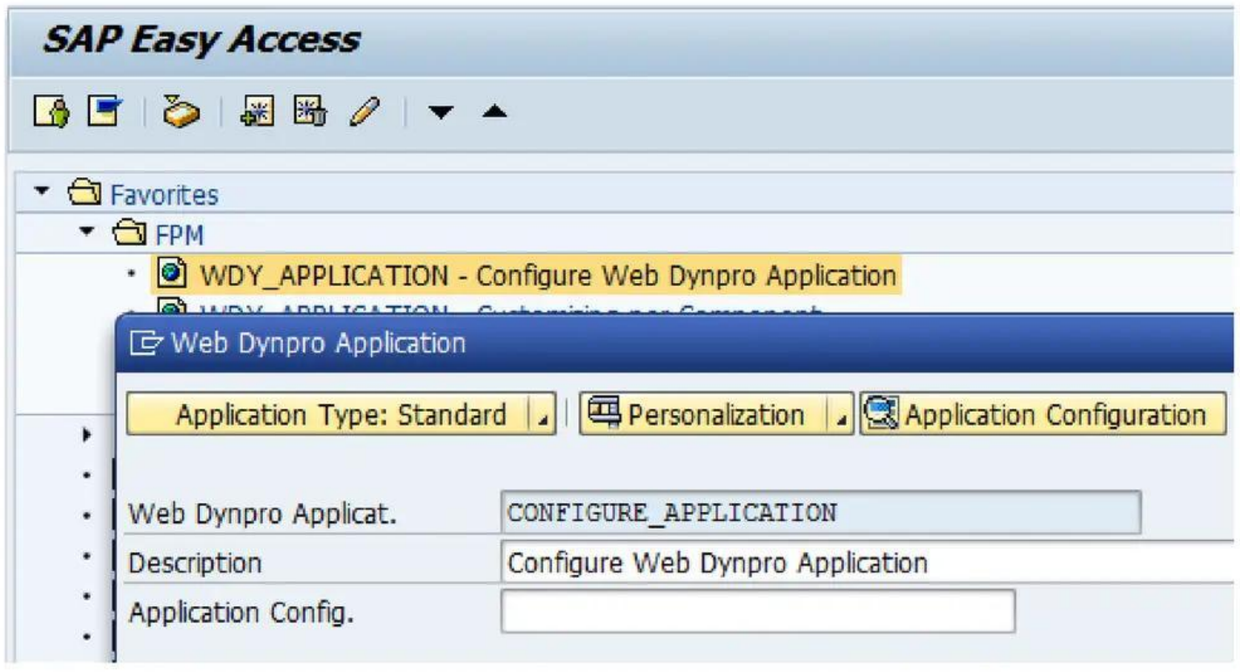
If, in spite of the recommendation, you copy the user interface anyway, ensure that you copy it completely. Remember that each user interface consists of an application configuration, a UI configuration and communicator settings.

3.47.2 Q: How can I start the user Web Dynpro application configurator for user interfaces?

You use the Web Dynpro application configurator to define your Web Dynpro based applications. It defines the main components of the user interfaces (for example the overview pages to be used) as well as application specific parameters.

The most efficient way to start the Web Dynpro application configurator for user interfaces is using a favorite in the SAP Menu.

1. Logon to your MDG Hub system.
2. Create a new entry in you SAP Menu Favorite List.
3. Choose *Web Dynpro Application*.
4. Define the *Web Dynpro Application* as **CONFIGURE_APPLICATION**.
5. Define any *Description*.

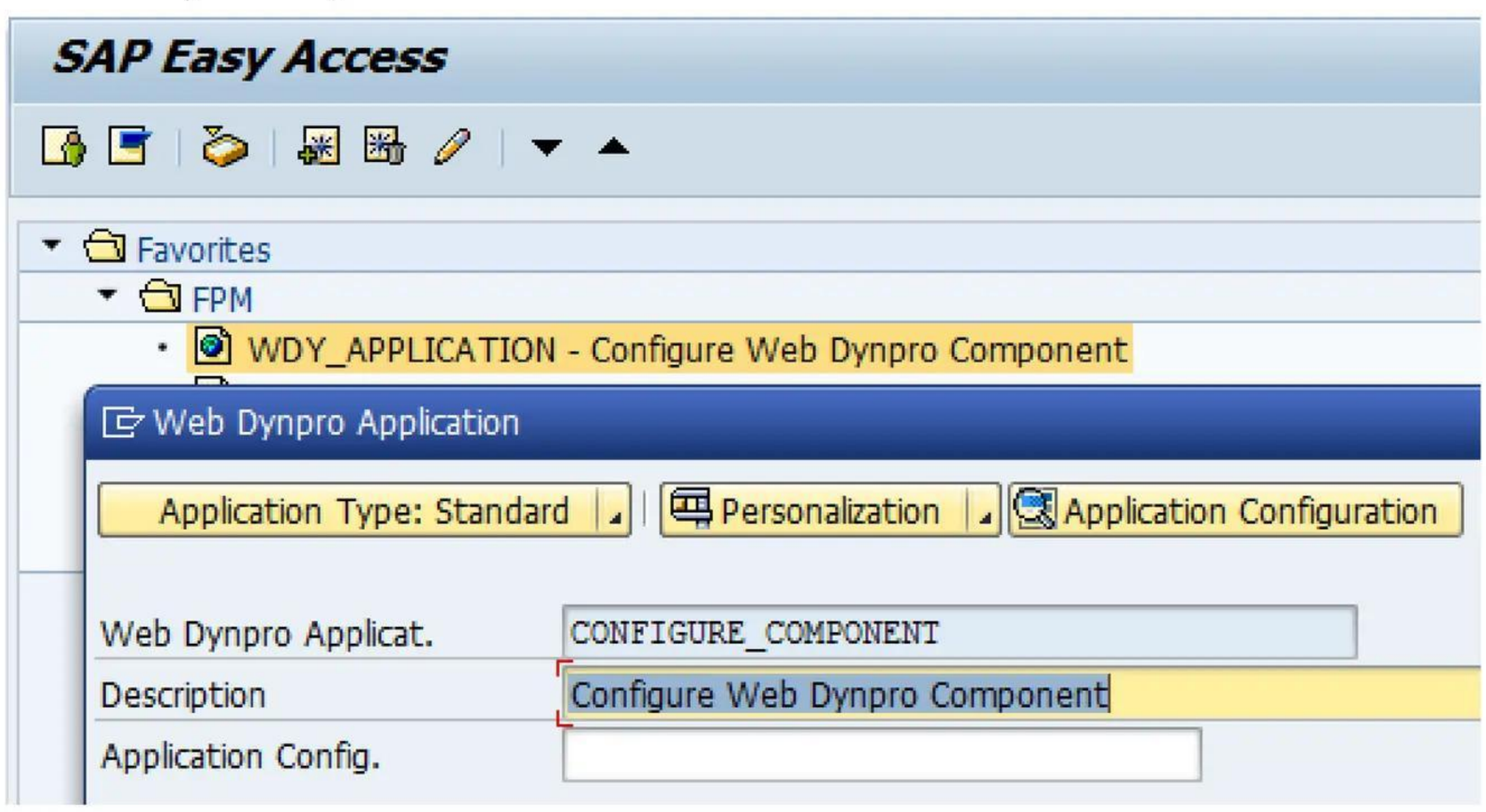

SAP Easy Access
Favorites
FPM
WDY_APPLICATION - Configure Web Dynpro Application
Web Dynpro Application
Application Type: Standard
Personalization
Application Configuration
Web Dynpro Applicat.
CONFIGURE_APPLICATION
Description
Configure Web Dynpro Application
Application Config.

3.47.3 Q: How can I start the user Web Dynpro component configurator for user interfaces?

You use the Web Dynpro component configurator to create your own Web Dynpro components, for example for MDG Communicator settings.

The most efficient way to start the Web Dynpro component configurator for user interfaces is using a favorite in the SAP Menu.

1. Favorite in the SAP Menu
    a. Logon to your MDG Hub system.
    b. Create a new entry in you SAP Menu Favorite List.
    c. Choose *Web Dynpro Application*.
    d. Define the *Web Dynpro Application* as **CONFIGURE_COMPONENT**.
    e. Define any *Description*.

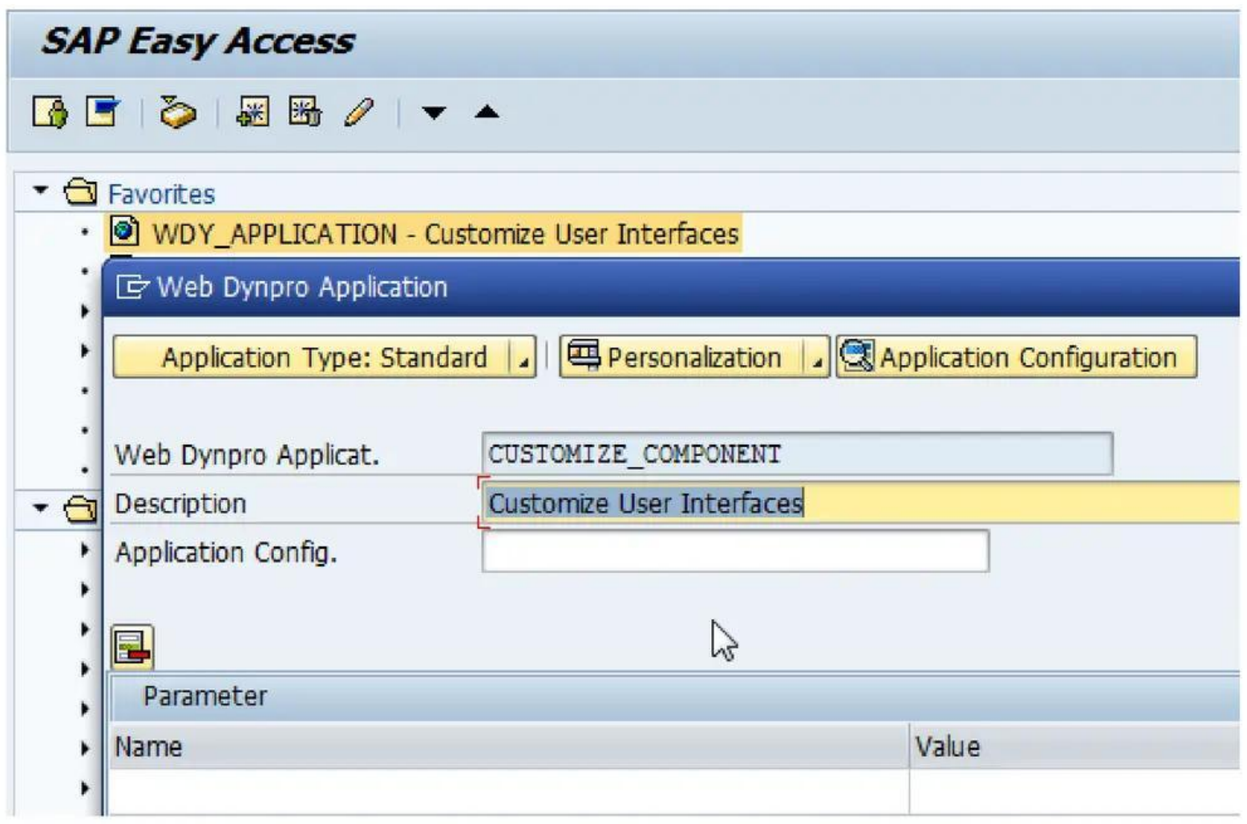


3.47.4 Q: How can I start the user Web Dynpro customizing configurator for user interfaces?

You use the Web Dynpro component customizing configurator to customize SAP defined user interface components like complete overview pages or single forms or lists.

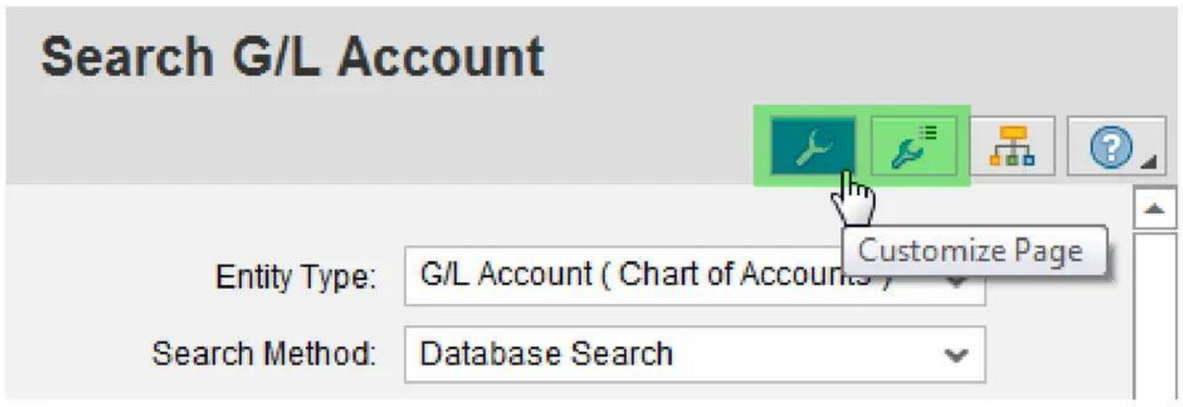
There are different ways to start the Web Dynpro component customizing configurator for user interfaces:

1. Favorite in the SAP Menu
    a. Logon to your MDG Hub system.
    b. Create a new entry in you SAP Menu Favorite List.
    c. Choose *Web Dynpro Application*.
    d. Define the *Web Dynpro Application* as **CUSTOMIZE_COMPONENT**.
    e. Define any *Description*.



2. Configure the own user as FPM Configuration Expert
    a. Logon to your MDG Hub system.
    b. In the top-menu bar choose *System* □ *User Profile* □ *Own Data*.
    c. Switch to tab *Parameters*.
    d. Enter *Parameter ID* **FPM_CONFIG_EXPERT**and set its value to **X**.
    e. Save the changes.

If you now start any user interface that is based upon FPM, you'll notice two new buttons in the upper right area of the user interface. You can use both buttons to start the customizing configurator. The screenshot below uses the search for account as

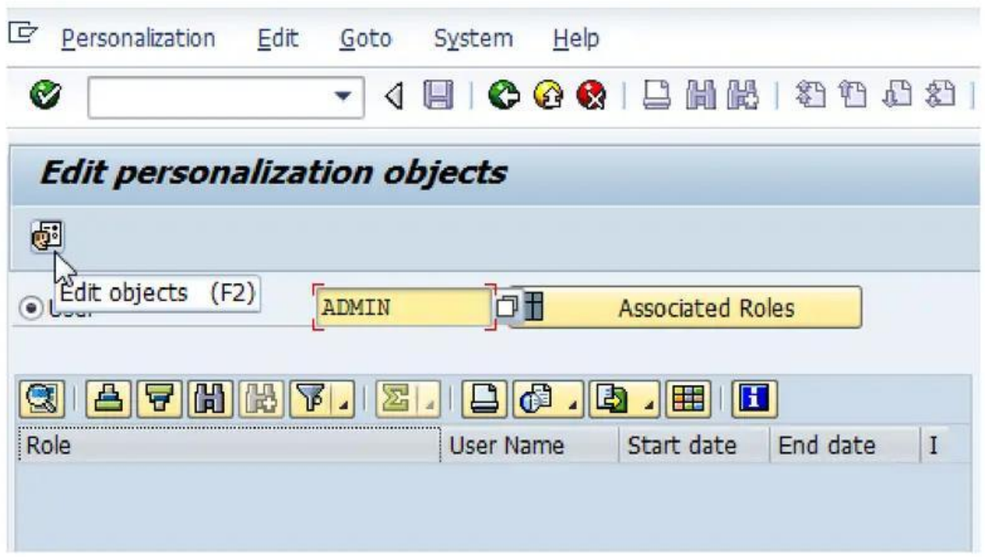


example.

3.47.5 Q: Why is there a short dump when I try to customize a user interface?

Before you can create or modify user interface customizing, you must define the **Standard Data Model** in your user personalization. The model must be set according to the user interface that you want to customize. Ensure that the value is set to **0G**for MDG-F related user interfaces:

1. Start transaction **SPERS_MAINT**.
2. Enter your user name and choose *Edit objects (F2)* above your user name.



3. Use the search with search term *MDM* to locate the entry *SAP Master Data Governance – **R_FMDM_MODEL***in the table of personalization objects.
4. Double-click on the entry.
5. In the pop-up *Edit personalization objects* set the *Standard Data Model* to *0G*.

6. Save your changes.

3.47.6 Q: What does the error message “Feeder class does not provide a valid field catalog” mean?

This error message might be displayed if you try to open the web dynpro component configuration for any UIBB. The root cause is the same as for question “Why is there a short dump when I try to customize a user interface?”. The SAP New Weaver Basis release 7.40 displays the error message instead of causing a short dump. To resolve the issue you need to define the **Standard Data Model** as **0G** as described above.

3.47.7 Q: Why is there still a dump / error message although I have defined the Standard Data Model as 0G?

Technically the assignment of the standard data model to your user is stored in data base table **USMD2000**. You could either check by yourself or ask your system administrator to check, if there is a valid entry for your user in the table.

In systems that have been upgraded from an older MDG-F release to MDG-F 7.0 (or newer), it might be possible that:

a) Your user is still assigned to the old MDG-F roles.
b) The old MDG-F roles define a standard data model via customizing table **USMD2001**.

If this is the case, the standard data model being displayed by transaction **SPERS_MAINT** is determined according to your roles. This does not work for the user interface configuration of MDG-F 7.0. The invalid customizing settings (especially, the role assignment for your user to outdated MDG-F roles) have to be deleted. Afterwards you can define the standard data model.

3.47.8 Q: Why does the context based adaptation (CBA) not change the layout of the overview page (OVP)?

You can only use CBAs to change the layout of an OVP during the startup of the application. It is not possible to change the OVP (for example, the sequence of UIBBs) using a CBA during UI round-trips. CBAs can only change the layout of single UIBBs for each round-trip.

3.47.9 Q: Why are recent changes applied to the user interface not visible for other users?

The actual UI that is being displayed to a user in the web browser is determined from various components of the UI configuration:

- Personalization
- Enhancements
- Context Based Adaptations
- Base Configuration

The general rule is that the personalization is the strongest component. This is best explained with an example:

The base configuration defines the overview page as a list of UIBBs. Since a user does not want to scroll, he or she creates a personalization of the page introducing a stacking of the UIBBs in tab-strips. A UI designer decides to create a context-based adaptation that sets a single UIBB to “hidden and excluded from event loop”. All users not having a personalization no longer see this UIBB anymore. The user with the personalization set is unaffected by this change. This is because the UIBBs that are hidden and excluded from event loop still belong to the OVP. They can be added to the OVP using personalization. Since the user has created a personalization that shows the UIBB (the personalization was created before the CBA), the UIBB is still visible. To exclude the UIBB, you must either reset personalization or delete the UIBB in the CBA.

System administrators might use the ABAP Web Dynpro application **WD_ANALYZE_CONFIG_USER** to check and/or reset user personalization centrally.

3.47.10 Q: Create a new User Interface for custom Entity with Storage and Use Type 1?

If you have enhanced the standard data model 0G with a custom entity type with SUT 1, you need to create a new user interface, too. Follow the steps mentioned in the SAP Help for Creating User Interfaces for Single Object Processing. Of course you can use any of the existing MDG-F user interfaces as template.

3.48 Data Maintenance

3.48.1 Q: What does the error message “Relation <ABC> is not supported by Object <XYZ> mean?”

This error message is thrown by the genIL layer if the user interfaces requests a relation that is unknown in genIL.

This could be caused by an invalid Wire Configuration between UIBBs of the Overview Page. You should first check if both the genIL model and the OVP wiring are consistent.

In productive customer systems it could be the case that the shared memory component used by genIL is outdated or corrupt (genIL uses shared memory only in productive systems but not in development or test system). You can use transaction **SHMM** to invalidate the shared memory area **CL_CRM_GENIL_MODEL_SHM_AREA**. Afterwards simply restart the user interface.

3.48.2 Q: Why does the Undo button not work for key fields on the Overview Page?

All pre-defined MDG-F entities with **SUT1** do not allow key changes. This rule applies to choosing the ***Undo*** button as well. After you enter a valid value for a key of the main entity, you cannot undo this action. You can also not undo any attribute change you have done at the same time as the key change.

The MDG-F 7.0 Feature Pack offers changing IDs. The how to guide Enable changeable IDs in MDG-F explains the new functionality in more details.

3.48.3 Q: How can I search for an Edition instead of a Valid On date?

All database searches of the MDG-F user interfaces support both searching for specific validity dates or editions. The pre-defined user interface configurations use only the validity dates. If you want to switch to the edition instead, you need to change the **application configuration** of the related user interface.

1. Copy a pre-defined application configuration (for example **MDGF_0G_OVP_CCTR**for Cost Centers) to your custom namespace.
2. Edit the application parameter ***USMD_SEARCH_EDITION_MODE***and set its value to X. Note that you might have to adapt existing links (for example in the work centers) accordingly.

3.48.4 Q: Why does the Search not find my Object that I have saved as Draft?

The minimal requirements for saving an object as a draft are the object keys. Once the keys are defined, saving is possible. Nevertheless, MDG does not create an inactive record in that case but stores the object only within the object list of a change request. The object list is not accessed by the search. The search can only display active and inactive records. You have to access a draft that contains only the object key via the My Change Request application. Alternatively ensure to maintain at least a single object attribute besides the object keys before saving the object as draft.

3.48.5 Q: Why does the Print Preview not work in my copied Application Configuration?

The print forms that are used for the print preview functionality are bound to the application configuration. If you have copied the application configuration to a custom version, you need to create new entries in the related customizing table:

1. Start transaction *MDGIMG* and navigate to *General Settings* □ *UI Modeling* □ *Assign Print Forms for Single Processing*.
2. Mark data model *0G* and double click on *Forms*.
3. Add entries using the name of your custom application configuration. The screenshot below shows the default values pre-defined by SAP.

Change View "Forms": Overview

New Entries | BC Set: Change Field Values

Dialog Structure
- Data Models
  - Forms

Data Model: 0G

Forms

UI Configuration	Entity Type	Form
MDGF_0G_OVP_BDC	BDC	USMD_0G_BDC_PDF
MDGF_0G_OVP_BDCSET	BDCSET	USMD_0G_BDCSET_PDF
MDGF_0G_OVP_CCTR	CCTR	USMD_0G_CCTR_PDF_02
MDGF_0G_OVP_CCTRG	CCTRG	USMD_0G_CCTRG_PDF
MDGF_0G_OVP_CCTRH	CCTRH	USMD_0G_CCTRH_PDF
MDGF_0G_OVP_CELEM	CELEM	USMD_0G_CELEM_PDF
MDGF_0G_OVP_CELEMG	CELEMG	USMD_0G_CELEMG_PDF
MDGF_0G_OVP_CELEMH	CELEMH	USMD_0G_CELEMH_PDF
MDGF_0G_OVP_COMPANY	COMPANY	USMD_0G_COMPANY_PDF
MDGF_0G_OVP_CONSCHAR	CONSCHAR	USMD_0G_CONSCHAR_PDF
MDGF_0G_OVP_CONSGRP	CONSGRP	USMD_0G_CONSGRP_PDF
MDGF_0G_OVP_CONSGRPH	CONSGRPH	USMD_0G_CONSGRPH_PDF
MDGF_0G_OVP_CONSUNIT	CONSUNIT	USMD_0G_CONSUNIT_PDF
MDGF_0G_OVP_CO_ACCOUNT	FSI	USMD_0G_FSI_PDF
MDGF_0G_OVP_CO_REPORT	FSIH	USMD_0G_FSIH_PDF
MDGF_0G_OVP_CO_REPORT	FSIT	USMD_0G_FSIT_PDF
MDGF_0G_OVP_FI_ACCOUNT	ACCCCDET	USMD_0G_ACCCCDET_PDF
MDGF_0G_OVP_FI_ACCOUNT	ACCOUNT	USMD_0G_ACCOUNT_PDF
MDGF_0G_OVP_FI_REPORT	FRS	USMD_0G_FRS_PDF
MDGF_0G_OVP_FI_REPORT	FRSI	USMD_0G_FRSI_PDF
MDGF_0G_OVP_PCTR	PCTR	USMD_0G_PCTR_PDF_02
MDGF_0G_OVP_PCTRG	PCTRG	USMD_0G_PCTRG_PDF
MDGF_0G_OVP_PCTRH	PCTRH	USMD_0G_PCTRH_PDF
MDGF_0G_OVP_SUBMPACK	SUBMPACK	USMD_0G_SUBMPACK_PDF
MDGF_0G_OVP_TRANSTYPE	TRANSTYPE	USMD_0G_TRANSTYPE_PDF

3.49 Data Replication

3.49.1 Q: How does the MDG Hub handle key mapping for MDG-F entities?

Key mapping is not written automatically, for example it is not written during data import or data replication. If you require key mapping in your replication scenarios, you have to define the key mapping for each object manually. The key mapping maintenance can be accessed via the common user interface using its ***Show*** button.

3.49.2 Q: Why are attachments not replicated to ERP?

Most of the MDG-F entities support attachments. This enables adding documents and/or links to the master data. The attachments are a MDG specific functionality. They are not related to the SAP ECC document features and thus not part of data replication. Furthermore the IDOC and service interfaces of the financial objects do not support attachments.

3.49.3 Q: Why don’t ERP transactions show a change history for master data changed in MDG?

When object changes in MDG are replicated to ERP via IDOCs or services the ERP inbound works with a simple and efficient logic: all existing data are being deleted before the changed data coming from MDG are inserted. Unfortunately, no change documents are created for the deletion and the insert.

For this reason, we recommend using the change tracking in MDG when you run central master data maintenance.

3.49.4 Q: Replicate MDG-F entities into the same client of my system?

MDG-F entities are completely stored within the generated tables of MDG. If you want to use the same system and client for productive usage of the data within SAP ERP, too, you need to replicate the MDG-F data into the same system and client. How-to guide “ALE Replication from MDG Hub to ERP using the Same Client in MDG-F” explains the required configuration.

3.49.5 Q: Why are some of my Account IDocs not processed immediately?

The ALE IDoc based replication might show some IDocs with *status* **64**and the following messages in the target system:

- **IDoc: <IDoc Number> Status: IDoc ready to be transferred to application**
- **Immediate Processing was Canceled**

This might happen if the replication contained both the creation and change of accounts or accounts in company codes at the same time. If IDocs are processed in parallel by the receiving system, locking issues may occur.

We recommend using transaction **WE02**in the target system to check the status of the IDocs. You can also use transaction **BD87**to post-process the IDoc.

3.50 SAP Notes

- 1637249 specifying required information for OSS support
- 2105467 specifying required information for Performance Issues

3.51 ALE Replication using the same Client in MDG-F

For the replication of master data created or changed in MDG into the client systems (ERP) SAP recommends the usage of enterprise services or ALE IDocs. The standard replication scenarios require that MDG hub and MDG client are two different systems, or at least two different physical clients within the same system.

For MDG-F, however, customers might want to run the MDG hub in the same physical client as the ERP client due to the nature of the MDG-F data model. Since all MDG-F entity types use the flexible mode their data is stored only in the MDG generated tables and not within ERP tables.

This document explains how-to setup an ALE scenario for data replication where MDG hub and ERP client share the same physical client of the same system.

We recommend that you study the following how-to guide before working with the current guide:

Extensibility Options for SAP Master Data Governance for Financial Data Overview about MDG-F

Strong ALE configuration expertise is required, too.